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McGarrigle SA, Hanhauser YP, Mockler D, Gallagher DJ, Kennedy MJ, Damen JAAG, Bennett K, Connolly EM, Supported by the Cochrane Breast Cancer Review Group

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[Prognosis Review]

Risk prediction models for familial breast cancer

Sarah A McGarrigle^{1,2a}, Yvonne P Hanhauser², David Mockler³, David J Gallagher⁴, Michael J Kennedy⁵, Johanna AAG Damen⁶, Kathleen Bennett⁷, Elizabeth M Connolly^{1,2}, Supported by the Cochrane Breast Cancer Review Group

¹Department of Surgery, Trinity College Dublin, Trinity St James's Cancer Institute, Dublin, Ireland. ²Breast Care Department, St James's Hospital, Dublin, Ireland. ³John Stearne Library, Trinity Centre for Health Sciences, St James's Hospital, Dublin, Ireland. ⁴HOPE Directorate, St James's Hospital and Trinity College Dublin, Dublin, Ireland. ⁵Trinity St James's Cancer Institute, Dublin, Ireland. ⁶Cochrane Netherlands, Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht University, Utrecht, Netherlands. ⁷School of Population Health, RCSI University of Medicine and Health Sciences, Dublin, Ireland

^aORCID iD: <https://orcid.org/0000-0003-4404-6564>

Contact: Sarah A McGarrigle, smcgarrigle@stjames.ie.

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ABSTRACT

Background

Women with a family history of breast cancer have an elevated risk of developing the disease. In clinical practice, the probability of developing breast cancer over a specified timeframe is frequently estimated using breast cancer risk prediction models. It is currently unclear which of the available models performs best in women with a breast cancer family history.

Objectives

To identify, describe, and appraise breast cancer risk prediction models developed or validated in women with a family history of breast cancer, and to meta-analyse their performance in predicting breast cancer occurrence.

Search methods

We searched MEDLINE, Embase, Cumulative Index to Nursing and Allied Health Literature, and the Institute of Scientific Information Web of Science to February 2022, with a targeted top-up search of MEDLINE to 19 December 2024 to capture additional validation studies of included models. We also screened reference lists of included studies.

Selection criteria

We included studies that developed or validated a breast cancer risk prediction model(s) in women with a family history of breast cancer, if the model(s) in question included family history of breast cancer among its predictors.

Data collection and analysis

We based our data extraction form on the CHARMS checklist. Two authors independently screened references, extracted data, and assessed risk of bias using the Prediction model Risk Of Bias ASsessment Tool (PROBAST). We summarised risk prediction models developed or validated in the target population. Where performance statistics were reported by at least four studies, we pooled model performance measures using random-effects meta-analyses. We assessed model performance using calibration (agreement between predicted risks and observed breast cancer occurrences) and discrimination (ability to distinguish between women who did and did not develop breast cancer). We did not apply GRADE because guidance for prognostic model reviews is not yet available.

Main results

We included 45 studies and listed 17 as 'awaiting classification'. We identified 12 externally validated models in the target population. We meta-analysed four models that had at least four external validation studies.

Reporting of studies varied. Several did not adequately report follow-up time or handling of missing data. Most validation studies reported at least one performance measure (calibration or discrimination), though some did not report both together. Basic details, such as validated model versions, were often missing.

Most studies had a high or unclear risk of bias based on PROBAST ratings, although concerns about applicability were generally low.

We report results for four models which had data available from at least four external validation studies in the target population.

Gail/Breast Cancer Risk Assessment Tool (BCRAT)

Calibration: the pooled observed (O)/expected (E) ratio of the Gail model (combined versions) in the target population was 1.06 (95% confidence interval (CI) 0.91 to 1.25), indicating that the model is well calibrated in this population. The 95% prediction interval (PI) was 0.65 to 1.74.

Discrimination: the pooled estimate for the C statistic of the Gail model (combined versions) in the target population was 0.61 (95% CI 0.57 to 0.66). The 95% PI was 0.47 to 0.74.

Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS)

Calibration: the pooled estimate for the O/E ratio of the Tyrer-Cuzick model in the target population was 0.86 (95% CI 0.74 to 0.98) (combined versions) and 0.90 (95% CI 0.76 to 1.06) (version 8), indicating that the model overpredicts breast cancer risk in this population. The 95% PI was 0.56 to 1.33 (combined versions) and 0.55 to 1.47 (version 8).

Discrimination: the pooled estimate for the C statistic of the Tyrer-Cuzick model (combined versions) in the target population was 0.62 (95% CI 0.58 to 0.66). The 95% PI was 0.49 to 0.74. The pooled C statistic for version 8 of the Tyrer-Cuzick model was 0.64 (95% CI 0.58 to 0.71). The 95% PI was 0.46 to 0.79.

Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm (BOADICEA)

Calibration: the pooled estimate for the O/E ratio of BOADICEA (combined versions) in the target population was 0.98 (95% CI 0.90 to 1.17), indicating that the model is well calibrated in this population. The 95% PI was 0.89 to 1.09.

Discrimination: the pooled estimate for the C statistic of BOADICEA (combined versions) in the target population was 0.65 (95% CI 0.58 to 0.71). The 95% PI was 0.44 to 0.81.

BRCAPRO

Calibration: the pooled estimate for the O/E ratio of the BRCAPRO model (combined versions) in the target population was 1.44 (95% CI 1.25 to 1.62), indicating that the model underpredicts breast cancer risk in this population. The 95% PI was 1.02 to 2.04.

Discrimination: the pooled estimate for the C statistic of BRCAPRO (combined versions) in the target population was 0.64 (95% CI 0.54 to 0.73). The 95% PI was 0.37 to 0.84.

Authors' conclusions

Our meta-analyses showed that the Gail (BCRAT) and BOADICEA models are well calibrated in women with a family history of breast cancer. The Tyrer-Cuzick (IBIS) model overpredicts risk, while BRCAPRO underpredicts risk in this population.

In terms of discriminatory accuracy in the target population, no model was clearly superior. Tyrer-Cuzick version 8, BOADICEA, and BRCAPRO showed similar modest discrimination in our meta-analyses, which was slightly better than that of the Gail model.

Considering both calibration and discrimination together, our findings suggest that the BOADICEA model is well calibrated in this population and shows similar (modest) discriminatory accuracy to Tyrer-Cuzick (version 8) and BRCAPRO, suggesting that it may be useful for patient management in the familial breast cancer risk setting. However, this cannot be interpreted as conclusive: we judged most included studies to have a high or unclear risk of bias; the number of validation studies included in the meta-analyses was small (≤ 10 for each model); and the contributing studies were heterogeneous in terms of prediction time horizons and case-mix.

Room for improvement remains in terms of the discriminatory ability of existing breast cancer risk prediction models in women with a family history of breast cancer. Reporting of prognostic model studies is currently suboptimal.

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Registration

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PLAIN LANGUAGE SUMMARY

How accurate are breast cancer risk prediction models in women with a family history of breast cancer?

Key messages

- We found four breast cancer risk prediction models that had been tested enough times to evaluate in detail. These were the Gail, Tyrer-Cuzick, BOADICEA, and BRCAPRO models.
- The BOADICEA model was one of the more reliable tools for estimating future breast cancer risk in women with a family history of the disease, meaning it may help them and their doctors to decide on treatment.
- Further research is needed to improve the accuracy of existing prediction models in distinguishing between women with a family history of breast cancer who will and will not develop the disease.

Why is it helpful to be able to predict a woman's risk of breast cancer?

Women who have a history of breast cancer in their family have a higher likelihood of developing breast cancer themselves during their lifetime. In clinics, a woman's chance of developing breast cancer in a given time period is often estimated using statistical tools known as breast cancer risk prediction models.

Being able to estimate the risk of developing breast cancer accurately for a woman with a family history of breast cancer helps doctors and the woman decide how to manage or reduce her risk of breast cancer. Management may include:

- regular imaging with mammograms or magnetic resonance imaging (MRI; a type of scan that creates detailed pictures of the breast tissue) to detect breast cancer at an early stage;
- taking risk-reducing medications; or
- in some cases, surgical removal of both breasts to prevent breast cancer.

Currently, it is not clear which of the available breast cancer risk prediction models works best in women with a family history of breast cancer.

What did we want to find out?

We wanted to:

- identify breast cancer risk prediction models that have been developed or tested (or both) in women with a family history of breast cancer; and
- assess how accurately they predict future risk of developing breast cancer in these women.

What did we do?

We searched for studies that developed or tested these models. We looked at how accurately the models predicted breast cancer risk, focusing on:

- whether the predicted number of breast cancer cases was similar to the number that actually occurred (calibration); and
- whether the model could distinguish between women who did and did not develop breast cancer (discrimination).

When enough studies assessed the same model, we combined their results statistically.

What did we find?

We identified 12 models that estimate future breast cancer risk that had been tested in studies where all or most women had a family history of breast cancer. The models were tested using information from as few as 134 women to as many as 130,058. Most of the women lived in North America, Europe, or Australia, with a small number from Asia.

The studies were funded by governments (25 studies), universities (24), non-profit organisations (21), and industry (3). Six studies did not report funding sources, and some received funding from more than one source.

We were able to combine results from several studies for four models: the Gail, Tyrer-Cuzick, BOADICEA, and BRCAPRO models.

Calibration: was the predicted number of breast cancer cases similar to the actual number?

- The Gail and BOADICEA models accurately estimated the numbers of women in the included studies who would develop breast cancer in a given timeframe:
 - for every 100 breast cancers Gail predicted would occur, about 106 actually occurred in reality;
 - for every 100 breast cancers BOADICEA predicted, about 98 actually occurred.
- The Tyrer-Cuzick model estimated that more women in the studies would develop breast cancer than actually did. For every 100 breast cancers it predicted, only about 86 actually occurred.
- The BRCAPRO model estimated that fewer women in the studies would develop breast cancer than actually did. For every 100 breast cancers it predicted, about 144 actually occurred.

Discrimination: how well did the models distinguish between women who develop and do not develop breast cancer?

- All four models were moderately accurate in distinguishing between women who would and would not develop breast cancer in a given timeframe, but none did a great job.
- The Tyrer-Cuzick (version 8), BOADICEA, and BRCAPRO models correctly distinguished women who would develop breast cancer from those who would not about 64 to 65 times out of 100.
- The Gail model performed slightly less well, correctly distinguishing women who would develop breast cancer 61 times out of 100.

What are the limitations of the evidence?

We rated the quality of most studies included in our review as poor or unclear, which means we cannot be confident that these results are reliable. Our confidence was reduced for several reasons, including:

- in some studies, too few women developed breast cancer, making it harder to judge how accurate the prediction models were;
- not all studies reported the model performance information we sought;
- some studies had missing information or did not explain how they handled missing information.

How up to date is this evidence?

The review includes studies published up to December 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings 1: Gail (BCRAT) model (all versions combined)

Population: women (pre- and postmenopausal) with a family history of breast cancer
Intervention (model): all breast cancer risk prediction models that include family history of breast cancer among its predictors
Comparator: models were compared to each other
Outcome: breast cancer occurrence
Timing: timespan of prediction varies and includes: lifetime risk, risk to a certain age, 5- or 10-year risk, etc.
Setting: models were used in asymptomatic adult females with a family history of breast cancer to predict their future personal risk of developing breast cancer

Outcome	Measure	No. of participants (studies)	Prediction horizon	Summary measure	Pooled result (95% CI)	95% PI	Overall risk of bias summary
Breast cancer	Calibration	7 studies	2-year risk intervals (1 study)	O/E ratio of external validation studies in target population	1.06 (0.91 to 1.25)	0.65 to 1.74	High for 6 studies; unclear for 1 study
		> 60,785 participants ^a	5-year risk (3 studies)				
			10-year risk (2 studies)				
		2455 breast cancer events	15-year risk (1 study)				
	Discrimination	10 studies	2-year risk intervals (1 study)	C statistic of external validation studies in target population	0.61 (0.57 to 0.66)	0.47 to 0.74	High for 9 studies; unclear for 1 study
		> 54,184 participants ^a	5-year risk (3 studies)				
			10-year risk (2 studies)				
		> 2939 breast cancer events ^b	15-year risk (1 study)				
			Lifetime risk (2 studies)				
			Risk predictions adjusted to follow-up period (1 study)				

CI: confidence interval; **PI:** prediction interval

We did not evaluate the evidence with GRADE as no formal GRADE guidance for prognostic models currently exists.

^aWe extrapolated the total number of participants (n = 24,511) for McCarthy 2021, as the study did not explicitly report this information. In addition, Glynn 2019 reported the number of breast cancers over 86,892 two-yearly intervals, without reporting actual numbers of individuals.

^bBased on nine studies that provided numbers of breast cancers observed. Clendenen 2019 did not clearly present the numbers.

Summary of findings 2. Summary of findings 2: Tyrer-Cuzick (IBIS) model (all versions combined)

Population: women (pre- and postmenopausal) with a family history of breast cancer

Intervention (model): all breast cancer risk prediction models that include family history of breast cancer among its predictors

Comparator: models were compared to each other

Outcome: breast cancer occurrence

Timing: timespan of prediction varies and includes: lifetime risk, risk to a certain age, 5- or 10-year risk, etc.

Setting: models were used in asymptomatic adult females with a family history of breast cancer to predict their future personal risk of developing breast cancer

Outcome	Measure	No. of participants (studies)	Prediction horizon	Summary measure	Pooled result (95% CI)	95% PI	Overall risk of bias summary
Breast cancer	Calibration	9 studies	2-year risk intervals (1 study)	O/E ratio of external validation studies in target population	0.86 (0.74 to 0.98)	0.56 to 1.33	High for 7 studies; unclear for 2 studies
		> 184,068 participants ^a	5-year risk (1 study)				
		2889 breast cancer events	6-year risk (1 study)				
			10-year risk (5 studies) ^b				
			Unclear (1 study) ^c				
	Discrimination	10 studies	2-year risk intervals (1 study)	C statistic of external validation studies in target population	0.62 (0.58 to 0.66)	0.49 to 0.74	High for 8 studies; unclear for 2 studies
		> 47,829 participants ^d	5-year risk (2 studies)				
		2597 breast cancer events	6-year risk (1 study)				
			10-year risk (5 studies) ^b				
			Lifetime risk (1 study)				

CI: confidence interval; **PI:** prediction interval

We did not evaluate the evidence with GRADE as no formal GRADE guidance for prognostic models currently exists.

^aBased on eight studies (out of nine) that reported participant numbers. Of these, McCarthy 2020 reported results separately for participants with one and with two or more first- or second-degree family members with breast or ovarian cancer. We pooled the results on O/E ratio across those with one and those with two or more first- or second-degree family members with breast or ovarian cancer using the method described by Vandenbroucke [1]. In addition, Glynn 2019 reported the number of breast cancers over 86,892 two-yearly intervals, without reporting actual numbers of individuals.

^bBrooks 2023 predicted 10-year risk adjusted to the number of years of follow-up.

^cPredicted risks appear to have been adjusted to the follow-up period for Mabey 2024.

^dBased on nine studies (out of 10) that reported participant numbers. Of these, McCarthy 2020 reported results separately for participants with one and with two or more first- or second-degree family members with breast or ovarian cancer. We pooled the results on the C statistic across those with one and those with two or more first- or second-degree family members with breast or ovarian cancer using a weighted average (inverse-variance) method. Glynn 2019 reported the number of breast cancers over 86,892 two-yearly intervals, without reporting actual numbers of individuals.

Summary of findings 3. Summary of findings 3: Tyrer-Cuzick (IBIS) model (version 8)

Population: women (pre- and postmenopausal) with a family history of breast cancer

Intervention (model): all breast cancer risk prediction models that include family history of breast cancer among its predictors
Comparator: models were compared to each other
Outcome: breast cancer occurrence
Timing: timespan of prediction varies and includes: lifetime risk, risk to a certain age, 5- or 10-year risk, etc.
Setting: models were used in asymptomatic adult females with a family history of breast cancer to predict their future personal risk of developing breast cancer

Outcome	Measure	No. of participants (studies)	Prediction horizon	Summary measure	Pooled result (95% CI)	95% PI	Overall risk of bias summary
Breast cancer	Calibration	5 studies	5-year risk (1 study)	O/E ratio of external validation studies in target population	0.90 (0.76 to 1.06)	0.55 to 1.47	High for 4 studies; unclear for 1 study
		166,696 participants ^a	6-year risk (1 study)				
		1491 breast cancers	10-year risk (2 studies)				
			Unclear (1 study) ^b				
	Discrimination	5 studies	5-year risk (2 studies)	C statistic of external validation studies in target population	0.64 (0.58 to 0.71)	0.46 to 0.79	High for 4 studies; unclear for 1 study
		38,405 participants ^c	6-year risk (1 study)				
		1292 breast cancer events	10-year risk (2 studies)				

CI: confidence interval; **PI:** prediction interval

We did not evaluate the evidence with GRADE as no formal GRADE guidance for prognostic models currently exists.

^aMcCarthy 2020 reported results separately for participants with one and with two or more first- or second-degree family members with breast or ovarian cancer. We pooled the results on O/E ratio across those with one and those with two or more first- or second-degree family members with breast or ovarian cancer using the method described by Vandenbrouke [1].

^bPredicted risks appear to have been adjusted to the follow-up period for Mabey 2024.

^cMcCarthy 2020 reported results separately for participants with one and with two or more first- or second-degree family members with breast or ovarian cancer. We pooled the results on the C statistic across those with one and those with two or more first- or second-degree family members with breast or ovarian cancer using a weighted average (inverse-variance) method.

Summary of findings 4. Summary of findings 4: BOADICEA model (all versions combined)

Population: women (pre- and postmenopausal) with a family history of breast cancer
Intervention (model): all breast cancer risk prediction models that include family history of breast cancer among its predictors
Comparator: models were compared to each other
Outcome: breast cancer occurrence
Timing: timespan of prediction varies and includes: lifetime risk, risk to a certain age, 5- or 10-year risk, etc.
Setting: models were used in asymptomatic adult females with a family history of breast cancer to predict their future personal risk of developing breast cancer

Outcome	Measure	No. of participants (studies)	Prediction horizon	Summary measure	Pooled result (95% CI)	95% PI	Overall risk of bias summary
Breast cancer	Calibration	5 studies 23,596 participants 972 breast cancer events	10-year risk (4 studies) Risk predictions adjusted to follow-up period (1 study)	O/E ratio of external validation studies in target population	0.98 (0.90 to 1.17)	0.89 to 1.09	High for 3 studies; unclear for 1 study; low for 1 study
	Discrimination	4 studies 62,080 participants 5671 breast cancer events	10-year risk (2 studies) Lifetime risk (1 study) Risk predictions adjusted to follow-up period (1 study)	C statistic of external validation studies in target population	0.65 (0.58 to 0.71)	0.44 to 0.81	High for 2 studies; unclear for 1 study; low for 1 study

CI: confidence interval; **PI:** prediction interval

We did not evaluate the evidence with GRADE as no formal GRADE guidance for prognostic models currently exists.

Summary of findings 5. Summary of findings 5: BRCAPRO model (all versions combined)

Population: women (pre- and postmenopausal) with a family history of breast cancer

Intervention (model): all breast cancer risk prediction models that include family history of breast cancer among its predictors

Comparator: models were compared to each other

Outcome: breast cancer occurrence

Timing: timespan of prediction varies and includes: lifetime risk, risk to a certain age, 5- or 10-year risk, etc.

Setting: models were used in asymptomatic adult females with a family history of breast cancer to predict their future personal risk of developing breast cancer

Outcome	Measure	No. of participants (studies)	Prediction horizon	Summary measure	Pooled result (95% CI)	95% PI	Overall risk of bias summary
Breast cancer	Calibration	5 studies 52,427 participants 1299 breast cancer events	5-year risk (2 studies) 10-year risk (1 study) 15-year risk (1 study) Risk predictions adjusted to follow-up period (1 study)	O/E ratio of external validation studies in target population	1.44 (1.25 to 1.62)	1.02 to 2.04	High for 1 study; unclear for 2 studies; low for 2 studies
	Discrimination	5 studies 52,427 participants	5-year risk (2 studies) 10-year risk (1 study)	C statistic of external validation studies in target population	0.64 (0.54 to 0.73)	0.37 to 0.84	High for 1 study; unclear for 2 studies; low for 2 studies

1299 breast cancer events	15-year risk (1 study) Risk predictions adjusted to follow-up period (1 study)
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CI: confidence interval; **PI:** prediction interval

We did not evaluate the evidence with GRADE as no formal GRADE guidance for prognostic models currently exists.

BACKGROUND

Description of the health condition and context

Breast cancer is the most common malignancy affecting women and the second most common cancer worldwide. It accounts for 11.6% of all new cancers diagnosed and ranks fourth in the leading causes of cancer death [2]. Breast cancer is a heterogeneous group of diseases with complex risk factors, including age, exposure to exogenous hormones, late age at first pregnancy, early age at menarche, late age at menopause, breast density, and family history.

Most breast cancers are sporadic, meaning they arise by chance. However, approximately 20% are considered familial, in that they occur in the context of a significant family history of the disease. Familial breast cancers often occur at an early age and may arise due to inherited mutations in known breast cancer susceptibility genes. Breast cancer predisposition alleles have been categorised into three groups based on the level of risk that they confer. High-risk genes include *BRCA1*, *BRCA2*, *TP53*, *PTEN*, *PALB2*, and *STK11*, with pathogenic mutations in these genes conferring a greater than 30% to 85% lifetime risk of breast cancer [3, 4, 5]. Moderate-risk genes include *ATM* and *CHEK2*, which confer approximately 20% to 30% lifetime risk [4]. In addition, genome-wide association studies and evidence from an Illumina collaborative oncological gene-environment study (iCOGS) have revealed a host of common lower-risk single nucleotide polymorphisms (SNPs) that contribute to the heritability of breast cancer predisposition [6, 7]. Familial breast cancers may also occur due to shared environmental or lifestyle factors in families, either alone or in combination with genetic alterations [8].

Women suspected of being at higher risk of breast cancer than the general population based on their family history are frequently referred to a 'family risk' clinic for risk assessment and management [9]. Reliable estimation of an individual's risk is important as it aids clinicians in recommending particular risk-management strategies and assists individuals themselves in reaching decisions about which, if any, risk-reducing measures (risk-reducing surgery, chemoprevention, surveillance, etc.) to opt for. It has been reported that the greatest benefit from breast cancer prevention strategies might be obtained through treating women who are at high risk of the disease [10, 11]. This further highlights the need for accurate risk assessment so that women can be appropriately stratified for breast cancer prevention strategies according to their estimated risk.

Management of women referred to a breast cancer family risk service is guided by their estimated individual risk of developing the disease. Management strategies in this setting are aimed at early detection and/or prevention of the disease. Early detection strategies aim to diagnose breast cancer at an early stage to improve clinical outcomes; these include radiologic surveillance at regular intervals by mammography and magnetic resonance imaging (MRI). Prevention strategies aim to reduce a woman's risk of developing breast cancer by means of prophylactic surgery (including prophylactic mastectomy and/or salpingo-oophorectomy) or risk-reducing medication with drugs such as tamoxifen, anastrozole, or raloxifene. Typically, radiologic screening is recommended as a management strategy for women considered 'moderate risk,' while additional risk-management

strategies such as prophylactic surgery may be appropriate for women deemed 'high risk' [12].

In the family risk setting, genetic testing for mutations in known breast cancer susceptibility genes represents the most accurate means of risk assessment. Criteria for genetic testing vary between countries and are often reserved for those who reach a given threshold risk score using various scoring systems based on family history of cancer [13]. In addition to genetic testing (e.g. for those who test negative for germline mutations) and in instances where genetic testing is not an option (e.g. for those who do not meet the criteria for genetic testing or decline testing), the risk of developing breast cancer over a given timeframe is frequently estimated using breast cancer risk prediction models [14].

Description of the prognostic models

A risk prediction model is a statistical tool for estimating the probability that an individual with specific risk factors (e.g. strong family history) will develop a condition such as breast cancer in the future [15, 16, 17]. Several such models have been developed to assess breast cancer risk [14]. Understanding the strengths and weaknesses of each model, as well as the clinical scenarios in which predictions from one model vary markedly from others, facilitates accurate breast cancer risk assessment [18]. There are three main categories of 'prognostic' models applicable to the breast cancer clinical setting: (1) models that predict an individual's risk of developing breast cancer over a specified timeframe (e.g. lifetime, 10 years, etc.); (2) models that assess the likelihood of detecting a germline mutation in a breast cancer susceptibility gene (e.g. *BRCA1* or *BRCA2*) in a given family or individual; and (3) models that estimate clinical outcome or response to therapy in a patient diagnosed with breast cancer.

This review is focussed solely on models in the first category: models that predict an individual's risk of developing breast cancer over time. Risk prediction models for breast cancer generally include multiple previously identified risk factors for the disease (identified by logistic regression and Cox proportional hazards regression), such as family history of breast cancer, breast density, reproductive and hormonal risk factors, history of benign breast disease, results from genetic testing and, in certain cases, modifiable risk factors such as alcohol consumption, body mass index, and physical activity [14]. Risk prediction models generate a numerical representation of the likelihood that an individual will develop breast cancer in a specified time period. To achieve this, a statistical algorithm is used to combine an individual's 'baseline risk of developing breast cancer' with their 'risk score'. Baseline risk is usually estimated from a prospective population-based cohort study and represents the underlying population risk for individuals in whom each risk factor included in the model is not present (i.e. has a value of 0). An individual's risk score is calculated from their set of risk factor values multiplied by their respective beta weights (e.g. log odds ratios) [16, 17, 19]. In recent years, other methods of prediction model development – such as machine learning and deep learning approaches – have also been used to develop breast cancer risk prediction models [20, Yala 2019 [21]].

Following the development of a risk prediction model in a sample from a population, it must then be validated to ensure its reliability. A model may be validated internally in another sample from the same population in which it was developed (using bootstrapping or cross-validation methods, for example), or preferably externally in

a sample from a different population [22]. The extent of validation varies considerably between different models. Often a model's predictive accuracy is not as good in the validation sample as in the development sample. Therefore, in this case, the model may be updated by modifying beta regression weights of risk factors, for example, to improve model performance [23].

Heterogeneity is typically high in prognostic model development and validation studies. Common sources of heterogeneity in these studies include case-mix and study design differences, including prediction horizons, follow-up time, and outcome definitions. It is important that this heterogeneity is considered when interpreting results from prognostic model evidence syntheses [24].

Most breast cancer risk prediction models were developed and validated for use in the general population. The performance of risk prediction models can vary according to the population to which they are applied. For example, a model may have good accuracy in a high-risk population but not in a low-risk population, or vice versa [16]. In clinical practice, the choice of risk prediction model varies between centres, and regional differences exist – for instance, some models are preferred in the USA, while others are more commonly used in Europe.

Health outcomes

The health outcome being predicted in the targeted population (women with a family history of breast cancer) by the prediction models under review is breast cancer occurrence. The time horizon of the prediction of breast cancer occurrence by the models under review varies: some models predict lifetime risk; some predict risk to a certain age; and some predict risk within five or 10 years, etc. We included all eligible models in the review irrespective of the time horizon of prediction.

Why it is important to do this review

This review attempts to answer this question: which breast cancer risk prediction model or models are reliable and appropriate for use in women with a family history of the disease? This question is important as it is unclear which of the currently available breast cancer risk prediction models is most accurate for use in the family risk setting. Risk prediction models are used to estimate an individual's risk of developing breast cancer, which aids clinicians and patients in reaching decisions about risk management. Therefore, accurate risk prediction is extremely important. Furthermore, the ability to accurately predict risk in this 'family risk' population can provide information about future disease burdens and future numbers of individuals that will require risk-reducing interventions. This information will benefit health policymakers.

Several systematic reviews of breast cancer risk prediction models have been published that have reported model performance in 'general' populations [16, 25, 26, 27, 28]. However, to our knowledge, no other systematic review has synthesised evidence on breast cancer risk prediction model performance specifically in women with a family history of the disease.

OBJECTIVES

To identify, describe, and appraise all available breast cancer risk prediction models developed or validated in women with a family

history of breast cancer, and to meta-analyse their performance in predicting breast cancer occurrence.

METHODS

Review structure and differences from the protocol

As prognosis reviews are a relatively new review type within Cochrane, little guidance on review structure was available when we developed the protocol [29]. In the meantime, the Cochrane Prognosis Methods Group developed a review template, and Kreuzberger and colleagues published an exemplar prognosis review [30]. We based the structure of our review on these. The methodology for this review was guided by the CHARMS checklist [31], recommendations reported by Debray and colleagues [32], and by the exemplar Cochrane prognostic review by Kreuzberger and colleagues [30].

Differences between the protocol and this review are detailed in [Supplementary material 9](#).

Criteria for considering studies for this review

Types of studies

- Cohort studies (prospective or retrospective) used in breast cancer risk prediction model development or validation.
- Case-control studies used in breast cancer risk prediction model development or validation.
- Randomised controlled trials used in breast cancer risk prediction model development or validation.

Targeted population

Target populations for this review were women of all ages and ethnicities considered to be at elevated risk of breast cancer based on having a family history of breast cancer. For the purpose of this review, we defined the target population as: (a) women enrolled in a screening programme for women at higher familial risk of developing breast cancer; (b) those with at least one first-degree relative with breast cancer; (c) those attending a breast cancer family risk service for risk assessment or management; or (d) study populations enriched for (oversampled for) women with a family history of breast cancer.

We excluded populations consisting solely of individuals with a known germline pathogenic variant (PV) in a breast cancer susceptibility gene, such as *BRCA1*, *BRCA2*, or *PALB2*.

Types of prognostic models

Included

- Studies describing development of a breast cancer risk prediction model in the target population that includes family history of breast cancer as a predictor, with or without external validation by independent data in the same study.
- External model validation in the target population with or without model updating.

Excluded

- Models predicting genetic mutations only, rather than breast cancer.
- Models that predict clinical outcome (prognosis) or response to therapy rather than breast cancer occurrence.

Types of outcomes to be predicted

Primary outcomes

Breast cancer occurrence, including occurrence of invasive breast cancer and/or ductal carcinoma in situ (DCIS).

Secondary outcomes

None.

Timespan of prediction

We included studies investigating model predictions over any timespan in the review.

Intended moment of model use

Models to be used in asymptomatic (breast cancer-free) adult females to predict their future personal risk of developing breast cancer.

We structured the research question and inclusion criteria according to the PICOTS framework, specifying the Population, Intervention (model), Comparator, Outcomes, Timing, and Setting of interest in [Table 1](#).

Search methods for identification of studies

Electronic searches

We developed the search strategies using search filters modified from those suggested by Geersing and colleagues [33], and Pace and colleagues [34]. No restrictions were applied for language or publication date.

We searched all records in the following electronic databases:

- MEDLINE via Ovid (from inception to 15 February 2022);
- Embase via Embase.com (from inception to 15 February 2022);
- Cumulative Index to Nursing and Allied Health Literature (CINAHL) (EBSCO host) (from 1982 to 15 February 2022);
- Institute of Scientific Information (ISI) Web of Science (Core collection from 1900 to 15 February 2022).

Please refer to [Supplementary material 1](#) for the search strategies. In a deviation from the protocol [29], we did not search the database of prognostic studies maintained by the Cochrane Prognosis Methods Group (PMG) as it has not yet been developed.

We performed a targeted top-up search of MEDLINE (via Ovid) on 19 December 2024 to capture additional validation studies of included models only (See [Supplementary material 1](#)). Additionally, to capture more recent validation studies of the models developed by Palmer and colleagues (Palmer 2021 [35]), we searched PubMed for articles published by the first and last authors of the article describing the development of these models from 15 February 2022 to 19 December 2024.

Searching other resources

As pre-specified in the protocol [29], we screened reference lists of included studies and non-Cochrane narrative, semi-systematic, and systematic reviews to identify additional studies for inclusion. Routine search and retrieval processes were also used to identify any retracted studies.

Data collection

Selection of studies

Working independently, two review authors (SMG, YH) screened the search results by title and abstract. We retrieved the full reports of potentially relevant studies. We used the Covidence platform to assess and document studies for inclusion and exclusion according to the above inclusion criteria [36]. Two review authors (SMG, YH) independently selected studies for inclusion. As recommended in the PRISMA statement, we documented the study selection process in a flow chart [37].

Resolving disagreements about study inclusion

During the initial screening and full-text review, we held regular team meetings to resolve any disagreements about study inclusion through discussion. A third review author (JAAGD) acted as the final arbiter if we were unable to reach consensus. We also consulted the Cochrane Breast Review Group when we required further guidance. The following issues emerged during our group discussions.

- It was not always obvious whether studies constituted a 'true' model validation. For example, in some cases, the study's primary aim was to use a model (or models) to determine the percentage of women in a population in various risk categories, rather than to validate the model in terms of its ability to predict breast cancer occurrence. These studies were not designed as validation studies and typically had inadequate follow-up for assessing predictive performance. Although some reported metrics such as sensitivity or specificity that could be interpreted as measures of model performance, model validation was not the focus. We excluded such studies because they do not provide reliable evidence on the real predictive performance of the models.
- Some studies reported model performance in risk quantiles (e.g. deciles of predicted risk). While it may be assumed that participants in the highest risk quantiles are likely enriched for those with a family history of breast cancer (our target population) as well as those at increased breast cancer risk for other reasons, we decided not to include these studies in this review unless it was explicitly stated that the population in question was enriched for or comprised solely of women with a family history of breast cancer.
- Several articles reported BOADICEA model updates. These articles, authored by the model's developers, typically described updates to the model narratively without reporting performance (re-validation) of the updated model. Other articles 'validated' the (updated) model by comparing model-predicted risks in a population with empirical estimates from separate epidemiological studies. Because 'predicted' and 'observed' breast cancers were not from the same participants, this cannot be considered a 'true' model validation. We decided not to include these studies for the purpose of data extraction and risk of bias (ROB) assessment, as they do not provide valuable information about the performance of the model. However, we summarised these studies in [Supplementary material 4](#) as they provide useful details about model predictions and updates to the BOADICEA model since its initial development.
- The protocol for this review did not pre-specify how we would handle conference abstracts. We decided to list conference abstracts as 'studies awaiting classification' if they met two

criteria: (1) in the review authors' judgement, the study appeared to meet the inclusion criteria for the review; (2) no full-text version of the study had yet been published. We excluded conference abstracts that did not meet these criteria. We excluded conference abstracts that contained insufficient information to gauge eligibility. The limited information reported in conference abstracts precluded completion of an accurate risk of bias assessment. For this reason, we did not extract data from studies awaiting classification, but instead listed them for the sake of completeness.

- Some studies developed a model using data from women in the general population or from other cohorts not oversampled for family history of breast cancer, and in the same study, validated the model in a subgroup of women with a family history of breast cancer. We included only the validation data from these studies.
- In line with the exemplar Cochrane prognostic review by Kreuzberger and colleagues [30], we decided to list model development studies for which no external validation studies in the target population have yet been published in the 'studies awaiting classification' category. These studies fit the formal inclusion criteria of the review. However, they have not yet been tested externally in the population of interest. Therefore, they cannot yet be evaluated regarding their clinical usefulness in the target population.

Data extraction and management

In performing data extraction from selected studies, we were guided primarily by the CHARMS checklist as well as by previous reports [15, 31, 38, 39, 40, 41, 42, 43, 44, 45].

Working independently, two review authors (SMG, KB) extracted study data. We resolved disagreements through discussion. We maintained extracted data in a pre-defined data extraction spreadsheet (Supplementary material 6), and piloted the data extraction forms to ensure clarity and consistency before full data extraction. We contacted corresponding authors of individual studies for additional information when required. We extracted information on the following items.

- Journal citation: study authors, journal name, year, volume, pages
- Source of data: e.g. cohort, nested case-control, case-cohort, registry data, etc.
- Participants: eligibility and recruitment method (e.g. consecutive participants, locations, number of centres, setting, inclusion and exclusion criteria, participant description, details of treatments received, details of genetic testing performed, if relevant, study dates)
- Outcome: definition and method of measurement; whether the same definition and measurement method were used for all participants; whether outcome assessment was blinded to candidate predictors; time of outcome occurrence or summary of follow-up duration
- Candidate predictors: number and type of predictors (e.g. demographics, family history of cancer, parity, breastfeeding, lifestyle factors, exogenous hormone use, genetic testing etc.); definition of and method for measurement of candidate predictors; timing of predictor measurement; whether predictors were assessed blinded for outcome; handling of predictors in the modelling (e.g. continuous, linear, non-linear transformations, categorised)

- Sample size: number of participants and number of events; number of events in relation to the number of candidate predictors (events per variable)
- Missing data: number of participants with any missing value (including predictors and outcomes); number of participants with missing data for each predictor; handling of missing data (e.g. complete case analysis, imputation, other methods)
- Model development: modelling method (e.g. logistical, survival, neural network, machine learning techniques); modelling assumptions satisfied; method for selection of predictors for inclusion in multi-variable modelling (e.g. all candidate predictors, pre-selection based on unadjusted association with the outcome); method for selection of predictors during multi-variable modelling (e.g. full model approach, backward or forward selection) and criteria used (e.g. P value, Akaike Information Criterion); shrinkage of predictor weights or regression coefficients (e.g. no shrinkage, uniform shrinkage, penalised estimation)
- Model performance: calibration (calibration plot, calibration slope, Hosmer-Lemeshow test) and discrimination (C statistic, D statistic, log-rank) measures with confidence intervals, classification measures (e.g. sensitivity, specificity, predictive values, net reclassification improvement) and whether a priori cut points were used
- Model evaluation: method used for testing model performance: development data set only (random split of data, re-sampling methods (e.g. bootstrap, cross-validation, none)) or separate external validation (e.g. temporal, geographical, different setting, different investigators); in the case of poor validation, whether model was adjusted or updated (e.g. intercept re-calibrated, predictor effects adjusted, new predictors added)
- Results: final and other multi-variable models (e.g. basic, extended, simplified) presented, including predictor weights or regression coefficients, intercept, model performance measures (with standard errors or confidence intervals); any alternative presentation of final prediction models (e.g. sum score, nomogram, score chart, predictions for specific risk subgroups with performance); comparison of the distribution of predictors (including missing data) for development and validation data sets
- Interpretation and discussion: interpretation of presented models (confirmatory, i.e. model useful for practice; exploratory, i.e. more research needed); comparison with other studies; discussion of generalisability, strengths and limitations, ease of use of model (length of time and level of skill required to use model)
- Author correspondence: details of correspondence with study authors for additional information or clarification of queries, if relevant
- References: additional relevant articles cited in reference list of article, if applicable

Risk of bias and applicability assessment

We assessed risk of bias for each included study using the prediction study risk of bias assessment tool (PROBAST) [46, 47]. The PROBAST tool utilises a domain-based structure supported by signalling questions grouped within these domains to assess risk of bias and applicability of prediction-modelling studies. Risk of bias refers to any flaw or shortcoming in the design, conduct, or analysis of a primary study that is likely to distort the predictive

performance of a model. Assessment of applicability examines whether the prediction-model development or validation study matches the systematic review question in terms of the target population, predictors, or outcome of interest. We used a PROBAST quality assessment spreadsheet to record quality assessments for each study. Working independently, two review authors (SMG, KB) performed PROBAST quality assessments for each study, drawing on guidance developed by Moons and colleagues [47]. We assigned an overall rating of low, high, or unclear risk of bias for each domain. We resolved any disagreements through discussion; a third review author (JAAGD) arbitrated, when required.

Measures of prediction model performance

The O/E statistic measures the calibration performance of the model. It compares expected (E) numbers to observed (O) numbers of events. For a well-calibrated model, the O/E ratio should be close to 1. Sometimes, these statistics are presented for quantiles (e.g. deciles, quintiles) of the population defined by predicted risk, to see whether O/E is close to 1 in all quantiles.

We assessed calibration using the ratio of observed to expected (O/E) breast cancers, based on the mean observed and mean expected breast cancer event rates [48]. When studies did not report the O/E variance, we estimated it to calculate 95% confidence intervals (CIs) [24, 48, 49]. We considered studies with O/E ratios close to 1 with overlapping 95% CIs to be well calibrated.

The C statistic (or Area Under the Curve (AUC)) measures the discriminatory performance of a model – for example, a breast cancer prognostic model's ability to distinguish between individuals who will and will not develop breast cancer. It represents the probability that, for a randomly selected pair consisting of one person who develops the condition and one who does not, the person who develops the condition has a higher predicted risk. A C statistic of 0.5 indicates no discrimination between people who develop the condition and those who do not, whereas 1.0 indicates perfect discrimination.

Dealing with missing data

We attempted to contact study authors to request any missing pertinent information. Where possible, we calculated missing performance statistics and their precision, using the methods described by Debray and colleagues [24, 32].

Assessment of heterogeneity

We planned to investigate sources of heterogeneity between studies, such as differences in study population or case-mix, using meta-regression. Performance measures (or their transformations) would serve as the dependent variables, with study-level and aggregate patient-level characteristics as independent variables, if the number of included studies was sufficiently large (> 10). However, this analysis was not possible as there were 10 or fewer included validation studies for each model. We assessed the magnitude of heterogeneity using the I^2 statistic, which describes the percentage of variation across studies that is due to heterogeneity rather than chance [50]. To further interpret the relevance of any between-study heterogeneity, we calculated an approximate 95% prediction interval. This interval provides a range for the potential model performance in a new validation study, although with fewer than 10 studies, it is usually very wide [32].

Assessment of reporting deficiencies

It is recommended that all prognostic model studies report both calibration and discrimination of the model(s) in question [22, 51]. However, it is recognised from several systematic reviews on methodological conduct and reporting of prognostic models that studies seldom report both measures or report them inadequately [52]. In addition, methods and reporting in prognostic research often do not adhere to current methodological recommendations [38, 53]. We evaluated reporting deficiencies in the included studies.

Data synthesis

Data synthesis and meta-analysis approaches

We qualitatively summarised all identified risk prediction models developed or validated in the target population. We meta-analysed performance statistics for each model that had been validated multiple times in independent populations. Originally, we planned to conduct meta-analysis if at least two studies reported performance statistics for a given model [29]. However, in line with the approach of Kreuzberger and colleagues [30], we raised this threshold to require at least four studies with data available for inclusion in the meta-analysis. Finally, we qualitatively synthesised evidence on model performance from studies that compared multiple models in the same dataset.

Qualitative synthesis

For each included study, we provide a qualitative overview of the model in question. Study characteristics were tabulated in 'overview of synthesis and included studies' tables and 'characteristics of included studies' tables using a schema similar to that used in the exemplar review by Kreuzberger and colleagues [30].

Meta-analysis

After we identified all relevant studies and extracted results on discrimination and calibration, where applicable, we calculated the pooled estimate based on a weighted average. Where more than one study used overlapping participant populations to validate a model or models, we included the study with the largest sample size in the meta-analysis. One exception to this was the meta-analysis of the BRCAPRO model performance: we selected Guan 2023 [54] rather than Guan 2022 [55] for meta-analysis despite a slightly larger sample size in the latter study. This was because Guan 2022 reported discriminatory performance (AUC) for a re-calibrated version of BRCAPRO rather than the standard BRCAPRO model, which was less appropriate for pooling in meta-analysis. However, we narratively summarised all included studies.

Variation between the study results is unlikely to be due solely to random error, as validation studies differ in design and population/case-mix. Therefore, we accounted for the additional heterogeneity between the studies by using a random-effects (rather than a fixed-effect) meta-analytic approach. This approach accounts for both within- and between-study variation, and weights for estimation of the overall effect are based on the within-study error variance. We presented all estimates with 95% confidence intervals (CIs). We used R software to collate the information from studies and perform meta-analyses [56].

To approximate normality, we applied a logarithmic transformation to the O/E ratio. We weighted study-specific O/E ratios by the inverse of their variance. When studies reported expected over observed (E/O) ratios, we converted them to O/E ratios to ensure consistency across studies included in the meta-analysis.

For studies reporting discrimination for a specific breast cancer prognostic model, we calculated a summary C statistic using inverse-variance weighting, deriving the variance from the reported 95% CIs where available. Because the C statistic is bounded between 0 and 1, we applied a logit transformation prior to meta-analysis, as recommended by Debray and colleagues [32].

When reported, we extracted data on model calibration in risk categories (e.g. age categories, number of first-degree affected relatives). We planned to perform meta-analysis of calibration statistics in risk categories if a sufficient number of studies reported calibration statistics in the same risk categories for a given model. However, an insufficient number of studies reported this information, precluding this step.

Subgroup analysis

According to our protocol, our planned primary analysis entailed identifying, qualitatively summarising, and meta-analysing the performance of all available breast cancer risk prediction models that include family history of breast cancer as a predictor. Our original target population for this analysis was all women (irrespective of family history of breast cancer) of all ages and ethnicities. Our planned subgroup analysis entailed repeating the above-mentioned analysis for models that were developed or validated in women with a family history of breast cancer (women with 'elevated risk' based on family history).

Following discussions with the Cochrane Breast and Prognosis Methods Review Groups, and given the large number of retrieved articles that developed or validated breast cancer risk prediction models in various populations, we refined the scope of the review. Specifically, we made the originally planned subgroup analysis the primary analysis and limited the review to models developed or validated in women with a family history of breast cancer. This change made the review feasible and resulted in a more focused review that better reflects the title.

No other subgroup analyses were performed.

Sensitivity analysis

Our protocol did not pre-specify any sensitivity analyses. We decided post hoc to perform the following model-specific meta-analyses or qualitative analysis.

- Sensitivity analysis of the C statistic for the Gail (BCRAT) model excluding three studies reporting no 95% CI. Estimating confidence intervals can introduce imprecision. Therefore, we aimed to explore the extent of differences in pooled C statistic estimates between studies that reported 95% CIs and those where we estimated the 95% CIs.
- Multiple versions of the main breast cancer risk prediction models exist. Our main analysis meta-analysed performance measures for all versions of each model combined. Where feasible, we have meta-analysed model performance measures separately for each version of a given model. This was only

possible for the Tyrer-Cuzick model version 8 due to an insufficient number of studies to pool for other distinct model versions.

- Included models were validated for the prediction of breast cancer over various timespans (5-year risk, 10-year risk, lifetime risk, etc.) in different studies. In the main meta-analyses of model performance, we pooled performance measures from non-overlapping studies irrespective of the timespan of prediction assessed. Where feasible, we conducted additional meta-analyses of model performance measures based on the timespan of prediction. This was only feasible for the 10-year risk for the Tyrer-Cuzick model (discrimination and calibration) and the BOADICEA model (calibration only) and for the 5-year risk for the Gail model (calibration only).

Direct model comparison using the same dataset(s)

During data extraction, it became apparent that some studies compared performance of two or more models in the same study participants. Meta-analysis of performance measures for an individual model gives an indication of the accuracy of that particular model across the studies in which it was validated. However, when the goal is to compare the performance of different risk prediction models in an effort to determine which model performs best in the target population, it is ideal to compare the performance of the various models amongst the same participants. For this reason, we decided post hoc to synthesise evidence on model performance from studies that compared multiple models in the same dataset. These multi-model studies provide for a more homogeneous comparison across models using similar cohorts.

Initially, we attempted a meta-analysis for each model using studies that compared multiple models within the same dataset. However, because specific methodological guidance for this approach is currently lacking, we instead conducted a qualitative synthesis of head-to-head model comparisons.

Rating the certainty of evidence and summary of findings

We had originally planned to rate the certainty of the evidence using an adapted GRADE framework for prognostic evidence. Although several methodological developments have aimed to adapt GRADE for evidence from meta-analyses of prognostic model performance – including work by Huguette and colleagues [57], and a concept paper by Foroutan and colleagues [58] – no formal guidance currently exists for grading the certainty of results from meta-analysis of prognostic models. Therefore, we did not apply the GRADE framework in this review.

RESULTS

Description of studies

Results of the search

We screened a total of 31,022 records (database searching yielded 30,661 records; reference list screening yielded three records; a targeted top-up search in December 2024 yielded 358 records). We removed 11,131 duplicates, leaving 19,891 records for screening. We excluded 19,480 records on the basis of title and abstract screening. We retrieved the remaining 411 articles potentially meeting our inclusion criteria as full texts, and excluded 355 articles. We documented the study selection process in a PRISMA flow diagram (Figure 1).

Figure 1. Study flow diagram

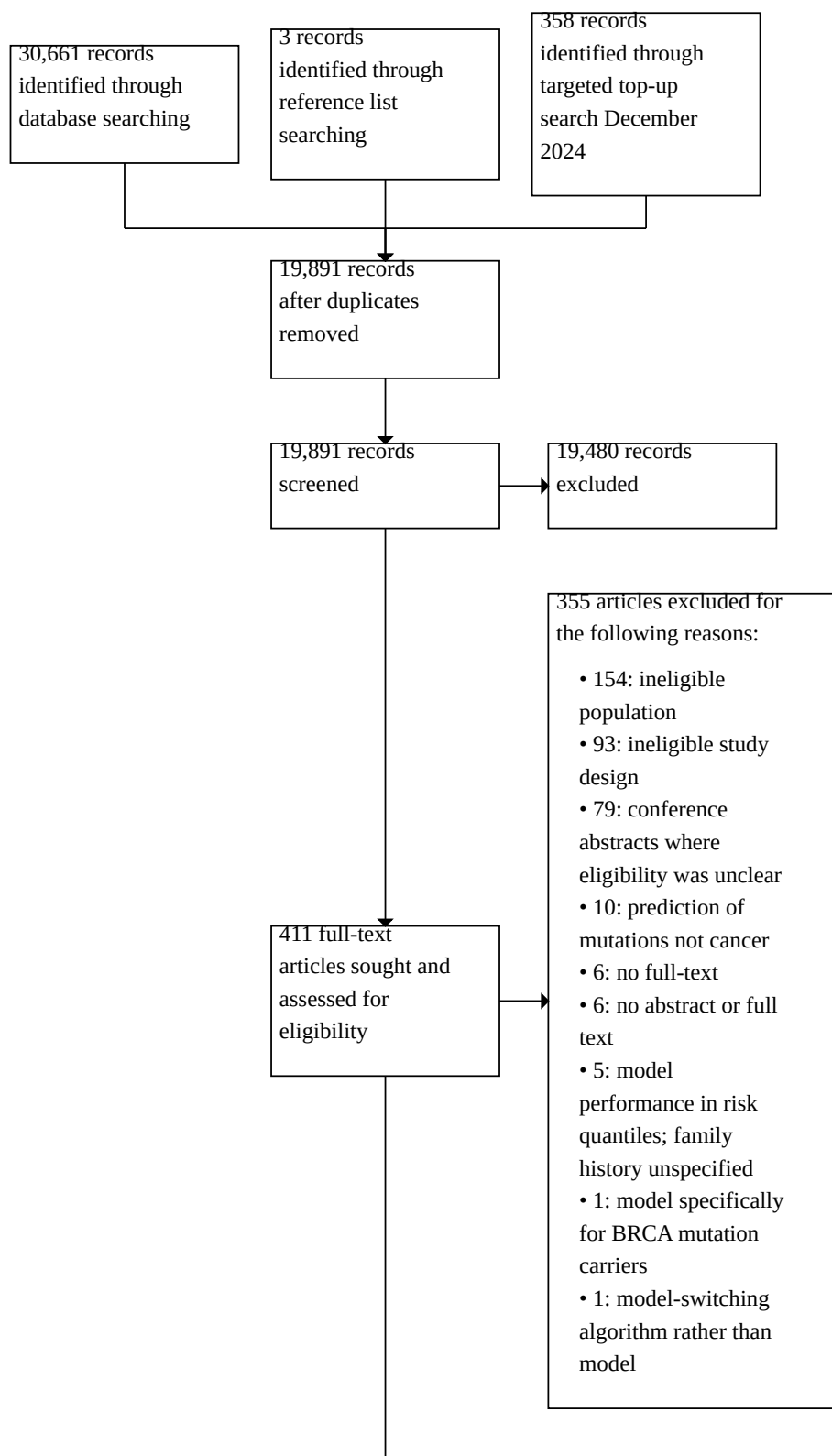
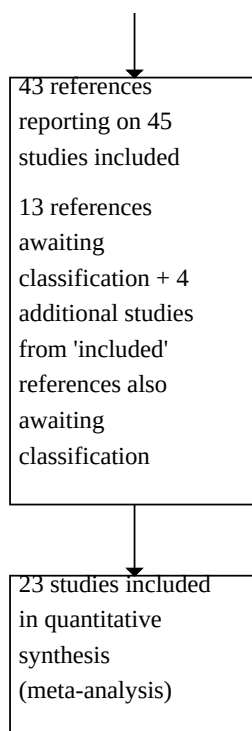


Figure 1. (Continued)



We included 43 articles, reporting on 45 studies, in the review. Three articles reported model development only. Fourteen articles reported validation of a single model. Of these, five articles updated the model with an additional predictor or predictors. Seventeen articles reported validation of multiple models. Eight articles (nine studies) developed/validated a new model (or models) and compared its performance to an existing model (or models) (thereby validating the existing model in the study population). One article reported on a large programme of research, most of which was not applicable to this review. However, we included two studies described in this article, labelled as Evans 2016 Chapter 2 (study 1a) [59] and Evans 2016 Chapter 3 (study 4) [60], which focussed on model validations.

We listed 13 articles, reporting on 17 studies, as 'awaiting classification'. We categorised studies as awaiting classification if they met the following criteria: (1) conference abstracts where no full-text version of the study had been published subsequently that in the authors' judgement appeared to meet the inclusion criteria for the review; (2) model development studies without any external validation studies in the target population; or (3) narrative reports of model updates that did not report performance (re-validation) of the updated model or did so inadequately (for example, where predicted and observed breast cancers were not from the same participants). We provide details of the studies awaiting classification in [Supplementary material 7](#).

Included studies

We have structured the description of the included studies below according to the breast cancer risk prediction model

they addressed. Several articles validated model(s) in general populations of women and reported model performance in women with a family history of breast cancer as subgroup analyses. In these cases, we included only data relating to the subgroup of women with a family history of breast cancer. Some articles 'validated' models using simulated data in addition to actual patient cohorts. In these cases, we included only model validations performed using real patient data.

Multiple versions of the main risk prediction models exist. It was not always obvious which model versions the validation studies used. Some publications contained web links to model(s) used that had expired. In other cases, when contacted, authors reported using the model version current at the time of the study. This made evidence synthesis and meta-analyses challenging. Where possible, we have indicated which model version was validated in each included study. Furthermore, when feasible, we have meta-analysed model performance measures separately for each version of a given model (in addition to meta-analysing performance measures for all versions of the model combined).

Models included in meta-analysis

As noted in the 'Methods > Data synthesis' section, we required at least four studies reporting performance statistics for a given model to undertake meta-analysis. The following four breast cancer risk prediction models met that threshold:

- Gail (BCRAT);
- Tyrer-Cuzick (IBIS);
- BOADICEA;

- BRCAPRO.

Model 1. Gail/Breast Cancer Risk Assessment Tool (BCRAT)

The Gail model was developed using case-control data from white women in the USA, who were participants in the Breast Cancer Detection Demonstration Project, to estimate relative risks for various breast cancer risk factors. These data were combined with American breast cancer incidence rates to predict absolute risks assuming a proportional hazards model. The original model (sometimes referred to as 'Gail 1') considered both invasive and in situ carcinomas as breast cancer events [61]. The model was later updated to predict only invasive breast cancers [62], and recalibrated using Surveillance, Epidemiology, and End Results (SEER) data (sometimes referred to as 'Gail 2' or 'modified Gail') [63]. The Gail model accounts for competing causes of death (risk of dying from non-breast cancer causes) in its calculation. The Gail model is an example of a so-called 'empirical' breast cancer risk prediction model in that it does not assume a specific genetic model but instead uses the number of affected first-degree relatives as a summary measure of family history of breast cancer [64]. The current version of the Gail/BCRAT model includes seven predictors: current age, age at menarche, ethnicity, number of previous biopsies, age at first live birth, number of first-degree relatives with breast cancer, and presence of atypical hyperplasia. It is intended for use by women in the general population who are 35 years or older. It is not suitable for women with a *BRCA1* or *BRCA2* mutation, women with a previous history of invasive or in situ breast cancer (ductal carcinoma in situ (DCIS) or lobular carcinoma in situ (LCIS)), or those who have had prior chest irradiation. Based on the dataset used to develop it, the Gail model is assumed to be most reliable for women undergoing regular (annual) breast examinations [61]. The model predicts the risk of developing invasive breast cancer over time up to age 90 years.

Included studies of the Gail/BCRAT model

We identified 18 references reporting external validation of the Gail model in the target population: Amir 2003 [65]; Bondy 1994 [66]; Clendenen 2019 [67]; Evans 2016 Chapter 2 (study 1a); Glynn 2019 [68]; Gong 2014 [69]; Guan 2023; Ho 2023 [70]; Lamb 2023 [71]; Li 2021a [72]; Mayo 2001 [73]; McCarthy 2020 [74]; McCarthy 2021 [75]; Ming 2019 (USA) [76]; Quante 2012 [77, 78]; Rockhill 2001 [79]; Sevdalis 2022 [80]; Terry 2019 [81, 82]. Of these, five validations reported calibration only (E/O or O/E ratio), five validations reported discrimination only (C statistic or AUC), and eight validations reported both model calibration and discrimination. We included 10 validations in the meta-analysis of discrimination because several validation studies used overlapping populations and one study only showed the E/O ratio graphically (Amir 2003; Clendenen 2019; Glynn 2019; Guan 2023; Li 2021a; Mayo 2001; McCarthy 2021; Ming 2019 (USA); Sevdalis 2022; Terry 2019). We included seven validations in the meta-analysis of calibration (Bondy 1994; Evans 2016 Chapter 2 (study 1a); Glynn 2019; Guan 2023; Li 2021a; McCarthy 2021; Terry 2019). See [Summary of findings 1](#). We did not include the development study for the Gail model in this review because the model was developed in the general population of women who have regular screening mammography, rather than in a population enriched for women with a family history of breast cancer. [Table 2](#) summarises the main characteristics of the studies validating the Gail model.

Model 2. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS)

The Tyrer-Cuzick model is a so-called 'genetic model'. The original development study for the model reports in detail the methods used to build the model but does not provide details of internal validation, exact sample size used for model development, or missing data (Tyrer 2004 [83, 84]). The model was developed using several published datasets, including data from a Swedish cohort of offspring of mothers with breast or ovarian cancer [85].

The model uses segregation analysis techniques based on Bayes' theorem to estimate familial risk. A woman's family history is used to calculate the likelihood of her carrying a breast cancer-predisposing gene variant, which in turn affects her likelihood of developing breast cancer. The model assumes the effects of *BRCA1* and *BRCA2* mutations and models the residual genetic component in terms of a third hypothetical, low-penetrance, dominantly inherited, breast cancer susceptibility gene. This hypothetical 'third' gene acts as a surrogate for the remaining breast cancer predisposing genetic component not accounted for by *BRCA1* or *BRCA2*. A proportional-hazards model is assumed for the incidence associated with the 'hypothetical' gene [86]. The risk from other hormonal and pathological risk factors are combined with the familial risk component. The model includes the following predictors: current age; body mass index (BMI); reproductive history (i.e. age at menarche, age at first live birth, parity, age at menopause); use of hormone replacement therapy; previous breast biopsies; presence of atypical hyperplasia or LCIS; family history, including first- and second-degree relatives with breast and/or ovarian cancer, incorporating age of onset and the presence of bilateral breast cancers; and Ashkenazi Jewish heritage. The most recent version of the model (version 8) also incorporates mammographic density and an optional polygenic single nucleotide polymorphism (SNP) score. The original model ignored competing causes of death, but more recent versions (versions 7 and 8) allow users to include competing causes of death in the risk calculation as an optional feature. The user can also select which baseline breast cancer incidence rates to include from UK, Swedish, or Slovenian population rates. The Tyrer-Cuzick model predicts a woman's 10-year and lifetime breast cancer risk up to age 85 years.

Included studies of the Tyrer-Cuzick model

We identified 24 external validations of the Tyrer-Cuzick model in the target population: Amir 2003; Brooks 2023 [87]; Eriksson 2023a [88]; Eriksson 2023b [89]; Evans 2016 Chapter 2 (study 1a); Evans 2016 Chapter 3 (study 4); Glynn 2019; Gong 2014; Guan 2023; Hughes 2021 [90]; Kenan 2018 [91]; Laitman 2013 [92]; Lamb 2023; Li 2021a; Li 2021b [93]; Mabey 2024 [94]; MacInnis 2020 [95]; McCarthy 2020; Quante 2012; Quante 2015 [96]; Sevdalis 2022; Terry 2019; Warwick 2014 [97]; Yala 2019. Of these, four validations reported calibration only (E/O or O/E ratio), four validations reported discrimination only (C statistic or AUC), and 10 validations reported both model calibration and discrimination measures. Two studies with overlapping populations reported discrimination as 'age-adjusted AUC' (Eriksson 2023a; Eriksson 2023b), making them unsuitable for pooling with the AUC/C statistic in meta-analysis. One study reported discrimination (AUC) and displayed model calibration using a calibration plot but did not report the O/E or E/O ratio (Quante 2015). One study showed the E/O ratio graphically only, but did not report a value for the E/O ratio (Lamb 2023). One

study did not report standard model performance measures, but instead evaluated discrimination accuracy using likelihood ratio χ^2 test statistics from age-adjusted logistic regression models as well as odds ratios (Hughes 2021). A further study reported that receiver operating character (ROC) analysis was performed, but data were not shown (Kenan 2018). We contacted the authors, who advised that these data were unavailable. Because several studies used overlapping populations, we included only 10 validations in the meta-analysis of discrimination (Brooks 2023; Evans 2016 Chapter 3 (study 4); Glynn 2019; Guan 2023; Li 2021b; McCarthy 2020; Sevdalis 2022; Terry 2019; Warwick 2014; Yala 2019) and nine validations in the meta-analysis of calibration (Brooks 2023; Evans 2016 Chapter 2 (study 1a); Glynn 2019; Guan 2023; Laitman 2013; Li 2021b; Mabey 2024; McCarthy 2020; Terry 2019). See [Summary of findings 2. Table 3](#) summarises the main characteristics of the studies validating the Tyrer-Cuzick model.

Model 3. Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm (BOADICEA)

The BOADICEA model is another example of a ‘genetic’ model. It was developed using combined data from a population-based series of breast cancer cases and high-risk families in the UK, whose *BRCA1* and *BRCA2* mutation status was known (Antoniou 2002 [98]; [99]). The model was derived using segregation analysis to simultaneously model the effects of *BRCA1*, *BRCA2*, and the residual familial clustering of breast cancer (modelled as the combined multiplicative effect of a large number of low-risk polymorphisms (polygenic component)). The BOADICEA algorithm can incorporate data (including age of breast or ovarian cancer onset) from all relatives of the individual being counselled rather than restricting family history data to first- and second-degree relatives. BOADICEA can also incorporate reduced BRCA test sensitivity into its calculations. BOADICEA predicts both the probability of carrying a *BRCA1/2* mutation and the risk of developing breast or ovarian cancer over time [99].

The BOADICEA model has been updated multiple times. In 2008, the model was extended to include data from additional UK population-based studies, to take into account the effects of male breast cancer, prostate and pancreatic cancer, and to allow for the risk of multiple cancers [100]. The model was again updated in 2014 to include updated cancer incidences, tumour pathology, and to implement a new web interface [101]. In 2016, the effects of truncating variants in *PALB2*, *CHEK2*, and *ATM* were included in the model [102]. The most recent version (version 6) has been incorporated into the web-based CanRisk tool, which, in addition to family history and genetic test results (including polygenic risk scores), includes mammographic density and other lifestyle and hormonal risk factors for breast cancer [103].

Included studies of the BOADICEA model

We designated Antoniou 2002 as the original development study for the BOADICEA model as this was the first description of the development of a comprehensive version of the model. However, analyses reported in a 2001 article by Antoniou and colleagues also contributed to the development of the model [104]. We identified 14 references reporting external validation of the BOADICEA model in the target population. Of these, two validations reported calibration only (E/O or O/E ratio), four validations reported discrimination only (C statistic or AUC), and six validations reported both model calibration and discrimination. In one validation (Li 2017 [105]), discriminatory accuracy was quantified using

hazard ratios rather than C statistics or AUC. One study reported discrimination (AUC) and displayed model calibration using a calibration plot but did not report the O/E or E/O ratio (Quante 2015). A further study reported that ROC analysis was performed, but data were not shown (Kenan 2018). We contacted the authors, who advised that these data were unavailable. Because several studies used overlapping populations, we included only four validations in the meta-analysis of discrimination (Li 2021b; Ming 2020 [106]; Stahlbom 2012 [107]; Terry 2019), and five in the meta-analysis of calibration (Evans 2016 Chapter 2 (study 1a); Laitman 2013; Li 2021b; Stahlbom 2012; Terry 2019). See [Summary of findings 4. Table 4](#) summarises the main characteristics of the studies validating the Tyrer-Cuzick model.

Model 4. BRCAPRO

The BRCAPRO model is another example of a ‘genetic’ model. The model was developed using published datasets. BRCAPRO has been updated multiple times. It calculates the probability of carrying a *BRCA1* or *BRCA2* mutation and the likelihood of developing invasive breast cancer. BRCAPRO derives risk estimates from family history data using Mendelian principles of inheritance, Bayes’ theorem, and age-dependent penetrances and prevalences based on a systematic review of the literature (age-specific probability of developing breast cancer for a given genotype). Penetrance estimates for non-carriers are based on rates from the US Surveillance, Epidemiology, and End Results (SEER) program (Guan 2023). The model assumes that *BRCA1/2* mutations account for all hereditary breast cancers; other breast cancer susceptibility genes or genetic variants are not considered [108]. The model is available as a Bayes Mendel R package, and has also been incorporated into several web-based interfaces and software tools that can be used more easily in clinical settings (<http://hughesriskapps.com/>; <https://progenygenetics.com/>). BRCAPRO incorporates an extensive family history including both cancer-affected and unaffected relatives. The following predictors are included in the model: for each relative, relationship to the individual being counselled, twin status, current age (or age at death), breast cancer and ovarian cancer status, age of cancer diagnosis if affected, tumour biomarkers for breast cancer-affected individuals, ethnicity, and risk-modifying interventions such as mastectomy or oophorectomy [109]. BRCAPRO does not consider other classical (non-familial) breast cancer risk factors. BRCAPRO can also predict the risk of ovarian cancer and contralateral breast cancer. The horizon used for risk prediction is five years; however, this can be modified by the user to any timeframe [109].

Included studies of the BRCAPRO model

The development study for the original version of the BRCAPRO model is reported in Parmigiani 1998 [110]. This study focused mainly on the prediction of *BRCA1* or *BRCA2* mutations by the model but also reports model predictions for breast cancer for a small number of individuals.

We identified nine external validations of the BRCAPRO model in the target population (Amir 2003; Li 2021a; McCarthy 2021; Terry 2019; Kenan 2018; MacInnis 2020; McCarthy 2020; Guan 2022; Guan 2023). Of these, eight validations reported both model calibration and discrimination measures. One study reported that ROC analysis was performed, but data were not shown (Kenan 2018). We contacted the authors, who advised that these data were unavailable. Because several validation studies used overlapping populations, we included only five validations in the meta-analysis

of discrimination (Amir 2003; Li 2021a; McCarthy 2021; Terry 2019; Guan 2023), and the same five studies in the meta-analysis of calibration (Amir 2003; Li 2021a; McCarthy 2021; Terry 2019; Guan 2023). See [Summary of findings 5. Table 5](#) summarises the main characteristics of the studies validating the Tyrer-Cuzick model.

Models not included in meta-analysis

We identified eight breast cancer risk prediction models that could not be included in the quantitative synthesis as each had only one to three external validation studies:

- Claus;
- BCSC;
- Manual (adjusted Claus);
- CARE;
- iCARE-Lit;
- Palmer and colleagues' model A and model B;
- BRCAPro-BCRAT.

We summarise the characteristics of the included studies for each of these models in [Supplementary material 11](#). The included studies that validated at least one of these models were: Amir 2003; Boggs 2013 [111]; Evans 2014 [112]; Evans 2016 Chapter 2 (study 1a); Gard 2024 [113]; Guan 2023; Li 2021a; McCarthy 2020; McCarthy 2021; and Palmer 2021.

Models without external validation studies

Four studies reported the development (with or without internal validation) of a new breast cancer risk prediction model or models (Louro 2021 [114]; Ming 2019 (Swiss) [115]; Ming 2020; Yala 2019). It is recognised that risk prediction models should not be used in clinical practice before they have been externally validated [47, 51, 116, 117]. Because no external validations for these models were identified in the target population for this review, these studies have been listed as 'awaiting classification' and summarised in [Supplementary material 4](#)). Importantly, of these four studies, three also validated existing breast cancer risk prediction models (Ming 2019 (USA); Ming 2020; Yala 2019), and we thus also classified them as included studies (see [Supplementary material 2](#)).

Models updated to include additional predictor(s)

We identified five articles that 'updated' an existing breast cancer risk prediction model by adding one or more additional predictors to the model. These incremental value studies compared the performance of the 'updated' model to that of the original model. Detailed methods and results for these studies are reported in [Supplementary material 12](#).

Excluded studies

We excluded 355 articles at the full-text screening stage for the following reasons:

- ineligible population: 154 articles;
- ineligible design: 93 articles;
- conference abstracts unlikely to meet inclusion criteria or with unclear eligibility: 79 articles;
- unavailable abstract/full-text; 12 articles
- study focused on mutation prediction rather than breast cancer: 10 articles;

- study reported performance in risk quantiles with family history of breast cancer unspecified: 5 articles;
- study focused on a model specific to *BRCA* mutation carriers: 1 article;
- study described a model-switching algorithm rather than a risk model: 1 article.

For further details, see [Supplementary material 3](#).

Risk of bias and applicability assessment of included studies

The risk of bias and applicability ratings described below relate to the entire study when the full study population was applicable to this review (i.e. composed solely of, or enriched for, women with a family history of breast cancer). When only a subgroup analysis was relevant (e.g. model performance evaluated in women with a family history of breast cancer), the ratings relate only to the subgroup of interest rather than the entire study. To avoid double rating, we assessed missing or incomplete predictor information in the 'Analysis' domain as missing data rather than in the 'Predictors' domain.

Models included in meta-analysis

Gail (BCRAT)

Across validation studies included in meta-analyses, we generally rated the risk of bias as low or unclear in the participants, predictors, and outcome domains. In contrast, we frequently rated the analysis domain as high or unclear risk, most commonly due to limited numbers of outcome events, incomplete evaluation of model performance measures, and concerns regarding the amount or handling of missing data. Validation studies not included in the meta-analyses showed a similar pattern, with the analysis domain frequently rated as high risk of bias. Refer to [Table 6](#).

Applicability concerns were largely low across all domains, with isolated exceptions related to inclusion of participants with a prior breast cancer diagnosis or unclear outcome definitions. Refer to [Table 7](#).

Tyrer-Cuzick (IBIS)

We rated the development study as having an unclear risk of bias for all PROBAST domains due to lack of reporting of pertinent information. For validation studies included in the meta-analyses, we deemed the risk of bias to be generally low or unclear for the participants, predictors, and outcome domains, and the analysis domain to be at predominantly high or unclear risk due to an inadequate number of participants with the outcome, relevant model performance measures not being evaluated, concerns over the amount and handling of missing data, or unclear reporting of these parameters. Similar patterns were observed in validation studies not included in the meta-analyses. Refer to [Table 8](#).

We judged applicability to be of low concern for most studies and domains, with a small number of studies rated as unclear or high risk, due to participant selection or outcome definition. Refer to [Table 7](#).

BOADICEA

We rated the development study as high risk of bias, for the participants, outcome, and analysis domains, reflecting the use of

retrospective datasets, with most participants having had a history of breast cancer at recruitment, and limited reporting. Among validation studies included in the meta-analyses, we generally rated the risk of bias as low or unclear for the participants, predictors, and outcome domains, and the analysis domain to be at predominantly high or unclear risk due to relevant model performance measures not being evaluated, an inadequate number of participants with the outcome, concerns over the amount and handling of missing data, or unclear reporting in relation to missing data. Validation studies not included in the meta-analyses demonstrated a similar pattern. Refer to [Table 9](#).

Applicability concern was generally low across studies, with unclear ratings mainly arising from inclusion of women with a prior breast cancer diagnosis or insufficient reporting of the extent of enrichment for family history of breast cancer in included cohorts. Refer to [Table 7](#).

BRCAPRO

We rated the development study as having a high or unclear risk of bias across all domains, due to heterogeneous retrospective datasets, lack of reporting of model performance measures, and lack of reporting of other pertinent information. For validation studies included in the meta-analyses, we rated the risk of bias as low for the participants, predictors, and outcome domains in most studies, while the analysis domain showed variable ratings, including high or unclear risk of bias due to an inadequate number of participants with the outcome or unclear details on the amount and handling of missing data. For validation studies not included in meta-analyses, again the analysis domain was the most common source of bias concern. Refer to [Table 10](#).

Applicability concerns were consistently low across all validation studies but unclear for the development study due to lack of reporting of participant details. Refer to [Table 7](#).

Detailed PROBAST assessments for these models are reported in [Supplementary material 13](#).

Risk of bias and applicability assessment of models with one to three external validation studies are detailed in [Supplementary material 10](#).

Deficiencies of prognostic model development and validation studies (models with PROBAST rating)

We identified several common issues among the model development and validation studies that were given a high or unclear PROBAST rating. Among the development studies, the most frequent concerns were the use of mixed retrospective datasets from multiple sources; predictor assessments that were typically not conducted without knowledge of the outcome;

and limited reporting, with pertinent information missing across several PROBAST domains.

We also identified several common issues among the model validation studies that received a high or unclear PROBAST rating. First, several studies reported only calibration or discrimination measures rather than both. Second, missing predictor data were common, with omission or incomplete capture of certain predictors. The extent of missing data was often unclear, particularly in studies where women with a family history of breast cancer were analysed as a subgroup of the overall study population, as missing data were frequently not reported separately for this subgroup. In addition, some models incorporate optional predictors, such as polygenic risk scores, which were often omitted in validation studies, limiting the ability to evaluate the performance of the full model. Finally, several studies did not report the median or mean length of follow-up.

Findings

Most identified studies reported calibration or discrimination measures. We report results from studies providing discrimination measures (C statistic or AUC) or calibration measures using observed-to-expected (O/E) ratios. Studies reporting both are presented separately by statistic type.

As noted above, we meta-analysed models that had at least four external validation studies for predicting breast cancer risk. This was to ensure that we could calculate prediction intervals from the studies available. Models meeting this threshold were:

- Gail (BCRAT) (all versions combined);
- Tyrer-Cuzick (IBIS) (all versions combined and version 8);
- BOADICEA (all versions combined); and
- BRCAPRO (all versions combined).

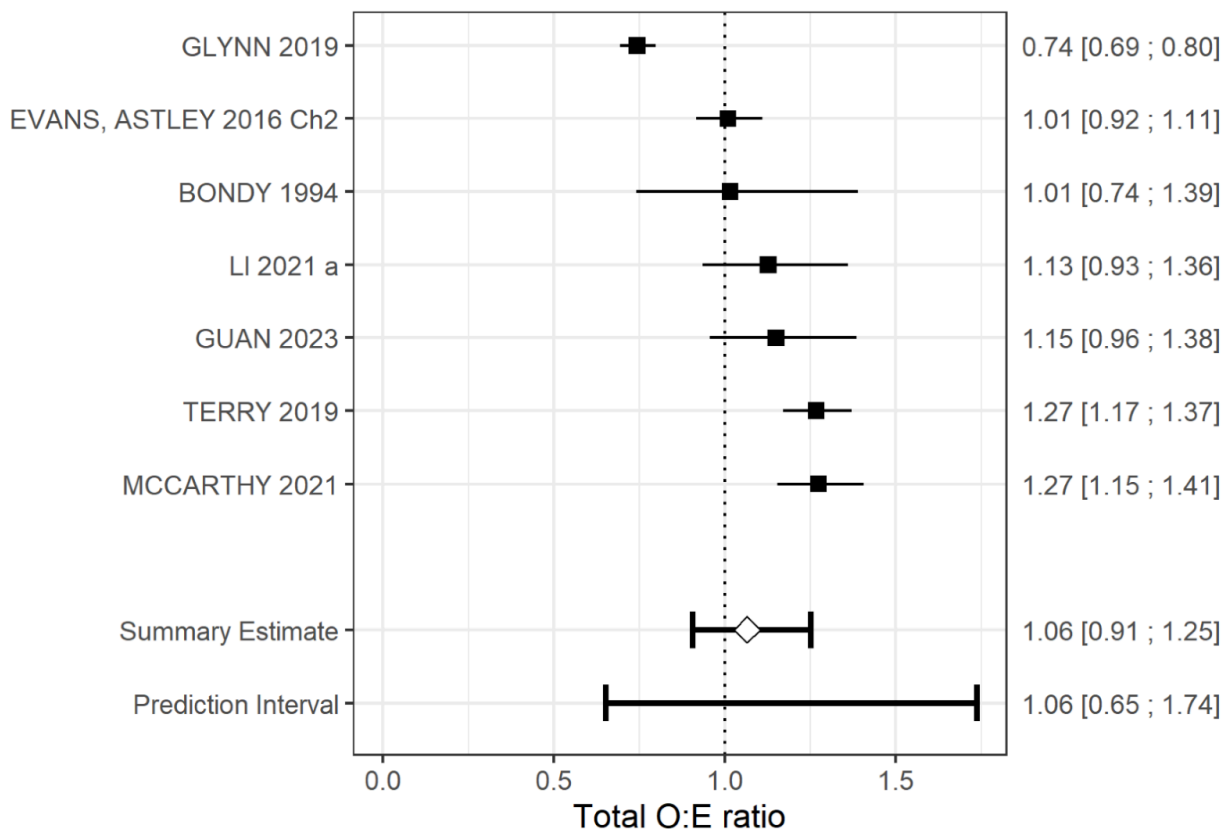
Models with meta-analysis

Gail (BCRAT)

Calibration

We identified 11 publications reporting external validation of the Gail model calibration. We excluded four overlapping populations for the purpose of meta-analysis, leaving seven studies. Six studies had a total of 60,785 individuals and 1690 breast cancers. We extrapolated the total number of participants ($n = 24,511$) for McCarthy 2021, as this information was not explicitly provided in the article. In addition, one study, Glynn 2019, reported 765 breast cancers over 86,892 two-yearly intervals, without reporting actual numbers of individuals. The pooled O/E of the seven studies was 1.06 (95% CI 0.91 to 1.25), with a high degree of heterogeneity (between-study standard deviation, $\text{Tau}^2 = 0.1756$; $I^2 = 90.26\%$). The 95% PI was 0.65 to 1.74 ([Figure 2](#); [Summary of findings 1](#)).

Figure 2. Meta-analysis of the O/E ratio for the Gail (BCRAT) model 95% confidence intervals (CIs) may differ slightly from those reported in the original studies because the R program used for meta-analysis recomputed them using the method described by Debray and colleagues, based on the O, E, and N values reported in each study [24]. The original studies used different methods to calculate 95% CIs, such as assuming Poisson distributions or bootstrap approaches.



Discrimination

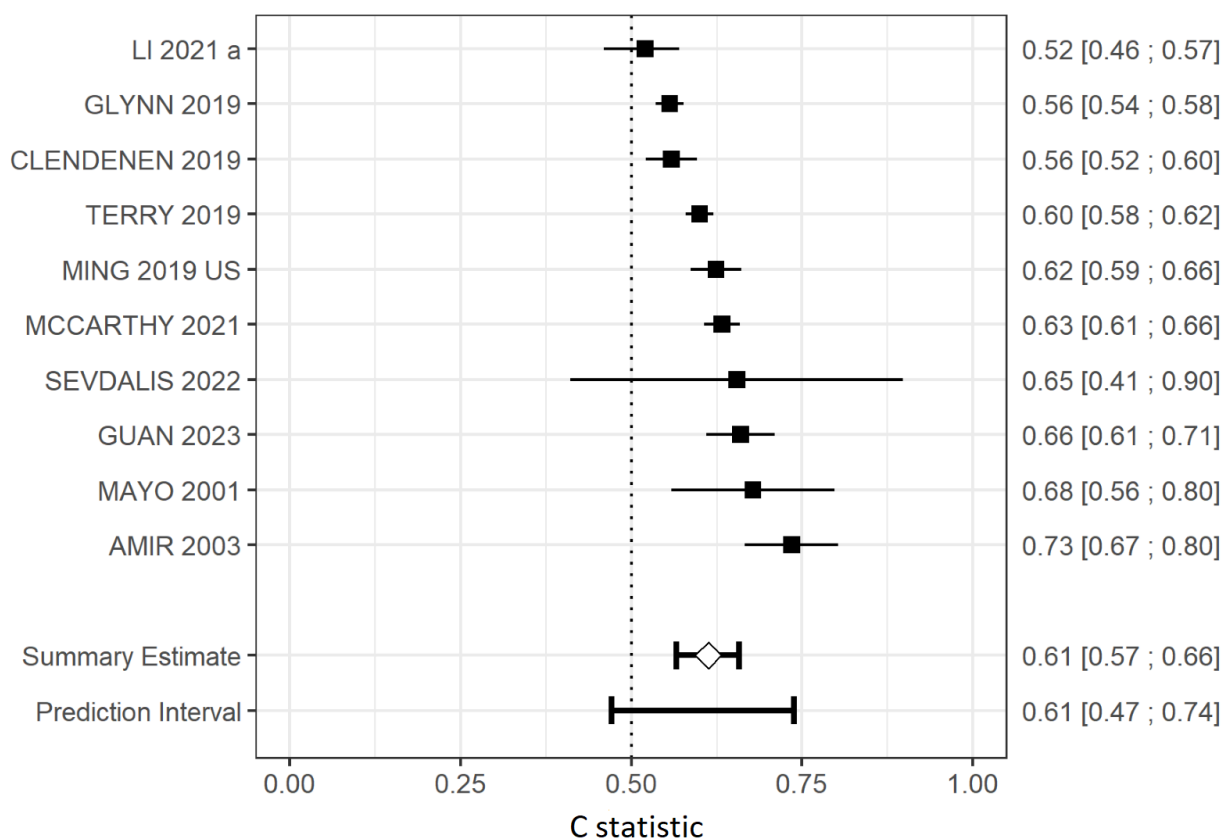
We identified 10 external validation studies in the target population. From eight studies that provided numbers of individuals, there were a total of 54,184 individuals; in nine studies providing numbers of breast cancers observed, there were 2939 during the overall observation time.

We extrapolated the total number of participants for McCarthy 2021. Clendenen 2019 did not clearly present the numbers, and Glynn 2019 reported 765 breast cancers over 86,892 two-yearly

intervals. Three studies that reported the C statistic did not report a measure of uncertainty for this measure. We estimated these missing measures of uncertainty for Ming 2019 (USA), Sevdalis 2022, and Mayo 2001 using the Newcombe method.

The pooled C statistic of the 10 studies (with an estimated 95% CI) was 0.61 (95% CI 0.57 to 0.66), with some degree of heterogeneity (between-study standard deviation, $\tau^2 = 0.0522$; $I^2 = 89.4\%$). The 95% PI was 0.47 to 0.74 (Figure 3; Summary of findings 1).

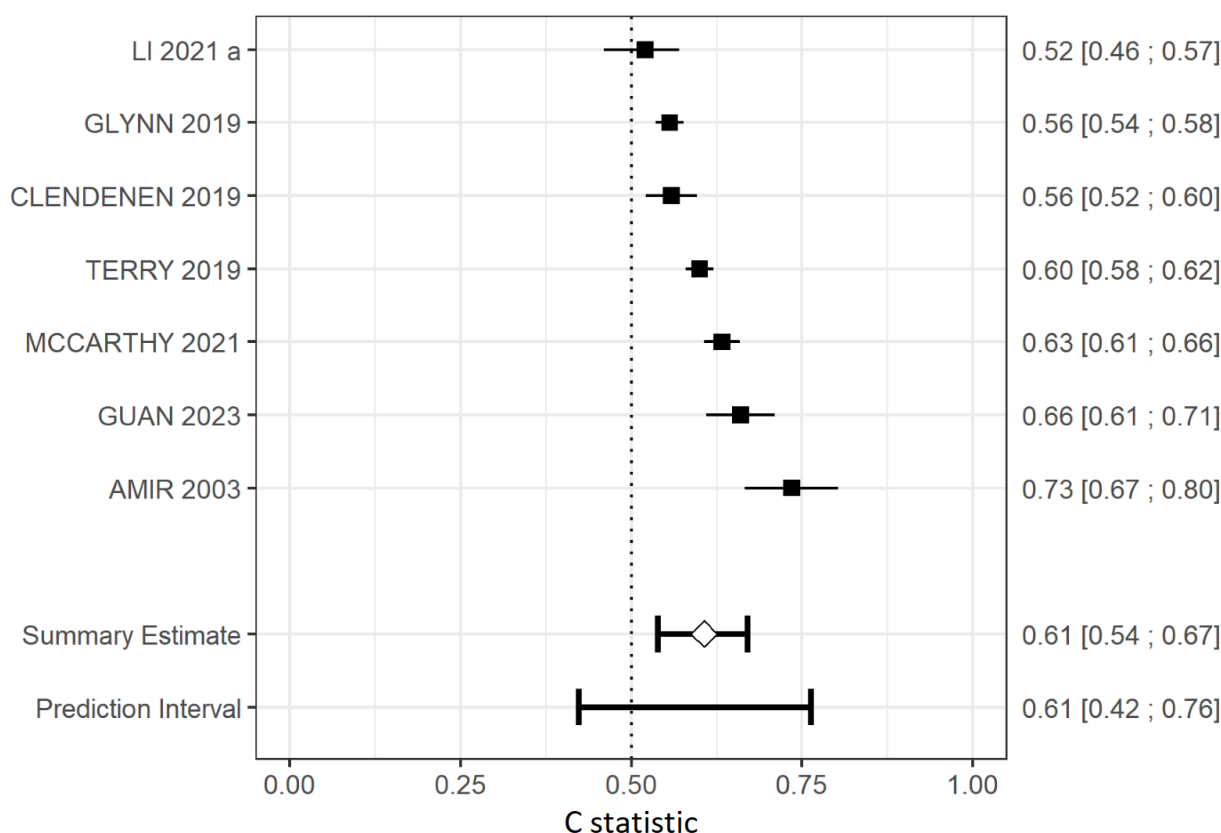
Figure 3. Meta-analysis of the C statistic for the Gail (BCRAT) model



We conducted a sensitivity analysis omitting the three studies for which we estimated a 95% CI, as the studies did not report one (Figure 4). The pooled C statistic of the seven studies was 0.61 (95% CI 0.54 to 0.67), with a high degree of heterogeneity (between-study

standard deviation, $\tau^2 = 0.0579$; $I^2 = 93.10\%$). The 95% PI was 0.42 to 0.76. Although the 95% CI for the sensitivity analysis result is wider, both analyses produce the same C statistic estimate.

Figure 4. Sensitivity analysis of the C statistic for the Gail (BCRAT) model excluding three studies reporting no 95% confidence interval



There was significant heterogeneity between the studies in relation to time horizons, from predictions within a 2-year window (Glynn 2019) to lifetime risk (Sevdalis 2022).

Tyrer-Cuzick (IBIS)

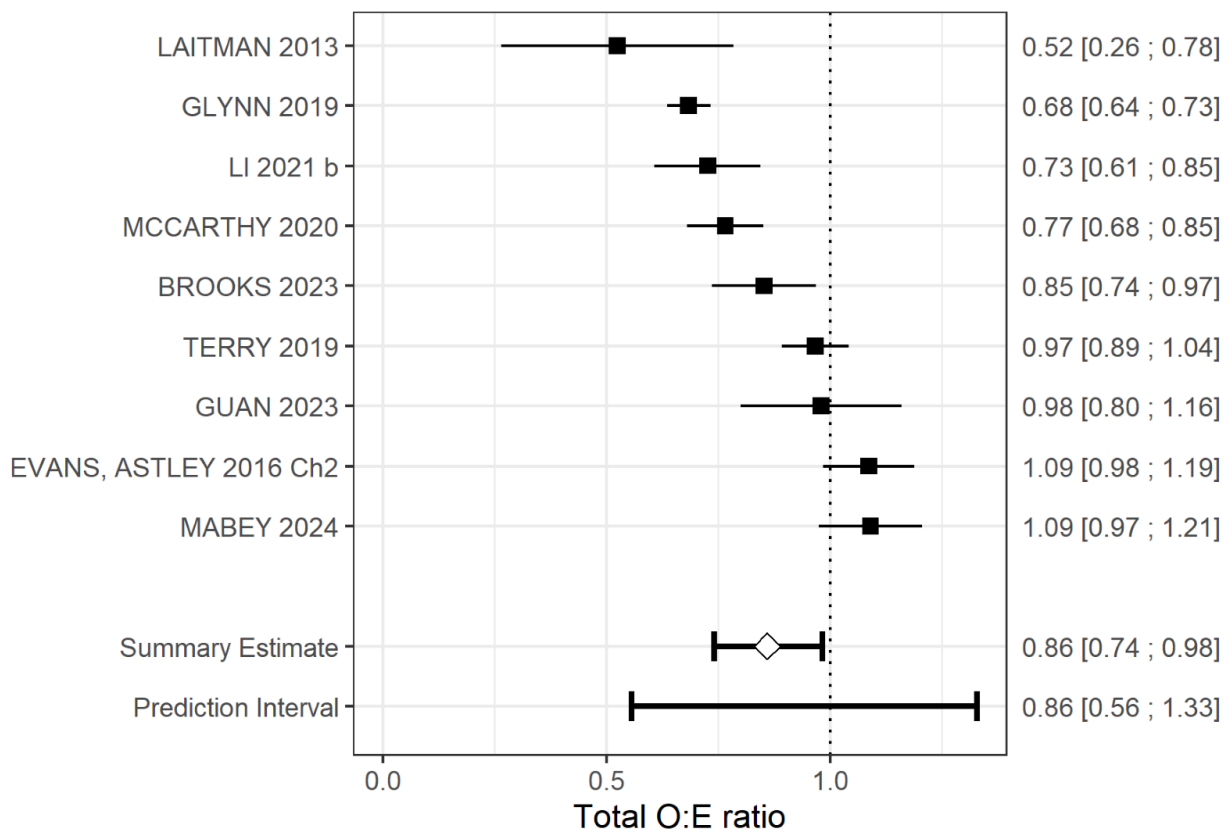
Calibration

We identified 14 external validation articles reporting Tyrer-Cuzick model calibration (O/E ratio). We excluded five studies whose population overlapped with other studies (Amir 2003; Gong 2014; Li 2021a; MacInnis 2020; McCarthy 2021), leaving nine studies for meta-analysis. Eight of the nine studies included a total of 184,068 individuals and 2124 breast cancers observed during the overall

observation time. One study, Glynn 2019, reported 765 breast cancers over 86,892 two-yearly intervals. One study, McCarthy 2020, included two versions of the Tyrer-Cuzick model (versions 7 and 8) but we considered only the latest version (v8) in the meta-analysis. McCarthy 2020 reported results separately for participants with one and with two or more first- or second-degree family members with breast or ovarian cancer. We pooled these results using the method described by Vandenbroucke [1].

The pooled O/E ratio of the nine studies was 0.86 (95% CI 0.74 to 0.98), with a high degree of heterogeneity (between-study standard deviation, $\tau^2 = 0.1569$; $I^2 = 91.14\%$). The 95% PI was 0.56 to 1.33 (Figure 5; Summary of findings 2).

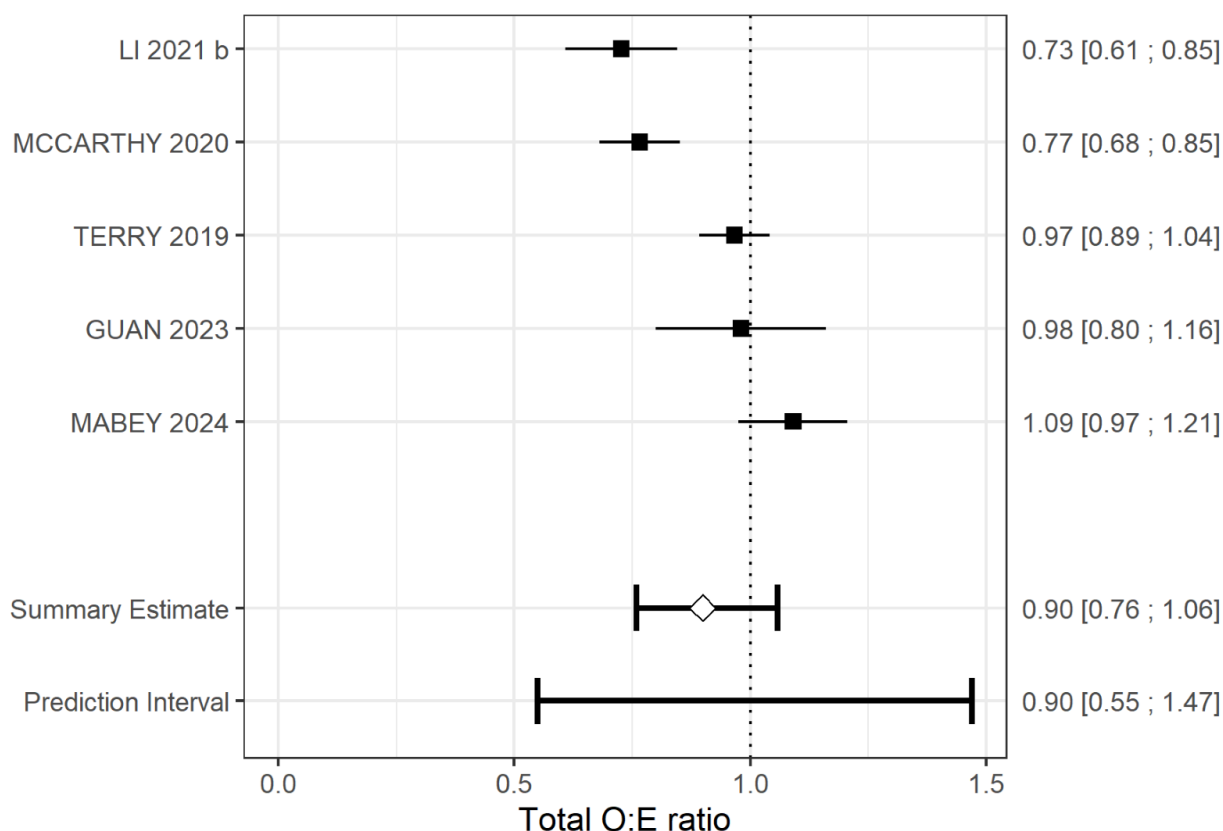
Figure 5. Meta-analysis of the O/E ratio for the Tyrer-Cuzick (IBIS) model 95% confidence intervals (CIs) may differ slightly from those reported in the original studies because the R program used for meta-analysis recomputed them using the method described by Debray and colleagues [24], based on the O, E, and N values reported in each study. The original studies used different methods to calculate 95% CIs, such as assuming Poisson distributions or bootstrap approaches.



The aforementioned analysis pooled several different versions of the Tyrer-Cuzick model. It is preferable to pool validation studies that report performance of the same model version in a meta-analysis where a sufficient number of studies exist to do so. We identified five external validation studies of Tyrer-Cuzick/IBIS version 8.0 that reported an O/E ratio (Guan 2023; Li 2021b; Mabey

2024; McCarthy 2020; Terry 2019). These included a total of 166,696 individuals and 1491 breast cancers observed during the overall observation time. The pooled O/E ratio of the five studies was 0.90 (95% CI 0.76 to 1.06), with a high degree of heterogeneity (between-study standard deviation, $\text{Tau}^2 = 0.1262$; $I^2 = 84.89\%$). The 95% PI was 0.55 to 1.47 (Figure 6; Summary of findings 3).

Figure 6. Meta-analysis of the O/E ratio for Tyrer-Cuzick model version 8.0



We identified three external validation studies of Tyrer-Cuzick/IBIS version 6.0 that reported an O/E ratio (Evans 2016 Chapter 2 (study 1a); Glynn 2019; Laitman 2013); these data were insufficient to combine in meta-analysis.

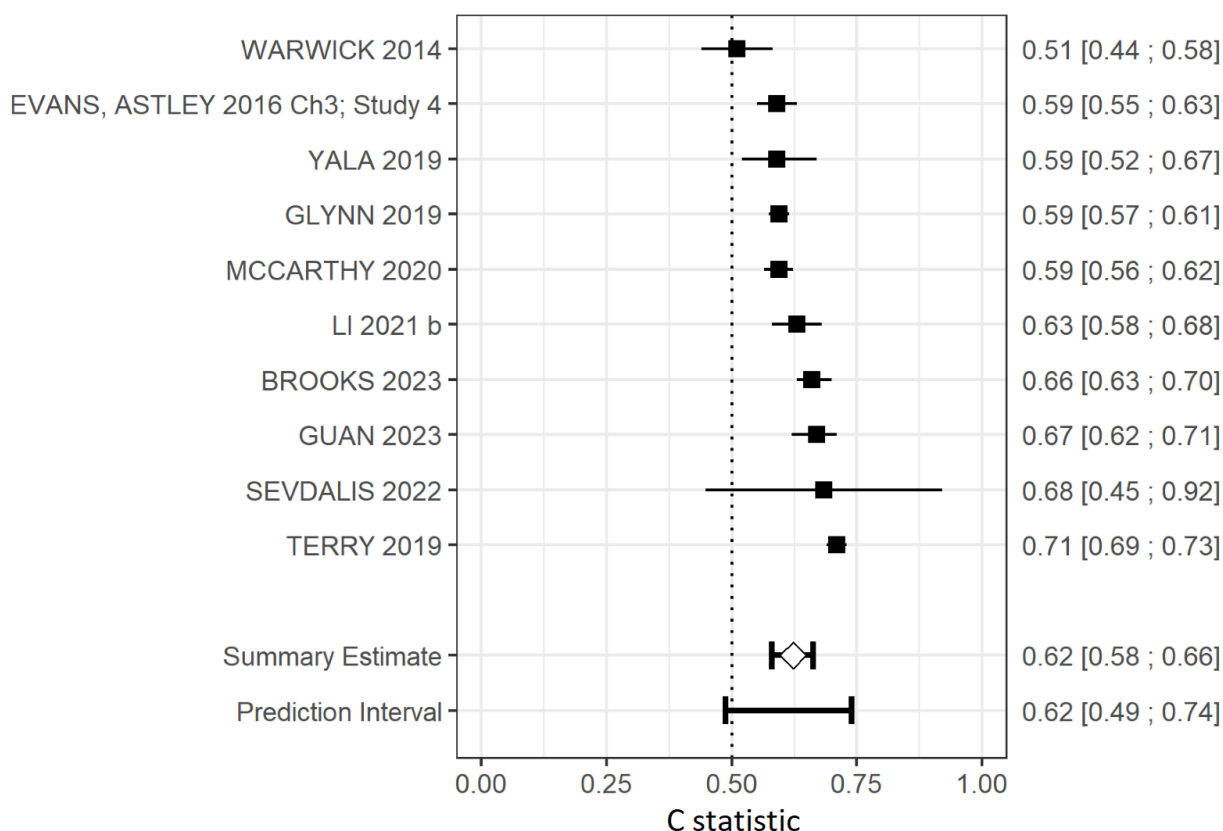
Discrimination

We identified 15 external validation articles reporting discrimination of the Tyrer-Cuzick model. We excluded five with overlapping populations (Amir 2003; Li 2021a; MacInnis 2020; MacInnis 2021 [118]; Quante 2012), leaving 10 studies for meta-analysis. Nine of the ten studies included a total of 47,829 individuals and 1832 breast cancers observed during the overall observation time. For Li 2021b, the number of participants included in the total are only those in the sub-cohort with a polygenic risk

score (PRS). Glynn 2019 reported 765 breast cancers over 86,892 two-yearly intervals. McCarthy 2020 included two versions of the Tyrer-Cuzick model (versions 7 and 8), but we only considered the latest version (v8) in the meta-analysis. For this study, we pooled the C statistic results of those with one and those with two or more first- or second-degree family members with breast or ovarian cancer using a weighted average (inverse-variance) method. We used the Newcombe method to estimate the 95% CIs for two studies (Sevdalis 2022; Warwick 2014).

The pooled C statistic of the 10 studies was 0.62 (95% CI 0.58 to 0.66). Heterogeneity statistics were as follows: between-study standard deviation, $\tau^2 = 0.048$; $I^2 = 87.65\%$. The 95% PI was 0.49 to 0.74 (Figure 7; Summary of findings 2).

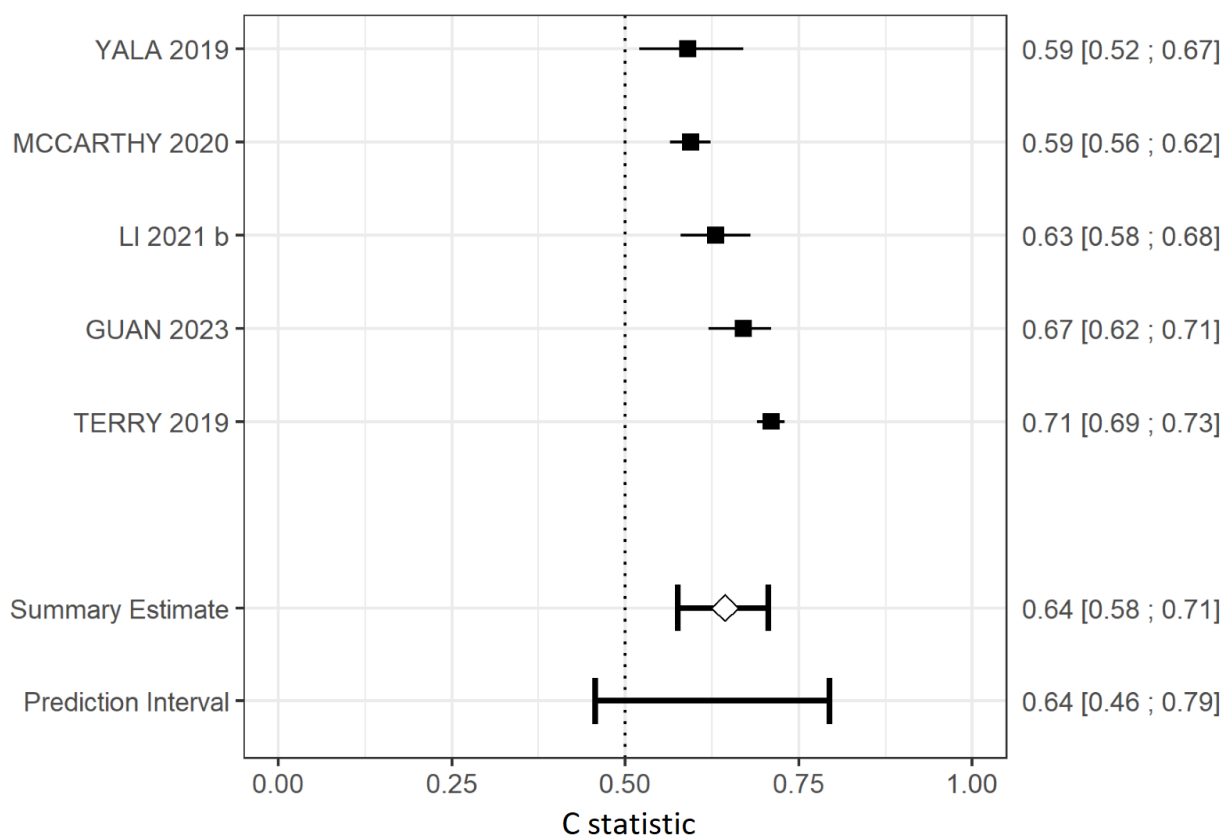
Figure 7. Meta-analysis of the C statistic for the Tyrer-Cuzick (IBIS) model



The aforementioned analysis pooled several different versions of the Tyrer-Cuzick model. It is preferable to pool validation studies that report performance of the same model version in a meta-analysis where a sufficient number of studies exist to do so. We identified three external validation studies using Tyrer-Cuzick version 6.0 that reported a C statistic (or AUC) (Evans 2016 Chapter 3 (study 4); Glynn 2019; Warwick 2014); this number was insufficient to combine in meta-analysis.

We identified five external validation studies using Tyrer-Cuzick (IBIS) version 8.0 that reported a C statistic (or AUC) (Guan 2023; Li 2021b; McCarthy 2020; Terry 2019; Yala 2019). The five studies included a total of 38,405 individuals and 1292 breast cancers observed during the overall observation time. The pooled C statistic of the five studies was 0.64 (95% CI 0.58 to 0.71), with some degree of heterogeneity (between-study standard deviation, $\tau^2=0.0433$; $I^2=84.48\%$). The 95% PI was 0.46 to 0.79 (Figure 8; Summary of findings 3).

Figure 8. Meta-analysis of the C statistic for Tyrer-Cuzick model version 8.0



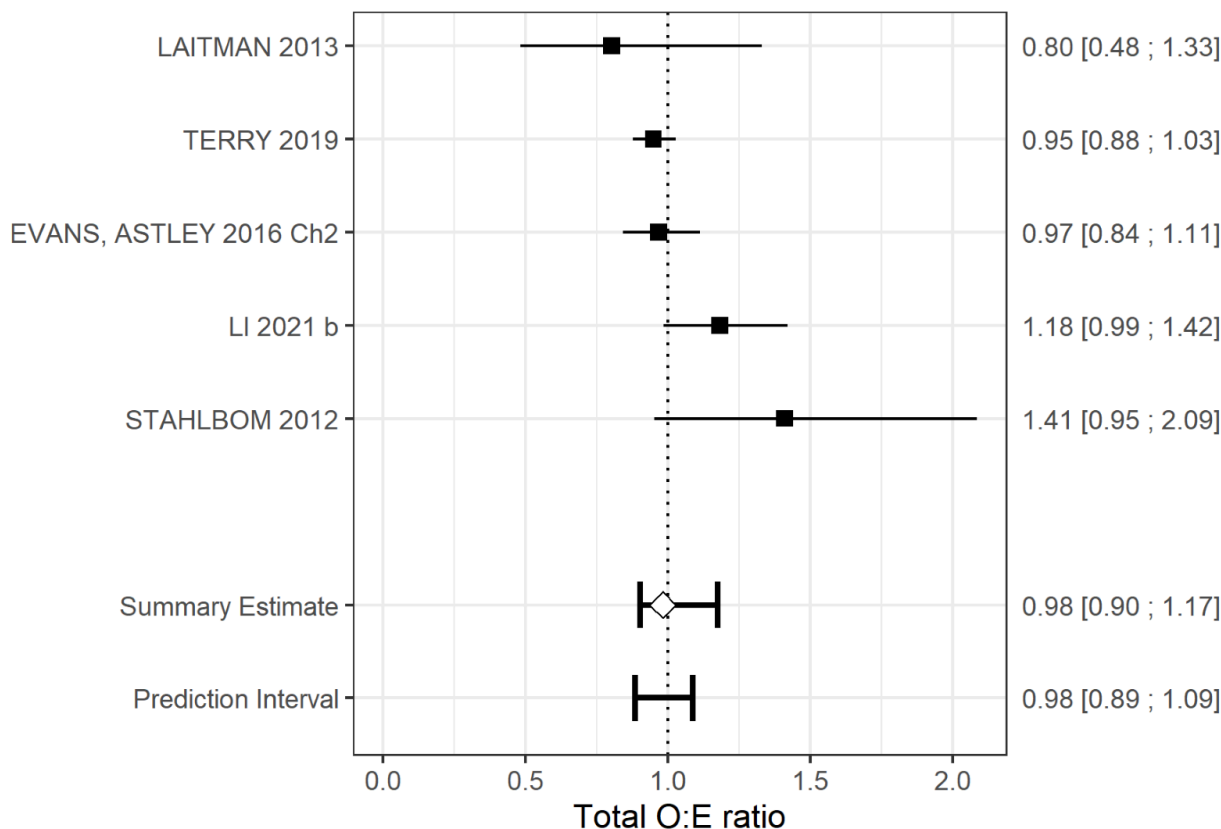
BOADICEA

Calibration

We identified eight external validation articles that reported calibration of the BOADICEA model. We excluded three articles whose populations overlapped with those of other included studies (Li 2021b; MacInnis 2013 [119, 120]; MacInnis 2020), leaving five studies for meta-analysis.

The five studies had a total of 23,596 individuals and 972 breast cancers observed during the overall observation time. The pooled O/E of the five studies was 0.98 (95% CI 0.90 to 1.17), with low heterogeneity (between-study standard deviation, $\tau^2 = 0.0009$; $I^2 = 0.01\%$). The 95% PI was 0.89 to 1.09 (Figure 9; Summary of findings 4).

Figure 9. Meta-analysis of the O/E ratio for the BOADICEA model 95% confidence intervals (CIs) may differ slightly from those reported in the original studies because the R program used for meta-analysis recomputed them using the method described by Debray and colleagues [24], based on the O, E, and N values reported in each study. The original studies used different methods to calculate 95% CIs, such as assuming Poisson distributions or bootstrap approaches.



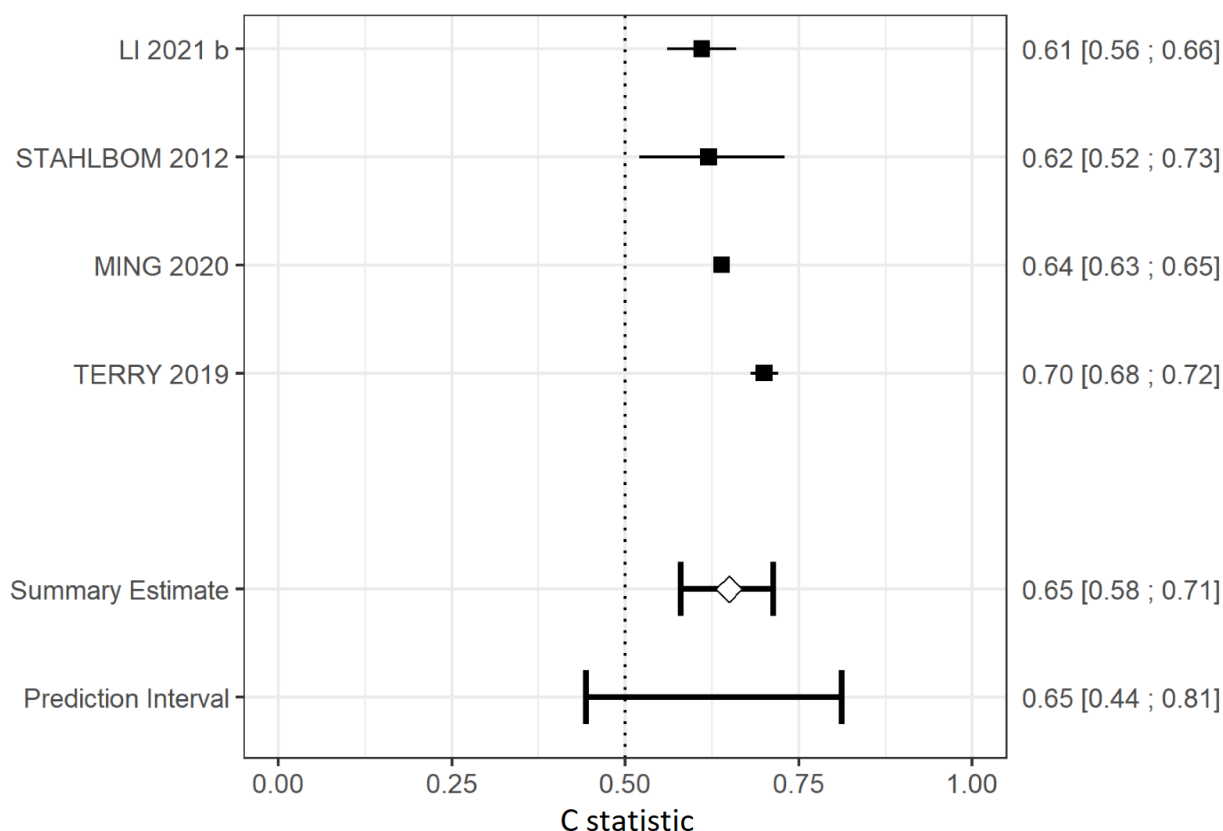
Discrimination

We identified nine external validation articles that reported discrimination of the BOADICEA model. We excluded five articles whose populations overlapped with those of other included studies (Li 2021a; MacInnis 2020; MacInnis 2021; Ming 2019 (Swiss); Quante 2015), leaving four studies for meta-analysis. The four studies had a total of 62,080 individuals and 5671 breast cancers observed during

the overall observation time. Ming 2020 did not report a 95% CI so we estimated it.

The pooled C statistic of the four studies (with the 95% CI estimated for Ming 2020) was 0.65 (95% CI 0.58 to 0.71), with a high degree of heterogeneity (between-study standard deviation, $\tau^2 = 0.0319$; $I^2 = 86.99\%$). The 95% PI was 0.44 to 0.81 (Figure 10; Summary of findings 4).

Figure 10. Meta-analysis of the C statistic for the BOADICEA model



We conducted a sensitivity analysis excluding Ming 2020, which did not report 95% CI. The overall summary estimate from the remaining studies was 0.65 (95% CI 0.51 to 0.77), with a wider 95% PI of 0.066 to 0.98.

BRCAPRO

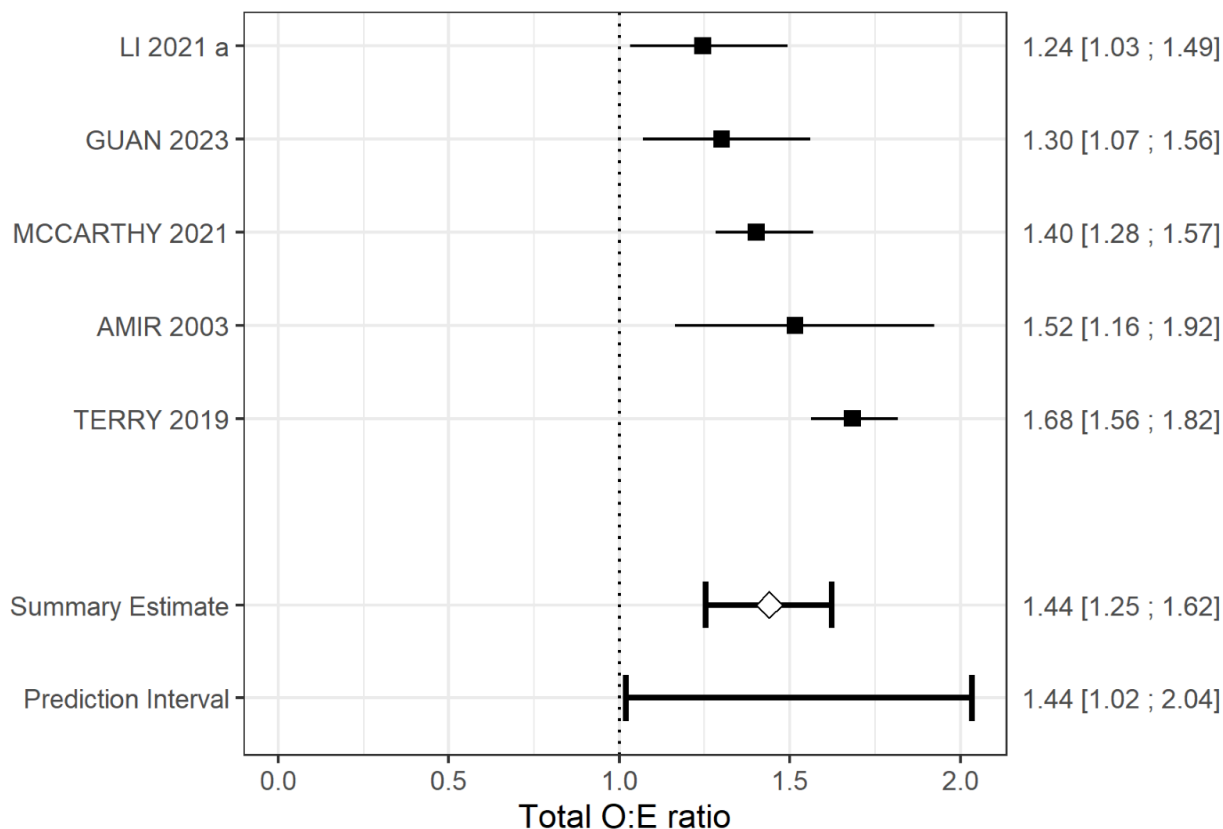
Calibration

We identified eight external validation articles reporting calibration (O/E ratio) of the BRCAPRO model. We excluded three articles whose populations overlapped with those of other included studies

(Guan 2022; MacInnis 2020; McCarthy 2020), leaving five studies for meta-analysis (Amir 2003; Guan 2023; Li 2021a; McCarthy 2021; Terry 2019). There were a total of 52,427 individuals and 1299 breast cancers observed during the overall observation time for the five studies.

The pooled O/E of the five studies was 1.44 (95% CI 1.25 to 1.62), with a moderate degree of heterogeneity (between-study standard deviation, $\tau^2 = 0.1395$; $I^2 = 67.64\%$). The 95% PI was 1.02 to 2.04 (Figure 11; Summary of findings 5).

Figure 11. Meta-analysis of the O/E ratio for the BRCAPRO model 95% confidence intervals (CIs) may differ slightly from those reported in the original studies because the R program used for meta-analysis recomputed them using the method described by Debray and colleagues [24], based on the O, E, and N values reported in each study. The original studies used different methods to calculate 95% CIs, such as assuming Poisson distributions or bootstrap approaches.



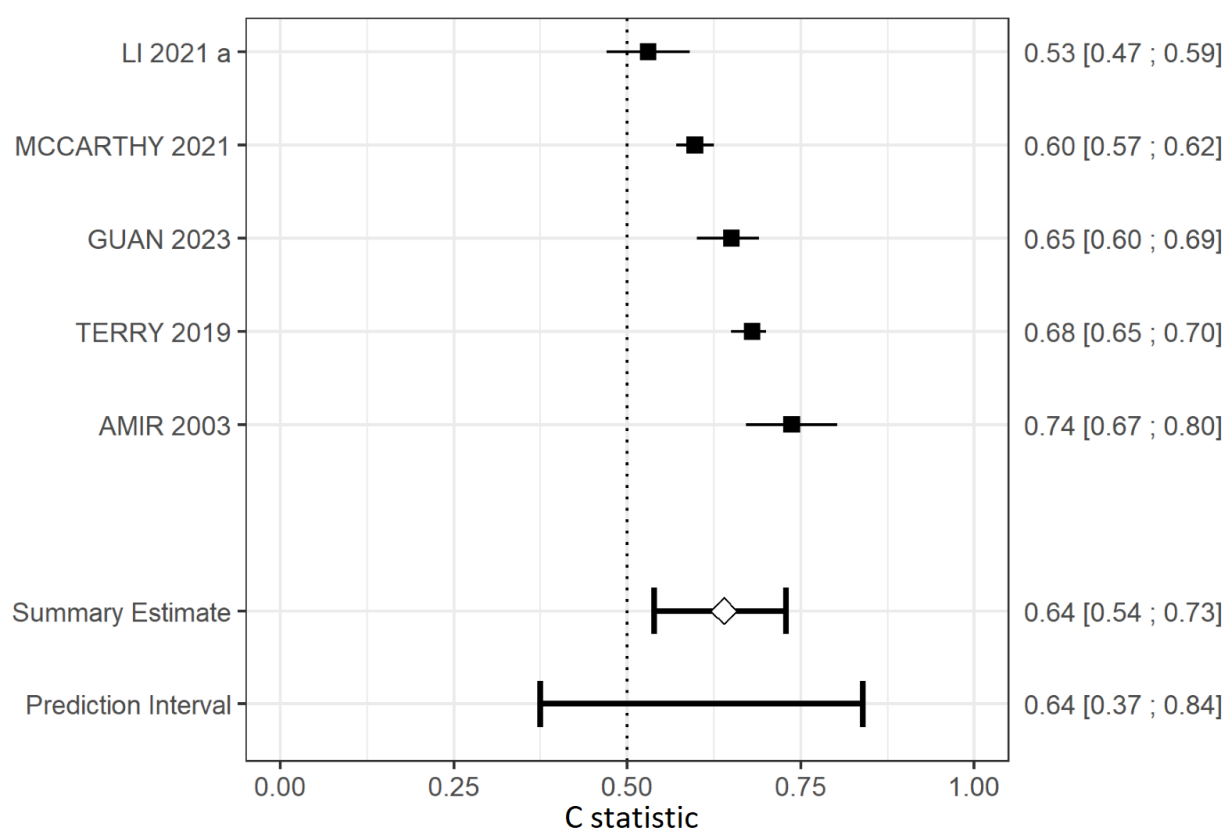
Discrimination

We identified eight articles reporting external validation (discrimination) of the BRCAPRO model. We excluded three articles whose populations overlapped with those of other included studies. We included the remaining five studies in the meta-analysis (Amir 2003; Guan 2023; Li 2021a; McCarthy 2021; Terry 2019). The

five studies that reported C statistics on unique populations had a total of 52,427 individuals and 1299 breast cancers observed.

The pooled C statistic of the five studies was 0.64 (95% CI 0.54 to 0.73), with a high degree of heterogeneity (between-study standard deviation, $\text{Tau}^2 = 0.0617$; $I^2 = 90.97\%$). The 95% PI was 0.37 to 0.84 (Figure 12; Summary of findings 5).

Figure 12. Meta-analysis of the C statistic for the BRCAPRO model



Models without meta-analysis

We could not meta-analyse performance statistics for the following seven breast cancer risk prediction models as they had only one to three external validation studies. We narratively summarise the individual results of the contributing validation studies for each model here.

Claus

We identified two external validations of the Claus model in the target population (Amir 2003; McCarthy 2020). Both validations reported both model calibration (E/O or O/E) and discrimination measures (C statistic/AUC).

Amir 2003 reported an E/O ratio of 0.76 (95% CI 0.59 to 0.99) for the total study population of 3150 women that were assessed in the family history clinic and an E/O ratio of 0.56 (95% CI 0.43 to 0.75) for the subgroup of 1933 individuals enrolled in a regular family history mammography screening programme. The C statistic for the Claus model in this study was 0.716 (95% CI 0.648 to 0.784) for the total study population of 3150 individuals. The C statistic was not reported separately for the screening population.

McCarthy 2020 reported an O/E ratio of 1.69 (95% CI 1.48 to 1.87) and a C statistic (AUC) of 0.59 (95% CI 0.56 to 0.62) based on 11,873 individuals for whom the Claus model was applicable (younger than 79 years with at least one first- or second-degree relative with breast cancer or at least one first-degree relative with ovarian cancer). Performance of the model was validated with respect to the risk of both DCIS or invasive breast cancer.

BCSC

We identified one external validation of the BCSC model in the target population (McCarthy 2021). This study validated the model in women attending for screening mammography at three centres in the USA, and reported performance in the subgroup of women with a family history of breast cancer (i.e. first- or second-degree relative with breast cancer). McCarthy 2021 reported both model calibration (O/E) and discrimination measures (AUC) for this subgroup. The O/E ratio for the BCSC model in the subgroup of interest was 1.509 (95% CI 1.38 to 1.686) and the C statistic (AUC) was 0.631 (95% CI 0.603 to 0.657).

The number of individuals with a family history was unclear, as McCarthy 2021 reported that 18,097 had at least one first-degree relative with breast cancer and 21,326 had at least one second-degree relative with breast cancer (but did not report how many of these also had a first-degree affected relative), so some participants may have been counted in both categories (i.e. first-degree and second-degree relatives).

A second study, Gard 2024, updated the BCSC model with additional predictors and more recent breast cancer incidence rates, and reported calibration (internal validation only) of the updated (v3) model in various subgroups of women with a family history of breast cancer enrolled in the BCSC cohort. As an example, 23,004 participants in the cohort had at least one first-degree and at least one second-degree relative with breast cancer. For this group, the combined E/O ratio (calculated as the average of 10 E/O values) was 1.00 (95% CI 0.92 to 1.08) for 5-year risk and 0.93 (95% CI 0.88 to

0.98) for 10-year risk in this internal validation. Version 3 of the BCSC model has not yet been externally validated in the target population for this review.

Manual (adjusted Claus)

We identified three references reporting external validation of the Manual model in the target population (Amir 2003; Evans 2014; Evans 2016 Chapter 2 (study 1a)). These articles had overlapping participants, with the more recent articles having larger numbers of participants and longer follow-up. All three articles reported model calibration measures (O/E or E/O ratio) but only Amir 2003 reported model discrimination (AUC). Amir 2003 reported the E/O ratio of the Manual model as 1.22 (95% CI 0.95 to 1.58) for the total study population of 3150 women that were assessed in the family history clinic, and an E/O ratio of 0.89 (95% CI 0.68 to 1.20) for the subgroup of 1933 individuals enrolled in a regular family history mammography screening programme.

In an overlapping population with more participants (8784 individuals), Evans 2014 reported an O/E ratio of 1.05 (95% CI 0.95 to 1.16). Evans 2014 also reported model calibration for different NICE risk categories (average, moderate, high, very high). Evans 2016 Chapter 2 (study 1a) reported an O/E ratio of 1.05 (95% CI 0.95 to 1.16) based on 8824 individuals from the same population.

In relation to Manual model discrimination, Amir 2003 reported a C statistic of 0.727 (95% CI 0.656 to 0.798) based on 3150 women that were assessed in a UK family history clinic.

CARE

We identified one external validation study of the CARE model in the target population (Boggs 2013). This study reported model calibration in the subgroup of participants with one and with two or more breast cancer-affected first-degree relatives with an E/O ratio of 0.85 (95% CI 0.72 to 1.0) based on 4298 participants with one breast cancer-affected first-degree relative. The E/O ratio for participants with two or more breast cancer-affected first-degree relatives was 1.0 (95% CI 0.52 to 1.92), based on 187 individuals. Model discrimination was not reported separately for the subgroup of women with a family history of breast cancer.

iCARE-Lit

We identified one external validation of the iCARE-Lit model in the target population for this review (Li 2021a). This study reported both model calibration (O/E ratio) and discrimination (C statistic) in the target population. Li 2021a reported an E/O ratio of 0.77 (95% CI 0.63 to 0.92) for 1720 women with a family history of breast cancer in first- or second-degree relatives. Li 2021a also performed an analysis censoring in situ breast cancer. In this case, the E/O ratio was 0.76 (95% CI 0.63 to 0.92).

With regard to the discriminatory performance of the iCARE-Lit model in the subgroup of women with a family history of breast cancer in first- or second-degree relatives, Li 2021a reported a C statistic of 0.53 (95% CI 0.47 to 0.59) for 1720 individuals without censoring at diagnosis of DCIS. The corresponding C statistic in this subgroup was identical 0.53 (95% CI 0.47 to 0.59) when censoring at diagnosis of DCIS was performed.

Palmer model A and model B

We identified one validation study of Palmer models A and B (Palmer 2021). These two models were externally validated in women enrolled in the Black Women's Health Study (BWHS) as part of the same publication reporting their development. Calibration (E/O ratio) of the models was reported separately for the subgroup with first-degree family history of breast cancer. However, discrimination was not reported separately for the family history subgroup. For model A, Palmer 2021 reported an E/O ratio of 1.08 (95% CI 0.94 to 1.22) for women with one affected relative who was 50 years or older at diagnosis, and an E/O ratio of 1.12 (95% CI 0.92 to 1.35) for individuals with either one affected relative who was diagnosed with breast cancer before 50 years of age or with at least two affected relatives.

For model B, Palmer 2021 reported an E/O ratio of 1.14 (95% CI 1.0 to 1.29) for women with one affected relative who was 50 years or older at diagnosis, and an E/O ratio of 1.16 (95% CI 0.96 to 1.41) for individuals with either one affected relative diagnosed with breast cancer before 50 years of age or at least two affected relatives.

Palmer 2021 did not report the numbers of individuals in the family history subgroups (there were 51,798 individuals in the overall cohort).

BRCAPRO-BCRAT

We identified three validation studies of the BRCAPRO-BCRAT hybrid model (Guan 2023; Li 2021a; McCarthy 2021). Li 2021a and McCarthy 2021 reported calibration and discrimination of the BRCAPRO-BCRAT model in a subgroup of women with a family history of breast cancer in first- or second-degree relatives. Guan 2023 reported both calibration and discrimination in the CGN cohort, which is enriched by family history, and also in the 'strong family history subgroup' within this cohort. With regard to model calibration, Li 2021a reported an E/O ratio of 0.8 (95% CI 0.67 to 0.97) based on 1720 women with a family history of breast cancer in first- or second-degree relatives. Li 2021a also performed an analysis censoring in situ breast cancer: in this case, the E/O ratio was 0.80 (95% CI 0.67 to 0.97). McCarthy 2021 reported an O/E ratio of 1.263 (95% CI 1.153 to 1.408) in the subgroup of women with a family history of breast cancer (i.e. first- or second-degree relative with breast cancer). The number of individuals with a family history was unclear in McCarthy 2021: it reported that 18,097 had at least one first-degree relative with breast cancer and 21,326 had at least one second-degree relative with breast cancer, but it did not report how many of these also had a first-degree affected relative, so some participants may have been counted in both categories (i.e. first-degree and second-degree relatives). Guan 2023 reported an O/E ratio of 1.03 (95% CI 0.84 to 1.23) for the overall (enriched for family history of cancer) cohort and 0.81 (95% CI 0.55 to 1.09) for the 'strong family history' subgroup within this cohort.

With regard to discriminatory performance of the BRCAPRO-BCRAT model in women with a family history of breast cancer, Li 2021a reported a C statistic of 0.52 (95% CI 0.47 to 0.58) for 1720 women with a family history of breast cancer in first- or second-degree relatives without censoring at diagnosis of DCIS. The corresponding C statistic in this subgroup was identical, at 0.53 (95% CI 0.47 to 0.59), when censoring at diagnosis of DCIS was performed. McCarthy 2021 reported a C statistic (AUC) of 0.636 (95% CI 0.611 to 0.666) for the model in the subgroup of women with a family history of breast cancer (i.e. first- or second-degree relative with breast

cancer). As mentioned above, McCarthy 2021 was unclear about the number of individuals with a family history. Guan 2023 reported a C statistic (AUC) of 0.68 (95% CI 0.63 to 0.72) for the overall (enriched for family history of cancer) cohort and 0.71 (95% CI 0.63 to 0.79) for the 'strong family history' subgroup within this cohort.

Direct model comparison using the same dataset(s)

We qualitatively synthesised evidence on model performance from studies that compared multiple models in the same dataset. These multi-model studies provide for a more homogeneous comparison across models using similar cohorts. Seven studies compared calibration of two or more models (Evans 2016 Chapter 2 (study 1a); Glynn 2019; Guan 2023; Laitman 2013; Li 2021a; McCarthy 2020; Terry 2019). Six studies compared discrimination of two or more models (Amir 2003; Glynn 2019; Guan 2023; Li 2021a; McCarthy 2020; Terry 2019). Four of these studies compared both calibration and discrimination of multiple models (Glynn 2019; Li 2021a; MacInnis 2020; Terry 2019). Findings from these direct model comparisons using the same dataset(s) are reported in [Supplementary material 14](#).

Additional analyses

Timespan of prediction

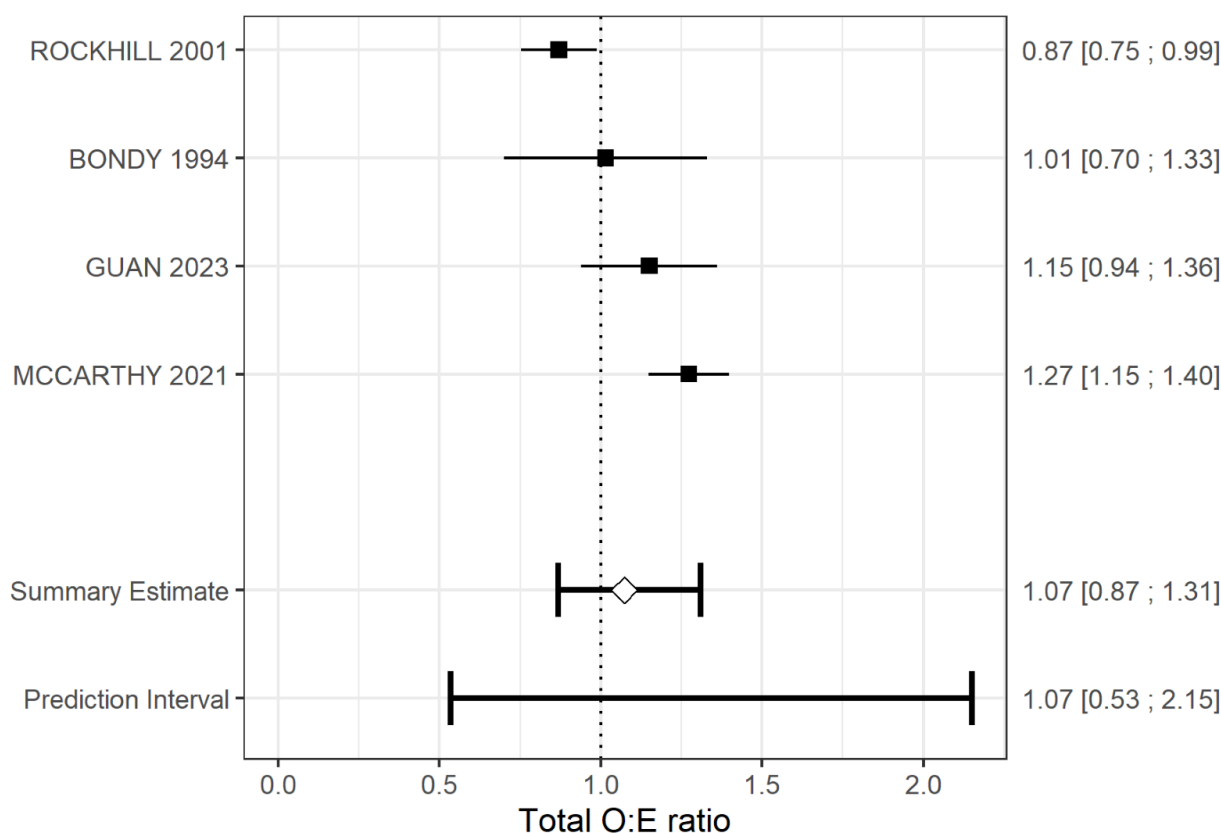
The included studies validated the various models for the prediction of breast cancer over varying timespans (5-year risk, 10-

year risk, lifetime risk, etc.). In the main meta-analyses of model performance reported here, we pooled performance measures from non-overlapping studies regardless of the prediction timespan assessed. Where feasible, we conducted additional meta-analyses of model performance measures based on prediction timespan. This was only feasible for the 5-year risk timespan for the Gail model (calibration only), and the 10-year risk timespan for the Tyrer-Cuzick model (discrimination and calibration) and the BOADICEA model (calibration only).

Gail (BCRAT) model 5-year risk prediction

We identified four external validation studies reporting Gail model calibration (O/E ratio) with regard to 5-year breast cancer risk prediction (Bondy 1994; Guan 2023; McCarthy 2021; Rockhill 2001). The four studies included a total of 41,970 individuals and 752 breast cancers. We extrapolated the total participant numbers for McCarthy 2021 ($n = 24,511$) as it did not explicitly provide this information in the article. The pooled O/E ratio of the four studies was 1.07 (95% CI 0.87 to 1.31), with a moderately high degree of heterogeneity (between-study standard deviation, $\text{Tau}^2 = 0.1486$; $I^2 = 75.75\%$). The 95% PI was 0.53 to 2.15 ([Figure 13](#)).

Figure 13. Meta-analysis of the O/E ratio for the Gail model for validation studies of 5-year breast cancer risk

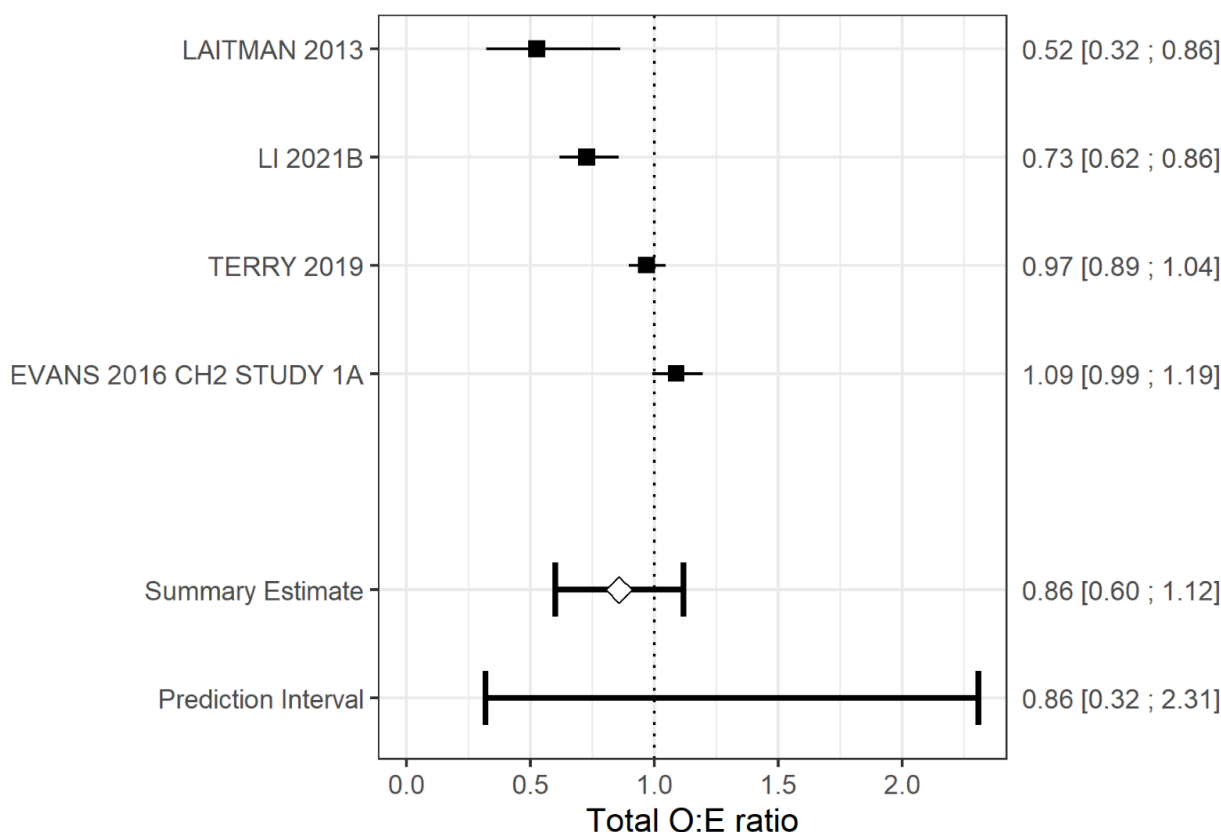


Tyrer-Cuzick model 10-year risk prediction

We identified eight external validation studies reporting Tyrer-Cuzick model calibration (O/E ratio) with regard to 10-year breast cancer risk prediction. We excluded four studies whose populations overlapped with those of other studies (Gong 2014; MacInnis 2020; Quante 2012; Quante 2015), leaving four studies for meta-analysis (Evans 2016 Chapter 2 (study 1a); Laitman 2013; Li 2021b; Terry 2019).

The four studies included a total of 26,215 individuals and 1166 breast cancers observed during the overall observation time. The pooled O/E ratio of the four studies was 0.86 (95% CI 0.60 to 1.12), with a high degree of heterogeneity (between-study standard deviation, $\text{Tau}^2 = 0.0346$; $I^2 = 91.14\%$). The 95% PI was 0.32 to 2.31 (Figure 14).

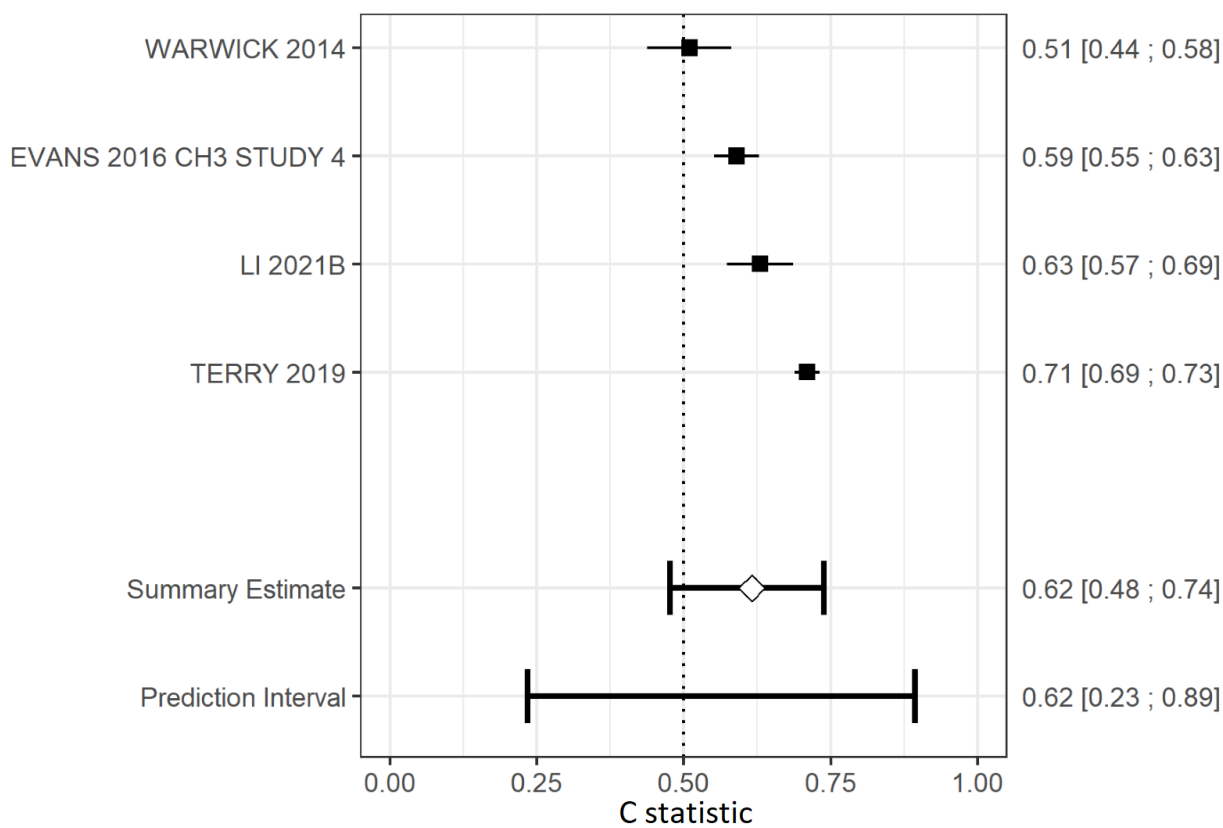
Figure 14. Meta-analysis of the O/E ratio for the Tyrer-Cuzick (IBIS) model for validation studies of 10-year breast cancer risk 95% confidence intervals (CIs) may differ slightly from those reported in the original studies because the R program used for meta-analysis recomputed them using the method described by Debray and colleagues [24], based on the O, E, and N values reported in each study. The original studies used different methods to calculate 95% CIs, such as assuming Poisson distributions or bootstrap approaches.



We identified seven external validation studies reporting Tyrer-Cuzick model discrimination (AUC or C statistic) with regard to 10-year breast cancer risk prediction. We excluded three studies whose populations overlapped with those of other studies (MacInnis 2020; Quante 2012; Quante 2015), leaving four studies for meta-analysis (Evans 2016 Chapter 3 (study 4); Li 2021b; Terry 2019; Warwick 2014).

The four studies included a total of 18,104 individuals and 1068 breast cancers observed during the overall observation time. The pooled C statistic of the four studies was 0.62 (95% CI 0.48 to 0.74), with a high degree of heterogeneity (between-study standard deviation, $\text{Tau}^2 = 0.0045$; $I^2 = 91.22\%$). The 95% PI was 0.23 to 0.89 (Figure 15).

Figure 15. Meta-analysis of the C statistic for the Tyrer-Cuzick (IBIS) model for validation studies of 10-year breast cancer risk

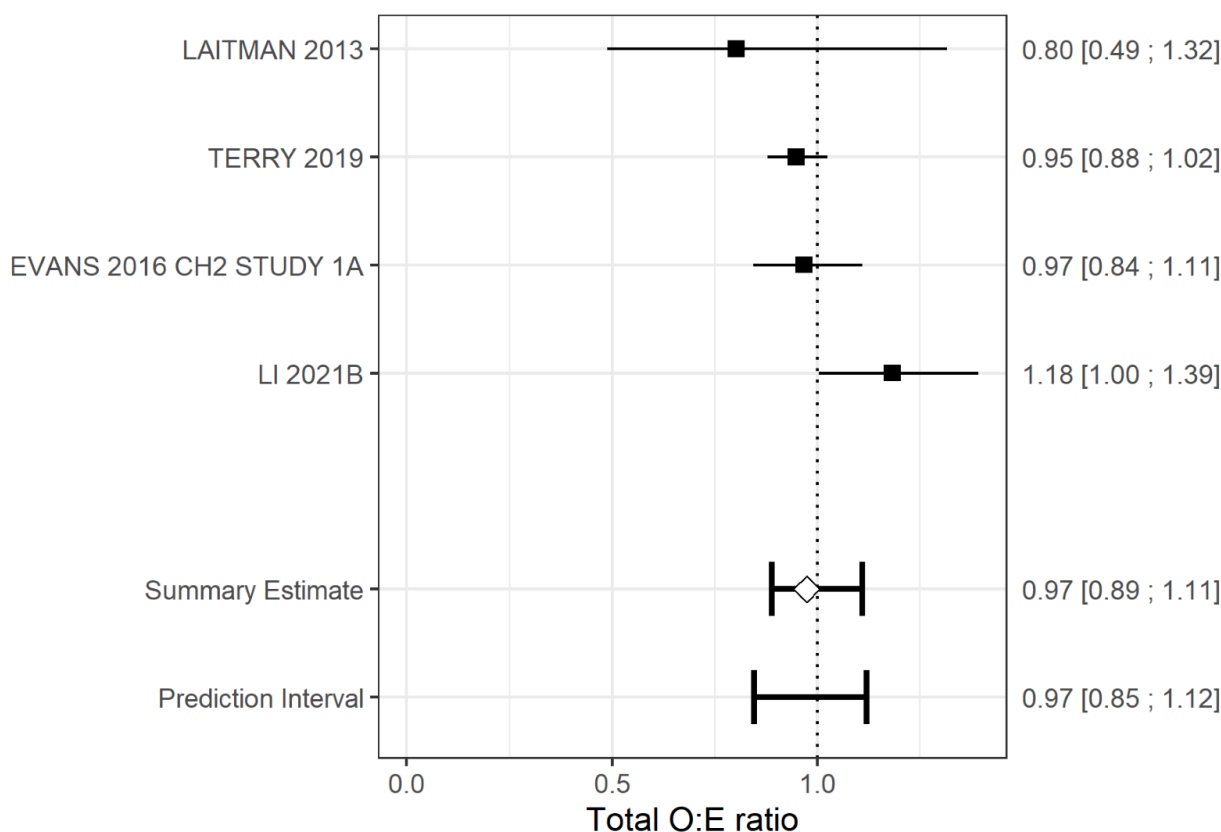


BOADICEA model 10-year risk prediction

We identified seven external validation studies reporting BOADICEA model calibration (O/E ratio) with regard to 10-year breast cancer risk prediction. We excluded three studies whose populations overlapped with those of other studies (MacInnis 2013; MacInnis 2020; Quante 2015), leaving four studies for meta-analysis (Evans 2016 Chapter 2 (study 1a); Laitman 2013; Li 2021b; Terry 2019).

The four studies included a total of 22,956 individuals and 947 breast cancers observed during the overall observation time. The pooled O/E ratio of the four studies was 0.97 (95% CI 0.89 to 1.11), with low heterogeneity (between-study standard deviation, $\tau^2 = 0.00$; $I^2 = 0.08\%$). The 95% PI was 0.85 to 1.12 (Figure 16).

Figure 16. Meta-analysis of the O/E ratio for the BOADICEA model for validation studies of 10-year breast cancer risk 95% confidence intervals (CIs) may differ slightly from those reported in the original studies because the R program used for meta-analysis recomputed them using the method described by Debray and colleagues [24], based on the O, E, and N values reported in each study. The original studies used different methods to calculate 95% CIs, such as assuming Poisson distributions or bootstrap approaches.



DISCUSSION

Summary of main results

We identified and evaluated breast cancer risk prediction models developed or validated specifically in women with a family history of breast cancer, and summarised their performance in this elevated-risk population. In total, we identified 12 models with at least one external validation study in this population. Of these, four models – Gail, Tyrer-Cuzick, BOADICEA, and BRCAPRO – had sufficient external validation data to enable meta-analysis of their calibration and discrimination. We also identified 15 models or model updates without external validation studies in the population of interest.

Summary of models with meta-analysed performance measures

The studies on which these meta-analyses were based were heterogenous in relation to time horizons of prediction and case-mix. Some studies reported model performance in subgroups consisting exclusively of women with at least one first-degree relative with breast cancer, while others assessed performance in populations with varying degrees of enrichment for family history of breast cancer. This likely contributed to the different performance of these models in different studies.

Model 1. Gail/Breast Cancer Risk Assessment Tool (BCRAT)

Our results indicate that the Gail model is well calibrated in women with a family history of breast cancer, suggesting that it provides risk estimates that are broadly consistent with observed outcomes. However, its ability to distinguish between individuals who did and did not develop breast cancer in the target population was only moderate. It is noteworthy that this model was intended for women in the general population who have regular screening mammography, rather than an elevated-risk population [61].

Model 2. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS)

For the Tyrer-Cuzick model, calibration differed across model versions. When we combined results from multiple versions (versions 6, 7, and 8), the model overpredicted breast cancer risk in women with a family history of breast cancer. Version 8 showed somewhat improved calibration compared to results for versions 6, 7, and 8 combined. The discriminatory performance of the Tyrer-Cuzick model was consistently moderate in the target population, with only marginal improvements observed with model version 8 compared to results when versions 6, 7, and 8 were combined.

Model 3. BOADICEA (Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm)

The BOADICEA model showed good agreement between predicted and observed risks among women with a family history of breast cancer, indicating robust calibration in this population. The discriminatory performance of the model was moderate in the target population and comparable to that of Tyrer-Cuzick version 8 and BRCAPRO.

Model 4. BRCAPRO

Our results indicate that the BRCAPRO model underpredicts breast cancer risk in women with a family history of breast cancer. The model showed moderate discriminatory accuracy in this target population, similar to that of Tyrer-Cuzick version 8 and BOADICEA.

Results from studies comparing multiple models

In an attempt to address the aforementioned issues relating to heterogeneous time horizons of prediction and case-mix, we examined the relative performance of models in studies that compared multiple models in the same dataset. Synthesis of evidence from these comparative studies did not demonstrate clear superiority of any model in the population of interest. However, the BOADICEA and Tyrer-Cuzick models most consistently achieved the best balance between calibration and discrimination. These head-to-head comparisons are particularly informative because they reduce confounding from differences in study populations.

Models with three or fewer external validation studies in the target population

Beyond the four principal models described above, eight additional models had limited external validation (one to three studies), and several other models had not yet been externally validated in this target population when we submitted this review. These models should therefore be interpreted cautiously and are not yet suitable for clinical use in women with a family history of breast cancer.

Overall completeness of the data

Reporting of included studies varied. In general, validation studies were reported more comprehensively than development studies. Several articles validated a model in cohorts that overlapped. For example, the studies used participants from the same population or registry recruited over a similar timeframe, often with different articles reporting different numbers of participants or different follow-up times. In the case of overlapping cohorts, we selected studies with the largest sample size for evidence synthesis. We have summarised the missing information per model for the development study (where applicable) and external validation studies in non-overlapping cohorts in [Supplementary material 8](#).

For the Gail model, we identified 13 separate model validations in non-overlapping cohorts. Eight of these reported model calibration (six with 95% CI) and 10 reported model discrimination (seven with 95% CI).

For the Tyrer-Cuzick model, we identified one model development study and 15 separate model validations in non-overlapping cohorts. Nine studies reported model calibration with 95% CI and 11 reported model discrimination (nine with 95% CI). Six of these studies reported both model calibration and discrimination in the same article/dataset. For one cohort,

calibration and discrimination were reported in separate study articles (overlapping cohorts but with different sample sizes). Four studies reported model discrimination only.

For the BOADICEA model, we identified one model development study and six separate model validations in non-overlapping cohorts. Five of these studies reported model calibration with 95% CI and four reported model discrimination (three with 95% CI).

For the BRCAPRO model, we identified one model development study and six separate model validations in non-overlapping cohorts. Five of these studies reported model calibration with 95% CI and five reported model discrimination with 95% CI.

Of the other eight models with 14 validation studies that were not meta-analysed, we identified 12 separate model validations in non-overlapping cohorts. Of these, 12 reported model calibration with 95% CI and eight reported model discrimination with 95% CI.

Several studies did not adequately report follow-up time. For example, several studies reported maximum length of follow-up rather than mean/median follow-up time. In other studies, follow-up time was reported for the full study cohort but not separately for the family history subgroup.

Several studies did not clearly report the amount and handling of missing data. In studies where women with a family history of breast cancer were a subgroup of the overall cohort, missing data were typically not reported separately for the family history subgroup. In addition, several of the included models can still be run without all the included predictors; that is, some predictors are optional (for example, polygenic risk scores used in Tyrer-Cuzick version 8 were rarely included in model validations) or missing predictor data can be accommodated in the model's algorithm. This is an important feature of these models as missing data are common in clinical practice. However, studies rarely reported details of the use of the model's missing data functions, which makes comparison of model performance across different studies challenging.

While most validation studies reported at least one model performance measure (calibration or discrimination), several studies did not report both together in the same article/dataset. This hampered our interpretation of overall model performance because both performance measures should be assessed to fully gauge the predictive performance of a model [47].

Some studies excluded women with missing predictor data at enrolment or less commonly excluded women with missing data from the analysis. These studies seldom provided a comparison of included and excluded individuals. It would be preferable for studies to impute data rather than exclude individuals with missing data.

Certainty of the evidence

When we submitted this review, no formal guidance was available for grading the certainty of results from meta-analysis of prognostic models. For this reason, we refrained from applying GRADE to rate the certainty of the evidence for this review.

Limitations of the evidence included in the review

Risk of bias (PROBAST) ratings were high or unclear for most of the included model validation studies. This was the case for all included models. For example, we rated 95.7%, 100%, 88.2%, and 71.4% of the Gail (BCRAT), Tyrer-Cuzick (IBIS), BOADICEA, and BRCAPRO model validation studies, respectively, as having an overall high or unclear risk of bias. Of the other models validated, we rated 92.9% of validation studies as having a high risk of bias, with the remaining one study rated as having an unclear risk of bias. While this does not confirm that bias was definitely present in these studies, review findings should be interpreted with this potential for bias in mind.

Similarly, PROBAST ratings for included model development studies were either unclear risk due to lack of reporting or high risk, mainly due to concerns over handling of missing data and lack of reporting on model performance measures in development studies. Again, while this does not confirm that bias was definitely present in these studies, review findings should be interpreted with this potential for bias in mind.

Another limitation that we frequently encountered in included studies was inadequate reporting of validated model versions. It was not always obvious which model versions the validation studies had used. Some publications contained web links to the model or models used that had expired. In other cases, when contacted, authors reported using the model version current at the time of the study. For models included in the meta-analyses, we pooled performance measures for all versions of each model, given the high similarity between versions. Where feasible, we also conducted sensitivity analyses of individual model versions, although this was only possible for Tyrer-Cuzick version 8.

Limitations of the review process

To prevent bias in the review process, we performed all relevant steps in duplicate and resolved conflicts in group discussions. As this is a relatively new review type, our methods evolved during the review process. Where applicable, we used the newest methods even where these deviated from the protocol. This may have introduced bias. Below, we outline key methodological decisions made in the design and conduct of the review and how we addressed the analytical challenges we encountered.

Searching for studies. We used a sensitive search strategy and also searched the reference lists of included articles. However, it is possible that other breast cancer risk prediction models may exist that may have been missed by the search, particularly if these models were unpublished.

Pre-specification of prognostic factors. Only models that included family history of breast cancer as a predictor(s) were eligible for inclusion, as these models are likely to be most applicable to the breast family risk setting. It is noteworthy that the vast majority of breast cancer risk prediction models include family history of breast cancer as a predictor. We did not apply additional limits in relation to which other predictors were included in the models.

Screening and study selection. Our target population was women with a family history of breast cancer. We selected studies that developed or validated a breast cancer risk prediction model in populations or subgroups of women composed solely of those with a family history of breast cancer. In these studies, the number

of breast cancer-affected relatives varied. In addition, several studies validated models in populations enriched for or purposely oversampled for women with a family history of breast cancer. We decided to also include studies that developed or validated a model or models in such 'enriched' populations. In these populations, the percentages of women with at least one affected first-degree relative varied between studies. This may have introduced bias.

We only included the development article for a model if the model was developed using datasets enriched for women with a family history of breast cancer rather than data from general populations of women (such as for the Gail model).

It was not always obvious whether studies constituted a 'true' model validation. For example, some studies used a model to determine the percentage of women in a population in various risk categories, rather than to validate them as breast cancer risk prediction models. These studies were not designed as validation studies and typically had inadequate follow-up for assessing predictive performance. Although some reported metrics such as sensitivity or specificity that could be interpreted as measures of model performance, model validation was not the focus. We excluded such studies because they do not provide reliable evidence on the real-world predictive performance of the models. This approach may have introduced bias.

Analysis. We focussed on model performance in the overall cohort of women with (or enriched for) a family breast cancer history. Some articles also reported model performance in quantiles of model-predicted risk. We considered attempting a separate subgroup analysis of these quantiles. However, following advice from the Cochrane Prognosis Methods Group, we decided that this would not be beneficial as a characteristic of the C statistic is that it will always decrease when a population is narrowed. Similarly, for this reason, we refrained from conducting subgroup analyses of model performance according to the number of breast cancer-affected relatives.

Multiple versions of the main breast cancer risk prediction models exist. Where possible, we meta-analysed performance measures for distinct model versions separately. However, this was often not feasible due to the limited number of validation studies for specific versions or uncertainty about which version was evaluated in a given study, even after attempts to contact the authors.

Models were validated over different prediction horizons in different studies. For the main meta-analyses, we combined studies regardless of the prediction timespan assessed. This may have introduced bias. However, where feasible, we conducted additional meta-analyses of model performance measures based on the prediction timespan. This was only feasible for the 10-year risk for the Tyrer-Cuzick model (discrimination and calibration) and the BOADICEA model (calibration only) and for the 5-year risk for the Gail model (calibration only). Results were similar to the main analyses.

For some studies, we estimated 95% confidence intervals for performance measures due to incomplete reporting. Although this may have introduced bias, sensitivity analyses suggested that these estimations did not materially influence the results.

We did not assess the likelihood of publication bias (preferential publication of models that performed well over those that

performed poorly), as there is currently no established method for evaluating this in reviews of prediction model studies.

Agreements and disagreements with other systematic reviews

There are several systematic reviews of breast cancer risk prediction models that assess model performance in 'general' populations [16, 25, 26, 27, 28]. However, to our knowledge, no other systematic review has synthesised evidence of breast cancer risk prediction model performance specifically in women with a family history of the disease.

Meads and colleagues synthesised evidence on breast cancer risk prediction models that contain modifiable factors [16]. This systematic review notably excluded studies in high-risk groups of women such as those who had close family relatives with breast cancer. The authors reported that none of the models evaluated demonstrated consistently outstanding ability to accurately discriminate between those who did and did not develop breast cancer.

Wang and colleagues evaluated the performance of different versions of the Gail model by means of systematic review and meta-analysis with trial sequential analysis (TSA) [25]. Various versions of the Gail model were well calibrated in American and European women while overpredicting risk in Asian women. Most versions of the Gail model showed poor or modest discriminatory accuracy in different populations except for the Gail 1 model in Asian women.

Louro and colleagues narratively synthesised the features and accuracy of breast cancer risk prediction models [26]. They concluded that it was a challenge to recommend any of the included models since they need further improvement in their quality and discriminatory capacity.

Zheng and colleagues narratively synthesised multivariable breast cancer risk prediction models in the general population [27]. They concluded that no model can be recommended for use in routine screening programmes.

Anothaisintawee and colleagues conducted a systematic review of breast cancer risk prediction models [28]. Based on a narrative synthesis, they reported that calibration was fair to good but discriminatory accuracy was poor to fair in general population samples of women.

Mertens and colleagues conducted a systematic review of breast cancer risk prediction models developed or validated in European females, from the general population, or those attending a preventive breast cancer screening with a focus on identifying the relative contribution of lifestyle predictors to the prediction of incident breast cancer [121]. The authors reported that all models leaned towards strong calibration, but low discriminatory accuracy in these populations.

Velentzis and colleagues conducted a systematic review assessing the accuracy of externally validated questionnaire-based risk assessment tools in women aged 40 and older undergoing population mammographic screening [122]. They reported that no model was consistently well-calibrated across multiple studies and breast density or polygenic risk scores did not improve calibration.

Overall, consistent with findings from our review in women with elevated familial risk of breast cancer, these systematic reviews in general populations of women also reported that model discriminatory accuracy was, at best, modest. When assessed, the risk of bias in included studies was frequently high.

AUTHORS' CONCLUSIONS

Implications for practice

This review attempted to determine which breast cancer risk prediction model or models are reliable and appropriate for use in the breast cancer family risk setting. These models are used in clinical practice to estimate an individual's risk of developing breast cancer, which helps clinicians and patients to reach decisions about risk management. Therefore, accurate risk prediction is extremely important.

Our meta-analysis showed that the Gail (Breast Cancer Risk Assessment Tool (BCRAT)) and BOADICEA (Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm) models are well calibrated in the population of women enriched for a family history of breast cancer. In contrast, the Tyrer-Cuzick (International Breast Cancer Intervention Study (IBIS)) model overpredicts risk in this population, and the BRCAPRO model underpredicts risk.

In terms of discriminatory accuracy in the target population, no model was clearly superior. Tyrer-Cuzick version 8, BOADICEA, and BRCAPRO showed similarly modest discrimination in our meta-analyses, which was slightly better than that of the Gail model in the target population. The modest discriminatory accuracy of these models in this population limits their utility for individual decision-making.

Similarly, collating evidence from direct model comparison studies indicated that in all but one study, the Tyrer-Cuzick and BOADICEA models showed (marginally) superior discriminatory accuracy compared to Gail or BRCAPRO in the population of women enriched for a family history of breast cancer.

Considering both calibration and discrimination together, our findings suggest that the BOADICEA model is well calibrated in this population and shows similar (though modest) discriminatory accuracy to Tyrer-Cuzick (version 8) and BRCAPRO. This suggests that BOADICEA may be useful for patient management in the breast cancer family risk setting. However, these conclusions should be interpreted cautiously, for several reasons: the risk of bias (PROBAST) ratings were high or unclear for most of the included model validation studies; the number of validation studies included in the meta-analyses was small (10 or fewer for each model); and the studies on which these meta-analyses were based were heterogeneous in relation to time horizons of prediction and case-mix.

Implications for research

Much room for improvement remains in terms of the discriminatory ability of existing breast cancer risk prediction models in women with a family history of breast cancer. This is a continually evolving field and future research might focus on updating these existing models with additional predictors. An updated version of the BOADICEA model has been developed (version 6, CanRisk) that includes mammographic density and other lifestyle and hormonal

risk factors for breast cancer, in addition to family history and genetic test results (including polygenic risk scores) [103]. At the time of our search, the updated version had not yet been externally validated specifically in women with a family history of breast cancer (our target population), though it has recently been validated in women with a known pathogenic *BRCA* variant (Yang 2024 [123]). External validations of this model version in the target population will be captured in future updates of this review.

In addition, certain predictors in existing models, such as polygenic risk scores (based on single-nucleotide polymorphisms (SNPs)) in the Tyrer-Cuzick (version 8) and BOADICEA (version 6) models, are considered optional, as these SNPs are rarely tested for in routine clinical practice. This is likely to change in the future as clinical decision-making increasingly incorporates genetic and genomic data. Additional research studies examining the incremental value of these polygenic risk scores in risk prediction specifically in the breast cancer family risk setting are warranted.

Furthermore, combination or 'hybrid' models have been developed; for example, the BRCAPRO-BCRAT model, which combines the predominantly 'genetic' predictors of BRCAPRO with the predominantly non-genetic predictors of BCRAT (Gail). It has been suggested that integrating information from different models may improve the accuracy of predictions (Guan 2023). We identified three external validations of the BRCAPRO-BCRAT model in the target population. However, this was an insufficient number to include in a meta-analysis. Additional validations of BRCAPRO-BCRAT in the target population will provide clarity on the usefulness of this model in the family risk setting.

All the models included in our quantitative analysis were developed using traditional methods. Prognostic model development using newer techniques, such as machine learning approaches, is becoming increasingly common. We identified several such models through our search. However, none of these had yet been externally validated in the target population. External validations of these models will provide valuable information on their performance and potential clinical utility in the breast cancer family risk setting.

Comparing accuracy across different models is inherently challenging. Our meta-analyses of performance measures for individual models gives an indication of the accuracy of each model across the studies in which it was validated. However, when the goal is to compare the performance of different risk prediction models in an effort to determine which model performs best in the target population, it is ideal to compare the performance of the various models in the same participants. To address this, we qualitatively synthesised evidence on model performance from studies that compared multiple models in the same dataset. These multi-model studies provide for a more homogeneous comparison across models using similar cohorts. However, studies that compared the performance of multiple models on the same dataset had an unclear or high risk of bias for at least two of the models included in the comparison. Furthermore, study populations in each of these comparison studies varied with regard to the extent of enrichment for family history of breast cancer. Additional direct model comparison studies, particularly in women attending breast family history clinics, will provide valuable information on which model (or models) is most accurate in this setting.

Finally, reporting of prognostic model studies is suboptimal. We recommend that authors adhere to reporting guidelines, such

as the TRIPOD statement [22, 124], to facilitate critical appraisal of prediction model studies, model evaluation, and evidence synthesis in this field.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD013185.pub2](https://doi.org/10.1002/14651858.CD013185.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Data package

Supplementary material 6 Data Extraction spreadsheet

Supplementary material 7 Characteristics of studies awaiting classification (models without external validation studies in the target population or conference abstracts that appear to meet eligibility criteria but for which no full text exists or narrative reports of model updates that did not report performance (re-validation) of the updated model)

Supplementary material 8 Reporting deficiencies in included studies of models with at least one external validation study in the target population (development + validation studies, listed per model. Where overlapping populations were used for model validation in separate articles the report with the largest sample size was used for evidence synthesis)

Supplementary material 9 Changes to the protocol

Supplementary material 10 Risk of bias and applicability assessment of models with 1 to 3 external validation studies

Supplementary material 11 Summary of characteristics of included studies for models without meta-analysis (1-3 external validations)

Supplementary material 12 Summary of characteristics and results of additional predictor(s) incremental value studies

Supplementary material 13 Detailed risk of bias and applicability assessment of models with more than three external validation studies

Supplementary material 14 Direct model comparison using the same dataset(s)

ADDITIONAL INFORMATION

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Contributions of authors

Conceiving the review: SMG, YH, EC
 Writing the protocol: SMG, KB, YH, EC
 Developing and running the search strategy: DM
 Selecting studies: SMG, YH
 Disagreement arbitration: KB, EC, JAAGD
 Extracting data from studies: SMG, KB
 Entering data into Review Manager 5: SMG
 Carrying out the analysis: KB, SMG
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Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (see [Supplementary material 5](#)): full search strategies for each database; full citations of each unique report for all studies (included, ongoing, or awaiting classification, or excluded at the full-text screening) in the final review; study data, including study information; consensus risk of bias assessments. Appropriate permissions have been obtained for such use. Template data extraction forms from Microsoft Word are available from the authors on reasonable request.

REFERENCES

1. Vandenbroucke JP. A shortcut method for calculating the 95 per cent confidence interval of the standardized mortality ratio. *American Journal of Epidemiology* 1982;**115**(2):303-4. [DOI: [10.1093/oxfordjournals.aje.a113306](https://doi.org/10.1093/oxfordjournals.aje.a113306)]
2. Bray F, Laversanne M, Sung H, Ferlay J, Lam F, Colombet M, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: a Cancer Journal for Clinicians* 2024;**74**(3):229-63. [DOI: [10.3322/caac.21834](https://doi.org/10.3322/caac.21834)]
3. Antoniou AC, Foulkes WD, Tischkowitz M. Breast-cancer risk in families with mutations in PALB2. *New England Journal of Medicine* 2014;**371**(17):1651-2.
4. Easton DF, Pharoah PD, Antoniou AC, Tischkowitz M, Tavtigian SV, Nathanson KL, et al. Gene-panel sequencing and the prediction of breast-cancer risk. *New England Journal of Medicine* 2015;**372**(23):2243-57.
5. Mavaddat N, Peock S, Frost D, Ellis S, Platte R, Fineberg E, et al. Cancer risks for BRCA1 and BRCA2 mutation carriers: results from prospective analysis of EMBRACE. *Journal of the National Cancer Institute* 2013;**105**(11):812-22.
6. Laloo F, Evans DG. Familial breast cancer. *Clinical Genetics* 2013;**82**(2):105-14.
7. Michailidou K, Hall P, Gonzalez-Neira A, Ghoussaini M, Dennis J, Milne RL, et al. Large-scale genotyping identifies 41 new loci associated with breast cancer risk. *Nature Genetics* 2013;**45**(4):353-61, 361e1-2.
8. Economopoulou P, Dimitriadis G, Psyrri A. Beyond BRCA: new hereditary breast cancer susceptibility genes. *Cancer Treatment Reviews* 2015;**41**(1):1-8.
9. Evans DG, Cuzick J, Howell A. Cancer genetics clinics. *European Journal of Cancer* 1996;**32A**(3):391-2.
10. Gail MH, Costantino JP, Bryant J, Croyle R, Freedman L, Helzlsouer K, et al. Weighing the risks and benefits of tamoxifen treatment for preventing breast cancer. *Journal of the National Cancer Institute* 1999;**91**(21):1829-46.
11. Powles TJ. Breast cancer prevention. *Oncologist* 2002;**7**(1):60-4.
12. National Institute for Health and Care Excellence (NICE). Familial breast cancer: classification, care and managing breast cancer and related risks in people with a family history of breast cancer. Clinical guideline [CG164]. NICE 2013.
13. Gadzicki D, Evans DG, Harris H, Julian-Reynier C, Nippert I, Schmidtko J, et al. Genetic testing for familial/hereditary breast cancer – comparison of guidelines and recommendations from the UK, France, the Netherlands and Germany. *Journal of Community Genetics* 2011;**2**(2):53-69.
14. Amir E, Freedman OC, Seruga B, Evans DG. Assessing women at high risk of breast cancer: a review of risk assessment models. *Journal of the National Cancer Institute* 2010;**102**(10):680-91.
15. Ahmed I, Debray TP, Moons KG, Riley RD. Developing and validating risk prediction models in an individual participant data meta-analysis. *BMC Medical Research Methodology* 2014;**14**:3.
16. Meads C, Ahmed I, Riley RD. A systematic review of breast cancer incidence risk prediction models with meta-analysis of their performance. *Breast Cancer Research and Treatment* 2012;**132**(2):365-77.
17. Pavlou M, Ambler G, Seaman SR, Guttman O, Elliott P, King M, et al. How to develop a more accurate risk prediction model when there are few events. *BMJ* 2015;**351**:h3868.
18. Domchek SM, Eisen A, Calzone K, Stopfer J, Blackwood A, Weber BL. Application of breast cancer risk prediction models in clinical practice. *Journal of Clinical Oncology* 2003;**21**(4):593-601.
19. Royston P, Moons KG, Altman DG, Vergouwe Y. Prognosis and prognostic research: developing a prognostic model. *BMJ* 2009;**338**:b604.
20. Ming C, Viassolo V, Probst-Hensch N, Chappuis PO, Dinov ID, Katapodi MC. Machine learning techniques for personalized breast cancer risk prediction: comparison with the BCRAT and BOADICEA models. *Breast Cancer Research* 2019;**21**(1):[no pagination].
21. Yala A, Lehman C, Schuster T, Portnoi T, Barzilay R. A deep learning mammography-based model for improved breast cancer risk prediction. *Radiology* 2019;**292**(1):60-6.
22. Collins GS, Reitsma JB, Altman DG, Moons KG. Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD). *Annals of Internal Medicine* 2015;**162**(10):735-6.
23. Van Calster B, Van Hoorde K, Vergouwe Y, Bobdiwala S, Condous G, Kirk E, et al. Validation and updating of risk models based on multinomial logistic regression. *Diagnostic and Prognostic Research* 2017;**1**(2):1-14.
24. Debray TP, Damen JA, Riley RD, Snell K, Reitsma JB, Hooft L, et al. A framework for meta-analysis of prediction model studies with binary and time-to-event outcomes. *Statistical Methods in Medical Research* 2019;**28**(9):2768-86. [DOI: [10.1177/0962280218785504](https://doi.org/10.1177/0962280218785504)]
25. Wang X, Huang Y, Li L, Dai H, Song F, Chen K. Assessment of performance of the Gail model for predicting breast cancer risk: a systematic review and meta-analysis with trial sequential analysis. *Breast Cancer Research* 2018;**20**(1):[no pagination]. [DOI: [10.1186/s13058-018-0947-5](https://doi.org/10.1186/s13058-018-0947-5)]
26. Louro J, Posso M, Hilton Boon M, Roman M, Domingo L, Castells X, et al. A systematic review and quality assessment of individualised breast cancer risk prediction models. *British Journal of Cancer* 2019;**121**(1):76-85.

27. Zheng Y, Li J, Wu Z, Li H, Cao M, Li N, et al. Risk prediction models for breast cancer: a systematic review. *BMJ Open* 2022;**12**(7):e055398.
28. Anothaisintawee T, Teerawattananon Y, Wiratkapun C, Kasamesup V, Thakkinstian A. Risk prediction models of breast cancer: a systematic review of model performances. *Breast Cancer Research and Treatment* 2012;**133**(1):1-10.
29. McGarrigle S, Hanhauser Y, Mockler D, Gallagher D, Kennedy M, Bennett K, et al. Risk prediction models for familial breast cancer. *Cochrane Database of Systematic Reviews* 2018, Issue 12. Art. No: CD013185. [DOI: [10.1002/14651858.CD013185](https://doi.org/10.1002/14651858.CD013185)]
30. Kreuzberger N, Damen JA, Trivella M, Estcourt LJ, Aldin A, Umlauff L, et al. Prognostic models for newly-diagnosed chronic lymphocytic leukaemia in adults: a systematic review and meta-analysis. *Cochrane Database of Systematic Reviews* 2020;**7**(7):CD012022.
31. Moons KG, De Groot JA, Bouwmeester W, Vergouwe Y, Mallett S, Altman DG, et al. Critical appraisal and data extraction for systematic reviews of prediction modelling studies: the CHARMS checklist. *PLoS Medicine* 2014;**11**(10):e1001744.
32. Debray TP, Damen JA, Snell KI, Ensor J, Hooft L, Reitsma JB, et al. A guide to systematic review and meta-analysis of prediction model performance. *BMJ* 2017;**356**:i6460.
33. Geersing GJ, Bouwmeester W, Zuithoff P, Spijker R, Leeflang M, Moons KG. Search filters for finding prognostic and diagnostic prediction studies in MEDLINE to enhance systematic reviews. *PLoS ONE* 2012;**7**(2):e32844.
34. Pace NL, Carlisle J, Eberhart LH, Kranke P, Trivella M, Lee A, et al. Prediction models for the risk of postoperative nausea and vomiting. *Cochrane Database of Systematic Reviews* 2014, Issue 9. Art. No: CD011318. [DOI: [10.1002/14651858.CD011318](https://doi.org/10.1002/14651858.CD011318)]
35. Palmer JR, Zirpoli G, Bertrand KA, Battaglia T, Bernstein L, Ambrosone CB, et al. A validated risk prediction model for breast cancer in US Black women. *Journal of Clinical Oncology* 2021;**39**(34):3866-77.
36. Covidence. Version accessed prior to 01 December 2025. Melbourne, Australia: Veritas Health Innovation, 2025. Available at [covidence.org](https://www.covidence.org).
37. Moher D, Liberati A, Tetzlaff J, Altman DG; PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Journal of Clinical Epidemiology* 2009;**62**(10):1006-12.
38. Bouwmeester W, Zuithoff NP, Mallett S, Geerlings MI, Vergouwe Y, Steyerberg EW, et al. Reporting and methods in clinical prediction research: a systematic review. *PLoS Medicine* 2012;**9**(5):1-12.
39. Collins GS, Mallett S, Omar O, Yu LM. Developing risk prediction models for type 2 diabetes: a systematic review of methodology and reporting. *BMC Medicine* 2011;**9**:103.
40. Damen JA, Hooft L, Schuit E, Debray TP, Collins GS, Tzoulaki I, et al. Prediction models for cardiovascular disease risk in the general population: systematic review. *BMJ* 2016;**353**:i2416.
41. Mallett S, Royston P, Dutton S, Waters R, Altman DG. Reporting methods in studies developing prognostic models in cancer: a review. *BMC Medicine* 2010;**8**:20.
42. Mallett S, Royston P, Waters R, Dutton S, Altman DG. Reporting performance of prognostic models in cancer: a review. *BMC Medicine* 2010;**8**:21.
43. Smit HA, Pinart M, Antó JM, Keil T, Bousquet J, Carlsen KH, et al. Childhood asthma prediction models: a systematic review. *Lancet. Respiratory Medicine* 2015;**3**(12):973-84.
44. Van Dieren S, Beulens JW, Kengne AP, Peelen LM, Rutten GE, Woodward M, et al. Prediction models for the risk of cardiovascular disease in patients with type 2 diabetes: a systematic review. *Heart (British Cardiac Society)* 2012;**98**(5):360-9.
45. Van Doorn S, Debray TP, Kaasenbrood F, Hoes AW, Rutten FH, Moons KG, et al. Predictive performance of the CHA2DS2-VASc rule in atrial fibrillation: a systematic review and meta-analysis. *Journal of Thrombosis and Haemostasis* 2017;**15**(6):1065-77.
46. Wolff R, Moons K, Riley R, Whiting P, Westwood M, Collins G, et al; for the PROBAST Group. PROBAST: a tool to assess the risk of bias and applicability of prediction model studies. *Annals of Internal Medicine* 2019;**170**:51-8. [DOI: [10.7326/M18-1376](https://doi.org/10.7326/M18-1376)]
47. Moons KG M, Wolff RF, Riley RD, Whiting PF, Westwood M, Collins GS, et al. PROBAST: a tool to assess risk of bias and applicability of prediction model studies: explanation and elaboration. *Annals of Internal Medicine* 2019;**170**(1):W1-W33.
48. Siregar S, Groenwold RH, De Heer F, Bots ML, Van der Graaf Y, Van Herwerden LA. Performance of the original EuroSCORE. *European Journal of Cardio-thoracic Surgery* 2012;**41**(4):746-54.
49. Hosmer DW, Lemeshow S. Confidence interval estimates of an index of quality performance based on logistic regression models. *Statistics in Medicine* 1995;**14**(19):2161-72.
50. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine* 2002;**21**(11):1539-58. [DOI: [10.1002/sim.1186](https://doi.org/10.1002/sim.1186)]
51. Moons KG, Altman DG, Reitsma JB, Ioannidis JP, Macaskill P, Steyerberg EW, et al. Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): explanation and elaboration. *Annals of Internal Medicine* 2015;**162**(1):W1-73.
52. Heus P, Damen JA, Pajouheshnia R, Scholten RJ, Reitsma JB, Collins GS, et al. Poor reporting of multivariable prediction model studies: towards a targeted implementation strategy of the TRIPOD statement. *BMC Medicine* 2018;**16**(1):120.
53. Peat G, Riley RD, Croft P, Morley KI, Kyzas PA, Moons KG, et al. Improving the transparency of prognosis research: the role

of reporting, data sharing, registration, and protocols. *PLoS Medicine* 2014;**11**(7):e1001671.

54. Guan Z, Huang T, McCarthy AM, Hughes K, Semine A, Uno H, et al. Combining breast cancer risk prediction models. *Cancers* 2023;**15**(4):1090. [DOI: [10.3390/cancers15041090](https://doi.org/10.3390/cancers15041090)]

55. Guan BZ, Parmigiani G, Braun DD, Trippa L. Prediction of hereditary cancers using neural networks. *Annals of Applied Statistics* 2022;**16**(1):495-520. [DOI: [10.1214/21-aos1510](https://doi.org/10.1214/21-aos1510)]

56. R: A language and environment for statistical computing. Version 4.4.0. Vienna, Austria: R Foundation for Statistical Computing, 2025. Available at <https://www.r-project.org>.

57. Huguot A, Hayden JA, Stinson J, McGrath PJ, Chambers CT, Tougas ME, et al. Judging the quality of evidence in reviews of prognostic factor research: adapting the GRADE framework. *Systematic Reviews* 2013;**2**:71.

58. Foroutan F, Guyatt G, Trivella M, Kreuzberger N, Skoetz N, Riley RD, et al. GRADE concept paper 2: concepts for judging certainty on the calibration of prognostic models in a body of validation studies. *Journal of Clinical Epidemiology* 2022;**143**:202-11.

59. Evans DG, Astley S, Stavrinou P, Harkness E, Donnelly LS, Dawe S, et al. Improvement in risk prediction, early detection and prevention of breast cancer in the NHS Breast Screening Programme and family history clinics: a dual cohort study. Chapter 2 Project 1: improvement of risk prediction algorithms for women at high risk of breast cancer Study 1a: cohort study. *NIHR Journals Library. Programme Grants for Applied Research* 2016;**08**(08):7-24.

60. Evans DG, Astley S, Stavrinou P, Harkness E, Donnelly LS, Dawe S, et al. Improvement in risk prediction, early detection and prevention of breast cancer in the NHS Breast Screening Programme and family history clinics: a dual cohort study. Chapter 3 Project 2: assessment of predictive value of new genetic variants. Study 4: FH-Risk case-control. *NIHR Journals Library. Programme Grants for Applied Research* 2016;**08**:08.

61. Gail MH, Brinton LA, Byar DP, Corle DK, Green SB, Schairer C, et al. Projecting individualized probabilities of developing breast cancer for white females who are being examined annually. *Journal of the National Cancer Institute* 1989;**81**(24):1879-86.

62. Anderson SJ, Ahn S, Duff K. NSABP breast cancer prevention trial risk assessment program, version 2. *NSABP Biostatistical Center Technical Report* 1992;**12**:[no pagination].

63. Costantino JP, Gail MH, Pee D, Anderson S, Redmond CK, Benichou J, et al. Validation studies for models projecting the risk of invasive and total breast cancer incidence. *Journal of the National Cancer Institute* 1999;**91**(18):1541-8.

64. Antoniou AC, Easton DF. Risk prediction models for familial breast cancer. *Future Oncology (London, England)* 2006;**2**(2):257-74.

65. Amir E, Evans DG, Shenton A, Lalloo F, Moran A, Boggis C, et al. Evaluation of breast cancer risk assessment packages in the

family history evaluation and screening programme. *Journal of Medical Genetics* 2003;**40**(11):807-14.

66. Bondy ML, Lustbader ED, Halabi S, Ross E, Vogel VG. Validation of a breast cancer risk assessment model in women with a positive family history. *Journal of the National Cancer Institute* 1994;**86**(8):620-5. [DOI: [10.1093/jnci/86.8.620](https://doi.org/10.1093/jnci/86.8.620)]

67. Clendenen TV, Ge W, Koenig KL, Afanasyeva Y, Agnoli C, Brinton LA, et al. Breast cancer risk prediction in women aged 35-50 years: impact of including sex hormone concentrations in the Gail model. *Breast Cancer Research* 2019;**21**(1):42.

68. Glynn RJ, Colditz GA, Tamimi RM, Chen WY, Hankinson SE, Willett WW, et al. Comparison of questionnaire-based breast cancer prediction models in the Nurses' Health Study. *Cancer Epidemiology, Biomarkers & Prevention* 2019;**28**(7):1187-94.

69. Gong G, Quante AS, Terry MB, Whittemore AS. Assessing the goodness of fit of personal risk models. *Statistics in Medicine* 2014;**33**(18):3179-90.

70. Ho PJ, Lim EH, Ri N, Hartman M, Wong FY, Li J. Will absolute risk estimation for time to next screen work for an Asian mammography screening population? *Cancers* 2023;**15**(9):2559. [DOI: [10.3390/cancers15092559](https://doi.org/10.3390/cancers15092559)]

71. Lamb LR, Mercaldo SF, Ghaderi K, Carney A, Lehman CD. Comparison of the diagnostic accuracy of mammogram-based deep learning and traditional breast cancer risk models in patients who underwent supplemental screening with MRI. *Radiology* 2023;**308**(3):e223077. [DOI: [10.1148/radiol.223077](https://doi.org/10.1148/radiol.223077)]

72. Li SX, Milne RL, Nguyen-Dumont T, English DR, Giles GG, Southey MC, et al. Prospective evaluation over 15 years of six breast cancer risk models. *Cancers* 2021;**13**(20):16.

73. Mayo MS, Kimler BF, Fabian CJ. Evaluation of models for the prediction of breast cancer development in women at high risk. *Journal of Applied Research* 2001;**1**(1):37-44.

74. McCarthy AM, Guan Z, Welch M, Griffin ME, Sippo DA, Deng Z, et al. Performance of breast cancer risk-assessment models in a large mammography cohort. *Journal of the National Cancer Institute* 2020;**112**(5):489-97.

75. McCarthy AM, Liu Y, Ehsan S, Guan Z, Liang J, Huang T, et al. Validation of breast cancer risk models by race/ethnicity, family history and molecular subtypes. *Cancers* 2021;**14**(1):23.

76. Ming C, Viassolo V, Probst-Hensch N, Chappuis PO, Dinov ID, Katapodi MC. Machine learning techniques for personalized breast cancer risk prediction: comparison with the BCRAT and BOADICEA models. *Breast Cancer Research* 2019;**21**(1):75.

77. Quante AS, Whittemore AS, Shriver T, Strauch K, Terry MB. Breast cancer risk assessment across the risk continuum: genetic and nongenetic risk factors contributing to differential model performance. *Breast Cancer Research* 2012;**14**(6):R144.

78. Quante AS, Whittemore AS, Terry MB. Breast cancer risk assessment: genetic and non-genetic risk factors contributing to differential model performance. *Cancer Prevention*

Research (Philadelphia, Pa) 2012;**5**(11):Abstract PR-02. [DOI: [10.1158/1940-6207.PREV-12-PR-02](https://doi.org/10.1158/1940-6207.PREV-12-PR-02)]

79. Rockhill B, Spiegelman D, Byrne C, Hunter DJ, Colditz GA. Validation of the Gail et al. model of breast cancer risk prediction and implications for chemoprevention. *Journal of the National Cancer Institute* 2001;**93**(5):358-66.

80. Sevdalis A, Deng X, Bandyopadhyay D, McGuire KP. The value of Tyrer-Cuzick versus Gail risk modeling in predicting benefit from screening MRI in breast cancer. *European Journal of Breast Health* 2022;**18**(1):79-84.

81. Terry MB, Liao Y, Whittemore AS, Leoce N, Buchsbaum R, Zeinomar N, et al. 10-year performance of four models of breast cancer risk: a validation study. *Lancet Oncology* 2019;**20**(4):504-17.

82. Terry MB, Phillips KA, Liao Y, MacInnis RJ, Dite GS, Daly MB, et al. Comparison of risk model recommendations for women at high-risk of breast cancer based on clinical thresholds using the Prospective Family Study Cohort (ProF-SC). *Cancer Epidemiology Biomarkers and Prevention* 2017;**26**(5 Suppl):Abstract PR07.

83. Tyrer J, Duffy SW, Cuzick J. A breast cancer prediction model incorporating familial and personal risk factors. *Statistics in Medicine* 2004;**23**(7):1111-30.

84. Tyrer J, Duffy SW, Cuzick J. A breast cancer prediction model incorporating familial and personal risk factors (vol 23, pg 1111, 2004). *Statistics in Medicine* 2005;**24**(1):156. [DOI: [10.1002/sim.1913](https://doi.org/10.1002/sim.1913)]

85. Anderson H, Bladstrom A, Olsson H, Moller TR. Familial breast and ovarian cancer: a Swedish population-based register study. *American Journal of Epidemiology* 2000;**152**(12):1154-63.

86. Evans DG, Howell A. Breast cancer risk-assessment models. *Breast Cancer Research* 2007;**9**(5):213.

87. Brooks JD, Mah A, Christensen RA, Arneja J, Eisen A, Chiarelli AM. Validation of the International Breast Cancer Intervention Study (IBIS) model in the high risk Ontario breast screening program: a retrospective cohort study. *Genetics in Medicine* 2023;**25**(6):100820. [DOI: [10.1016/j.gim.2023.100820](https://doi.org/10.1016/j.gim.2023.100820)]

88. Eriksson M, Czene K, Vachon C, Conant EF, Hall P. A clinical risk model for personalized screening and prevention of breast cancer. *Cancers* 2023;**15**(12):3246. [DOI: [10.3390/cancers15123246](https://doi.org/10.3390/cancers15123246)]

89. Eriksson M, Czene K, Vachon C, Conant EF, Hall P. Long-term performance of an image-based short-term risk model for breast cancer. *Journal of Clinical Oncology* 2023;**41**(14):2536-45. [DOI: [10.1200/JCO.22.01564](https://doi.org/10.1200/JCO.22.01564)]

90. Hughes E, Tshiaba P, Wagner S, Judkins T, Rosenthal E, Roa B, et al. Integrating clinical and polygenic factors to predict breast cancer risk in women undergoing genetic testing. *JCO Precision Oncology* 2021;**5**:307-16.

91. Kenan ES, Friger M, Shochat-Bigon D, Schayek H, Bernstein-Molho R, Friedman E. Accuracy of risk prediction models for

breast cancer and BRCA1/BRCA2 mutation carrier probabilities in Israel. *Anticancer Research* 2018;**38**(8):4557-63.

92. Laitman Y, Simeonov M, Keinan-Boker L, Liphshitz I, Friedman E. Breast cancer risk prediction accuracy in Jewish Israeli high-risk women using the BOADICEA and IBIS risk models. *Genetics Research* 2013;**95**(6):174-7.

93. Li SX, Milne RL, Nguyen-Dumont T, Wang X, English DR, Giles GG, et al. Prospective evaluation of the addition of polygenic risk scores to breast cancer risk models. *JNCI Cancer Spectrum* 2021;**5**(3):pkab021.

94. Mabey B, Hughes E, Kucera M, Simmons T, Hullinger B, Pederson HJ, et al. Validation of a clinical breast cancer risk assessment tool combining a polygenic score for all ancestries with traditional risk factors. *Genetics in Medicine* 2024;**26**(7):101128. [DOI: [10.1016/j.gim.2024.101128](https://doi.org/10.1016/j.gim.2024.101128)]

95. MacInnis RJ, Liao Y, Knight JA, Milne RL, Whittemore AS, Chung WK, et al. Considerations when using breast cancer risk models for women with negative BRCA1/BRCA2 mutation results. *Journal of the National Cancer Institute* 2020;**112**(4):418-22.

96. Quante AS, Whittemore AS, Shriver T, Hopper JL, Strauch K, Terry MB. Practical problems with clinical guidelines for breast cancer prevention based on remaining lifetime risk. *Journal of the National Cancer Institute* 2015;**107**(7):1-7.

97. Warwick J, Birke H, Stone J, Warren RM, Pinney E, Brentnall AR, et al. Mammographic breast density refines Tyrer-Cuzick estimates of breast cancer risk in high-risk women: findings from the placebo arm of the International Breast Cancer Intervention Study I. *Breast Cancer Research* 2014;**16**(5):451.

98. Antoniou AC, Pharoah PD, McMullan G, Day NE, Stratton MR, Peto J, et al. A comprehensive model for familial breast cancer incorporating BRCA1, BRCA2 and other genes. *British Journal of Cancer* 2002;**86**(1):76-83.

99. Antoniou AC, Pharoah PP, Smith P, Easton DF. The BOADICEA model of genetic susceptibility to breast and ovarian cancer. *British Journal of Cancer* 2004;**91**(8):1580-90.

100. Antoniou AC, Cunningham AP, Peto J, Evans DG, Lalloo F, Narod SA, et al. The BOADICEA model of genetic susceptibility to breast and ovarian cancers: updates and extensions. *British Journal of Cancer* 2008;**98**(8):1457-66.

101. Lee AJ, Cunningham AP, Kuchenbaecker KB, Mavaddat N, Easton DF, Antoniou AC, et al. BOADICEA breast cancer risk prediction model: updates to cancer incidences, tumour pathology and web interface. *British Journal of Cancer* 2014;**110**(2):535-45.

102. Lee AJ, Cunningham AP, Tischkowitz M, Simard J, Pharoah PD, Easton DF, et al. Incorporating truncating variants in PALB2, CHEK2, and ATM into the BOADICEA breast cancer risk model. *Genetics in Medicine* 2016;**18**(12):1190-8.

103. Lee A, Mavaddat N, Wilcox AN, Cunningham AP, Carver T, Hartley S, et al. BOADICEA: a comprehensive breast cancer risk

prediction model incorporating genetic and nongenetic risk factors. *Genetics in Medicine* 2019;**21**(8):1708-18.

104. Antoniou AC, Pharoah PD, McMullan G, Day NE, Ponder BA, Easton D. Evidence for further breast cancer susceptibility genes in addition to BRCA1 and BRCA2 in a population-based study. *Genetic Epidemiology* 2001;**21**(1):1-18.

105. Li H, Feng B, Miron A, Chen X, Beesley J, Bimch E, et al. Breast cancer risk prediction using a polygenic risk score in the familial setting: a prospective study from the Breast Cancer Family Registry and kConFab. *Genetics in Medicine* 2017;**19**(1):30-5.

106. Ming C, Viassolo V, Probst-Hensch N, Dinov ID, Chappuis PO, Katapodi MC. Machine learning-based lifetime breast cancer risk reclassification compared with the BOADICEA model: impact on screening recommendations. *British Journal of Cancer* 2020;**123**(5):860-7.

107. Stahlbom AK, Johansson H, Liljegren A, von Wachenfeldt A, Arver B. Evaluation of the BOADICEA risk assessment model in women with a family history of breast cancer. *Familial Cancer* 2012;**11**(1):33-40.

108. Berry DA, Iversen ES Jr, Gudbjartsson DF, Hiller EH, Garber JE, Peshkin BN, et al. BRCAPRO validation, sensitivity of genetic testing of BRCA1/BRCA2, and prevalence of other breast cancer susceptibility genes. *Journal of Clinical Oncology* 2002;**20**(11):2701-12.

109. Mazzola E, Blackford A, Parmigiani G, Biswas S. Recent enhancements to the genetic risk prediction model BRCAPRO. *Cancer Informatics* 2015;**14**(Suppl 2):147-57.

110. Parmigiani G, Berry D, Aguilar O. Determining carrier probabilities for breast cancer-susceptibility genes BRCA1 and BRCA2. *American Journal of Human Genetics* 1998;**62**(1):145-58. [DOI: [10.1086/301670](https://doi.org/10.1086/301670)]

111. Boggs DA, Rosenberg L, Pencina MJ, Adams-Campbell LL, Palmer JR. Validation of a breast cancer risk prediction model developed for Black women. *Journal of the National Cancer Institute* 2013;**105**(5):361-7.

112. Evans DG, Ingham S, Dawe S, Roberts L, Lalloo F, Brentnall AR, et al. Breast cancer risk assessment in 8,824 women attending a family history evaluation and screening programme. *Familial Cancer* 2014;**13**(2):189-96.

113. Gard CC, Tice JA, Miglioretti DL, Sprague BL, Bissell MC, Henderson LM, et al. Extending the Breast Cancer Surveillance Consortium model of invasive breast cancer. *Journal of Clinical Oncology* 2024;**42**(7):779-89. [DOI: [10.1200/JCO.22.02470](https://doi.org/10.1200/JCO.22.02470)]

114. Louro J, Roman M, Posso M, Vazquez I, Saladie F, Rodriguez-Arana A, et al. Developing and validating an individualized breast cancer risk prediction model for

women attending breast cancer screening. *PLoS ONE* 2021;**16**(3):e0248930.

115. Ming C, Viassolo V, Probst-Hensch N, Chappuis PO, Dinov ID, Katapodi MC. Machine learning techniques for personalized breast cancer risk prediction: comparison with the BCRAT and BOADICEA models. *Breast Cancer Research* 2019;**21**(1):75.

116. Moons KG, Altman DG, Vergouwe Y, Royston P. Prognosis and prognostic research: application and impact of prognostic models in clinical practice. *BMJ* 2009;**338**:b606.

117. Steyerberg EW, Moons KG, Van der Windt DA, Hayden JA, Perel P, Schroter S, et al. Prognosis Research Strategy (PROGRESS) 3: prognostic model research. *PLoS Medicine* 2013;**10**(2):e1001381.

118. MacInnis RJ, Knight JA, Chung WK, Milne RL, Whittemore AS, Buchsbaum R, et al. Comparing 5-year and lifetime risks of breast cancer using the prospective family study cohort. *Journal of the National Cancer Institute* 2021;**113**(6):785-91.

119. MacInnis RJ, Bickerstaffe A, Apicella C, Dite GS, Dowty JG, Aujard K, et al. Prospective validation of the breast cancer risk prediction model BOADICEA and a batch-mode version BOADICEACentre. *British Journal of Cancer* 2013;**109**(5):1296-301.

120. MacInnis R, Dite G, Bickerstaffe A, Dowty J, Aujard K, Apicella C, et al. Validation study of risk prediction models for female relatives of Australian women with breast cancer. *Hereditary Cancer in Clinical Practice* 2012;**10**(Suppl 2):A66.

121. Mertens, E, Barrenechea-Pulache, A, Sagastume, D, et al. Understanding the contribution of lifestyle in breast cancer risk prediction: a systematic review of models applicable to Europe. *BMC Cancer* 2023;**687**:1-15. [DOI: [10.1186/s12885-023-11174-w](https://doi.org/10.1186/s12885-023-11174-w)]

122. Velentzis LS, Freeman V, Campbell D, Hughes S, Luo Q, Steinberg J, et al. Breast cancer risk assessment tools for stratifying women into risk groups: a systematic review. *Cancers* 2023;**15**(4):1124. [DOI: [0.3390/cancers15041124](https://doi.org/0.3390/cancers15041124)]

123. Yang X, Mooij Thea M, Leslie G, Ficorella L, Andrieu N, Kast K, et al. Validation of the BOADICEA model in a prospective cohort of <ovid:i>BRCA1/2</ovid:i> pathogenic variant carriers [Validation of the BOADICEA model in a prospective cohort of <ovid:i>BRCA1/2</ovid:i> pathogenic variant carriers]. *Journal of Medical Genetics* 2024;**61**(8):803-9. [DOI: [10.1136/jmg-2024-109943](https://doi.org/10.1136/jmg-2024-109943)]

124. Collins GS, Moons KG, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 2024;**385**(e078378):1-14. Erratum In: *BMJ*. 2024 Apr 18;385:q902. [DOI: [10.1136/bmj-2023-078378](https://doi.org/10.1136/bmj-2023-078378)]

ADDITIONAL TABLES

Table 1. PICOTS framework

Population	Women (pre and post-menopausal) with a family history of breast cancer
Intervention (model)	All breast cancer risk prediction models that include family history of breast cancer as a predictor(s)
Comparator	Models will be compared to each other
Outcome	Breast cancer occurrence
Timing	Timespan of prediction will vary. Some models predict lifetime risk, some risk to a certain age, some risk within 10 years, etc.
Setting	Models to be used in asymptomatic adult females with a family history of breast cancer to predict their future personal risk of developing breast cancer

We drew on methodological guidance for systematic reviews of prediction modelling studies in our use of the PICOTS framework, including works by Moons and colleagues [31], and Debray and colleagues [32].

Table 2. Gail/Breast Cancer Risk Assessment Tool (BCRAT) model: overview of synthesis and included studies

Study	Characteristics of included studies							
	Participants	Study dates	Country	Number of participants/ number of events	Age	Source	Prediction horizon	Follow-up
Gail model studies included in meta-analyses of both C statistic & O/E ratio								
Glynn 2019	Participants of Nurses' Health Study with complete baseline risk factor data, cancer-unaffected at baseline. Of these, only subgroup with family hx relevant for this review	Recruitment between 1980 and 2006, incident invasive BCs identified by 2008	USA	<i>Numbers of participants not specified; rather the number of 2-year follow-up intervals</i> 86,892 intervals/765 events	Median age of women with family history 61 years (IQR 54 to 68)	Cohort	2-year risk intervals	Median follow-up 24 years; range 1 to 28 years for total cohort. Not reported separately for subgroup with family history.
Li 2021a	Australian women recruited to Melbourne Collaborative Cohort Study (MCCS). Of these, only subgroup with family BC hx applicable for this review	Baseline (recruited) 1990 to 1994	Australia, UK, USA	1720 participants/110 events	50 to 65 years at start of follow-up	Cohort	15-year risk	Median/mean follow-up NR Start of follow-up was 2003 to 2007. Follow-up ended when (i) diagnosis of invasive BC, (ii) follow-up time reached 15 years, (iii) age 80 years (maximum age for estimating risk in BOADICEA), or (iv) censor date of 1 March 2019, whichever came first.
McCarthy 2021	Women who underwent screening mammography. Of these, subgroup of women	Women presenting for screening mammography be-	USA	# participants in family history subgroup unclear; approx 24,511	Aged 40 to 84 at baseline	Cohort	5-year risk	Median/mean follow-up NR

Table 2. Gail/Breast Cancer Risk Assessment Tool (BCRAT) model: overview of synthesis and included studies (Continued)

	with family BC hx relevant for this review	tween 2006 and 2015		# events = 394				All participants had at least 5 years of FU.
Terry 2019	Women enroled in the ProF-SC cohort. This cohort is enriched for familial BC risk	Recruited between 17 March 1992 and 29 June 2011	USA, Canada, Australia	15,732 participants/619 events	20 to 70 years	Cohort	10-year risk	Median follow-up of 11.1 years (IQR 6.0 to 14.4)
Guan 2023	Women enroled in Cancer Genetics Network (CGN). This cohort represents a higher-risk population enriched for family hx of cancer.	Enrolment began in 1999 and ended in 2010	USA	7314 participants/112 events	Median age 47 years (IQR 38 to 57)	Cohort	5-year risk	Median follow-up 7.3 years (IQR 6.0 to 8.3)
Gail model studies included in meta-analyses of C statistic only								
Amir 2003	Women who attended a UK Family History clinic for evaluation	Data collected from 1987 to January 2001	UK	3150 participants (total study population)/64 events	21 to 73 years; median 44 years	Cohort	Predicted 10-year risk adjusted to no. of years of FU	5.27 years (range 0.10 to 15.00 years)
Clendenen 2019	Women recruited to one of 10 prospective cohort studies. All eligible women who developed incident BC included & were matched to controls within each cohort. Only women from subgroup analysis with ≥ 1 BC-affected FDR included for this review.	Ranged from 1977 to 2010	USA	NR for subgroup with a family BC hx	35 to 50 years	Nested case-control study including 10 prospective cohorts	5- year risk	Unclear
Mayo 2001	American women (95.2% were white) at high risk of BC because of a family BC hx, prior precancerous biopsy, and/or prior invasive cancer. 75.6% percent had at least one FDR or two SDRs with BC.	Enroled from August 1989 to January 1999	USA	480 participants/20 events	Mean age 44.31 years (SD 8.59)	Cohort	10-year risk	Average follow-up of 42.53 (SD 29.68) months
Ming 2019 (USA)	US population-based	Recruited between	USA	1143 participants/ 850 events	50.86 $\pm$ 6.22 (35 to 64)	Retrospective dataset	Lifetime risk	NR

Table 2. Gail/Breast Cancer Risk Assessment Tool (BCRAT) model: overview of synthesis and included studies (Continued)

	women diagnosed with BC younger than 45 years old and their cancer-free, first- and second-degree female relatives	1994 and 2008						
Sevdalis 2022	Females, aged 18 to 75, who underwent screening MRI between January 2005 and December 2015 within a VCU Health-affiliated hospital. Most (90.8%) participants had FDR with known BC	Women underwent screening MRI between January 2005 and December 2015	USA	134 participants/5 events	18 to 75 years	Cohort (retrospective)	Lifetime risk	NR/NA
Gail model studies included in meta-analyses of O/E ratio only								
Bondy 1994	Women who had at least one FDR with BC and who had participated in the American Cancer Society 1987 Texas Breast Screening Project	1987 to 1992	USA	1981 participants/39 events	30 to 75 years	Cohort	5-year risk	Follow-up period from 1987 to 1992 (5 years median or average follow-up NR)
Evans 2016 Chapter 2 (study 1a)	Women who had been referred to and assessed at the Family History Clinic (FHC) at the University Hospital of South Manchester	Assessed between 1987 and 2013	UK	9527 participants/416 events	Age range 14.5 to 81.3 years; IQR 33.9 to 46.0 years; median 39.5 years at entry	Cohort	10-year risk	Median FU of 10.2 years (IQR 5.0 to 15.5 years; range 0.1 to 26.0 years)
Gail model studies included in narrative synthesis only								
Gong 2014	Subset of women recruited and followed at New York site of the Breast Cancer Family Registry (BCFR) Cohort enriched for women with a family BC hx ~65% of cohort had FDR with BC, though ~87% had more distant relative(s) with BC.	Unclear	USA, Germany	1857 participants/83 events	20 to 70 years	Cohort	10-year risk	Mean FU 8.1 years (range 0.1 to 14.5)

Table 2. Gail/Breast Cancer Risk Assessment Tool (BCRAT) model: overview of synthesis and included studies (Continued)

McCarthy 2020	Women presenting for mammography screening at Newton-Wellesley Hospital. Of these, only subgroup of women with family BC hx applicable for this review	Recruited between 1 February 2007 and 31 December 2009	USA	12,994 participants (with at least one FDR or SDR with breast or ovarian cancer)/304 events	40 to 84 years	Cohort	6-year risk	Mean follow-up time was 6.7 years (IQR 6.3 to 7.1) <i>*This relates to the entire cohort; not reported separately for family hx subgroup</i>
Quante 2012	Subset of women recruited and followed at New York site of the Breast Cancer Family Registry. Cohort enriched for women with a family BC hx ~65% of cohort had FDR with BC, though ~87% had more distant relative(s) with BC.	Unclear	USA, Germany	1857 participants/83 events	20 to 70 years	Cohort	10-year risk	Mean follow-up 8.1 years (range 0.1 to 14.5)
Rockhill 2001	White women in the Nurses' Health Study (NHS) cohort. Of these, only women with at least one BC-affected FDR applicable	1992 to 1997	USA	8163 participants with 1+ affected FDRs/207 events	46 to 71 years	Cohort	5-year risk	Follow-up between 1992 and 1997 (5 years) No median/mean follow-up reported
Ho 2023	Asymptomatic Asian women recruited to the Singapore Breast Screening Project undergoing screening mammography Of these, only subgroup with family BC hx in a FDR applicable for review	Participants recruited between October 1994 and February 1997. Latest date of occurrence set at 31 December 2019	Singapore	689 participants/53 events <i>For family history subgroup (women with at least one FDR with BC)</i>	Median age at recruitment was 57 years (IQR 54 to 61). <i>*This relates to full study cohort, not reported separately for family hx subgroup</i>	Prospective cohort of randomised trial participants	5-, 10-, and 15-year risk	Participants recruited between October 1994 and February 1997. Latest date of occurrence set at 31 December 2019. Median /Mean FU not reported
Lamb 2023	High-risk women undergoing screening MRI Participants were considered high risk based on one or more of the following: genetic mutation carri-	Participants underwent screening MRI	USA	1941 MRI examinations (5 year)	Median age was 54 years (IQR 48 to 60 years)	Retrospective study using data from women un-	5-year, lifetime risk	Median/mean FU not reported It was an inclusion criterion that participants needed to

Table 2. Gail/Breast Cancer Risk Assessment Tool (BCRAT) model: overview of synthesis and included studies (Continued)

er (high-penetrance genes [e.g. <i>BRCA1/2</i> , <i>PTEN</i> , <i>TP53</i> , <i>CDH1</i>] or moderate-risk genes [e.g. <i>CHEK2</i> , <i>ATM</i> , <i>PALB2</i> , <i>BARD1</i>]); hx of chest radiation; personal BC hx; high-risk lesion hx; family BC hx.	between September 2017 and September 2020	2030 MRI examinations (lifetime) # participants unclear/# events unclear	dergoing MRI screening examination	have at least 1 year of follow-up after screening MRI.
Note: 76.9% [of examinations rather than individual participants] had a family hx of breast cancer				
25.4% [of examinations rather than individual participants] had a personal hx of BC				

BC: breast cancer; **FDR:** first-degree relative; **FU:** follow-up; **hx:** history; **IQR:** interquartile range; **LR:** lifetime risk; **MRI:** magnetic resonance imaging; **NR:** not reported; **ProF-SC:** Breast Cancer Prospective Family Study Cohort; **SD:** standard deviation; **SDR:** second-degree relative

Table 3. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS) model: overview of synthesis and included studies

Study	Characteristics of included studies							
	Participants	Recruitment	Country	Number of participants/ number of events	Age	Source	Prediction horizon	Follow-up
Model development study								
Tyrer 2004	Different published data sets that included: UK national statistics (Cancer Registry statistics from 1994) Cohort of offspring of mothers with breast or ovarian cancer diagnosed in 1958–1993, established using Swedish population-based registers.	Unclear - different study dates for each dataset used in model development	UK	NR	NR	Previously published data from several sources	10-year, various	NA
Tyrer-Cuzick model studies included in meta-analyses of both C statistic & O/E ratio								

Table 3. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS) model: overview of synthesis and included studies (Continued)

Brooks 2023	Women referred to the High Risk Ontario Breast Screening Program (OBSP). 60.4% had at least one FDR with BC	Participants referred to OBSP between July 1, 2011, and December 31, 2016, with follow-up until December 31, 2018	Canada	7487 participants/ 202 events	30 to 69 years	Routinely collected data for women screened within the High Risk OBSP	Predicted 10-year risks were adjusted to the follow-up period	Median (range) FU 4.3 (0.2-7.9) years
Guan 2023	Women enrolled in the Cancer Genetics Network (CGN). This cohort represents a higher-risk population enriched for family history of cancer	Enrolment began in 1999 and ended in 2010	USA	7314 participants/ 112 events	Median age 47 (IQR 38–57) years	Cohort	5-year risk	Median follow-up 7.3 years (IQR 6.0-8.3)
Glynn 2019	Participants of Nurses' Health study with complete baseline risk factor data, cancer-unaffected at baseline. Of these, only subgroup with family history relevant for review.	Recruitment between 1980 and 2006, incident invasive breast cancers identified by 2008	USA	<i>Numbers of participants not specified rather the number of 2-year follow-up intervals</i> 86,892 intervals/765 events	Median age of women with family history 61 years (IQR 54 to 68)	Cohort	2-year risk intervals	Median follow-up 24 years; range 1–28 years for total cohort. Follow-up not reported separately for subgroup with family history.
Li 2021b	Australian women recruited to the Melbourne Collaborative Cohort Study (MCCS) Of these, only the subgroup of women with a family history of BC are applicable for this review.	Baseline (1990-1994) and up to 2 additional waves of active follow-up (in 1995-1998 and/or in 2003-2007).	Australia, UK, USA	697 participants (with family history) in full case cohort 598 in sub-cohort with PRS/ 116 events (cases)	50 to 75 years at start of follow-up	Cohort (case cohort design)	10-year risk	Mean/median FU NR FU ended at 1) diagnosis of invasive BC, 2) follow-up time reaching 10 years, 3) age 80 years (maximum age for estimating risk in BOADICEA), or 4) censor date of

Table 3. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS) model: overview of synthesis and included studies (Continued)

								June 30, 2016, whichever came first.
McCarthy 2020	Women presenting for mammography screening at Newton-Wellesley Hospital. Of these, only subgroup of women with family history of breast cancer are applicable for this review.	Recruited between February 1, 2007 and December 31, 2009	USA	12,994 participants (with at least one FDR or SDR with breast or ovarian cancer)/ 304 events	Aged 40–84 years	Cohort	6-year risk	The mean follow-up time was 6.7 years (interquartile range = 6.3– 7.1 years) * This relates to the entire cohort. Not reported separately for family history subgroup
Terry 2019	Women enrolled in the ProF-SC cohort. This cohort is enriched for familial risk of breast cancer	Recruited between March 17, 1992, and June 29, 2011	USA, Canada, Australia	15732 participants/619 events	20-70 years	Cohort	10-year risk	Median follow-up of 11.1 years (IQR 6.0–14.4)
Tyrer-Cuzick model studies included in meta-analyses of C statistic only								
Evans 2016 Chapter 3 (study 4)	Women assessed at the FHC at the University Hospital of South Manchester BRCA1 carriers were excluded Controls: women who attended the clinic but were not diagnosed with BC to last FU matched approx 3 : 1 to BC cases	Recruitment between 1987 & 2012. Median year of enrolment for cases was 1999 (IQR 1995–2004); & for controls was 2002 (IQR 1996–2007)	UK	Prospective analysis: 1216 Participants/261 events	NR	Case-control Prospective analysis – where cases were women who developed BC during FU period.	10-year risk	Median FU 8.4 years
Sevdalis 2022	Females, aged 18 to 75, who underwent screening breast MRI between January 2005 to December 2015 within a VCU Health affiliated hospital. The majority (90.8%) of subjects had a first degree relative with known breast cancer	Women underwent screening breast MRI between January 2005 to December 2015	USA	163 participants/5 events	18-75 years	Cohort (retrospective)	Lifetime risk	NR/NA

Table 3. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS) model: overview of synthesis and included studies (Continued)

Warwick 2014	Subset of elevated risk women (based on family hx or hx of benign breast disease) from the placebo arm of the International Breast Cancer Intervention Study 1 (IBIS-1) with risk of developing BC estimated to be at least twice the population average.	Participants in trial randomised from April 1992 through March 2001	UK, Australia	558 participants/72 events (ie. 486 controls 72 cases)	35 to 70 years	Nested case-control study of women from the placebo arm of randomised controlled trial	10-year risk	Median follow-up for controls was 11.6 (10.7-12.3) years. Median time to diagnosis for cases was 5.1 (3.1-7.9) years
Yala 2019	Women who had screening mammograms at a single centre in USA. Of these, only subgroup of women with a family history of breast cancer applicable for review	Participants had screening mammograms between January 1, 2009, and December 31, 2012	USA	Test set: 1767 participants with any family history breast cancer/ 141 events in subgroup with any family history	56.20 years $\pm$ 10.04	Cohort (retrospective)	5-year risk	Mean/ median FU NR Diagnosis of breast cancer within 5 years, or imaging follow-up for at least 5 years from the date of index mammography
Tyrer-Cuzick model studies included in meta-analyses of O/E ratio only								
Evans 2016 Chapter 2 (study 1a)	Women who had been referred to and assessed at the Family History Clinic (FHC) at the University Hospital of South Manchester	Assessed between 1987 and 2013	UK	9527 participants / 416 events	Age range 14.5–81.3 years; interquartile range (IQR) 33.9–46.0 years; median 39.5 years at entry	Cohort	10-year risk	Median FU of 10.2 years (IQR 5.0–15.5 years; range 0.1–26.0 years)
Laitman 2013	Jewish Israeli high-risk women who tested negative for the predominant mutations in BRCA1 and BRCA2 in Jewish women 57.2% were of Ashkenazi origin.	Women counselled from 1 June 1996 to 31 May 2000.	Israel	358 participants/ 15 events	20–75 years (mean age 46.76 $\pm$ 9.8 years)	Cohort	10-year risk	Average FU 13.57 $\pm$ 1.45 years (range 11–16 years). total of 4.861 person-years of follow-up.

Table 3. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS) model: overview of synthesis and included studies (Continued)

Mabey 2024	US-based female patients who were referred for clinical germline genetic testing with a multigene hereditary cancer panel One-third of the study population reported a BC diagnosis in a first-degree relative, 75.2% reported a BC diagnosis in a first, second or third-degree relative	Referred for clinical germline genetic testing between January 2017 and February 2019. Follow-up until February 2020	USA	130,058 participants/340 events	Median age at the time of hereditary cancer testing was 46 years (range 18-84 years)	Cohort	Appears to have been adjusted to the follow-up period	Median follow-up of 12.1 months (range 4.0-29.5 months)
Tyrer-Cuzick model studies included in narrative synthesis only								
Amir 2003	Women who attended a UK Family History clinic for evaluation	Data collected from 1987 to January 2001	UK	3150 participants (Total Study Population)/ 64 events	21-73 years median 44 years	Cohort	Predicted 10-year risk adjusted to no. of years of FU	5.27 years (range 0.10–15.00 years)
Gong 2014	Subset of women recruited and followed at New York site of the Breast Cancer Family Registry (BCFR) Cohort enriched for women with a family history of breast cancer ~65% of cohort had a first degree relative with breast cancer though ~87% had a more distant relative(s) with breast cancer.	Unclear	USA, Germany	1857 participants/ 83 events	20 to 70 years	Cohort	10-year risk	mean FU 8.1 years (range 0.1 to 14.5).
Hughes 2021	Women of European ancestry who had undergone testing for hereditary cancer predisposition. Cohorts were enriched for women with family history of breast cancer	Accrual between June 1, 2017 and June 11, 2019	USA	Development 1 = 5489 participants/0 events Development 2 = 141,160 participants/28,928 events	Validation 1: 18-84 years. Median 48 years Validation 2: 19-84 years. Median 61 years	Cohort (a consecutive clinical testing cohort of patients submitted for hereditary cancer predisposition between June 1, 2017 and	5-year risk, RLR	Unclear

Table 3. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS) model: overview of synthesis and included studies (Continued)

				Validation 1 = 1,615 participants/988 events				June 11, 2019 divided into sets for development and validation)
				Validation 2 = 518 participants/256 events				and one prospectively collected case-control validation study (validation 2)
				Clinical Performance Population = 32,578 participants/ 0 events				
Kenan 2018	Women counselled and genotyped at an Israeli oncogenetics unit	2000 and 2005 for oncogenetic testing. Follow-up last updated for the calendar year 2016 in 2017.	Israel	374 participants/ 25 events	Age range at counselling was 19-85 years (mean age 50.9±11.4 years)	Cohort	5-year, 10-year & LR	Follow-up between 12 and 17 years (12 years for those counselled in 2005, 17 for those counselled in 2000)
Li 2021a	Australian women recruited to the Melbourne Collaborative Cohort Study (MCCS) aged. Of these, only subgroup with family history of BC applicable for this review	Baseline (recruited) 1990-1994	Australia, UK, USA	1720 participants/ 110 events	50 to 65 years at start of follow-up	Cohort	15-year risk	Median/Mean FUnr Start of follow-up 2003-2007 FU ended when (i) diagnosis of invasive breast cancer, (ii) follow-up time reaching 15 years, (iii) age 80 years (maximum age for estimating risk in BOADICEA), or (iv) censor date of 1 March 2019,

Table 3. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS) model: overview of synthesis and included studies (Continued)

								whichever came first.
MacInnis 2020	Women from Prospective Family Study Cohort (ProF-SC) Cohort enriched for women with family hx of breast cancer (96.1% had an affected relative)	NR	Australia, USA, Canada	14657 participants/482 events	20 to 70 years at baseline	Cohort	10-year risk	Median FU 10 years (interquartile range = 6.1–10.0 years)
MacInnis 2021	Women from ProF-SC Cohort is enriched for women with a family history of BC.	Enrolment between 1992 & 2011	Australia, USA, Canada	14,657 participants/482 events	20 to 70 years at baseline	Cohort	5-year risk, LR, RLR	Median FU 10 years
Quante 2012	Subset of women recruited and followed at New York site of the Breast Cancer Family Registry. Cohort enriched for women with a family hx of breast cancer ~65% of cohort had a first degree relative with breast cancer though ~87% had a more distant relative(s) with breast cancer.	Unclear	USA, Germany	1857 participants/ 83 events	20 to 70 years	Cohort	10-year risk	Mean follow-up 8.1 years (range 0.1 to 14.5)
Quante 2015	Women recruited to the New York site of the Breast Cancer Family Registry (BCFR). Participants enriched for a family history of breast cancer.	Participants recruited to BCFR since 1995. Study dates for this particular cohort unclear	USA, Germany, Australia	1764 participants/95 events (though of these only 79 developed BC within the 10-year FU timespan of interest for prediction of 10-year risk)	20 to 70 years	Cohort	10-year risk	Unclear
Eriksson 2023b	Women randomly selected from a Swedish screening cohort ('the sub-cohort') together with all incident breast cancers ('cases') that were diagnosed in the underlying cohort.	Participants underwent mammographic screening between October 2010	Sweden (with co-authors from USA)	1008 participants with a first degree relative with breast cancer in sub-cohort	40–74 years. At baseline, the mean age was 56.62(±9.50) years for cases and	Case-cohort	1, 2, 3, 4, 5, 6, 8, 10-year risk	The follow-up period from initial mammogram to breast cancer diagnosis ranged from

Table 3. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS) model: overview of synthesis and included studies (Continued)

	<i>*Of these, only participants in the [FDR] family hx subgroup relevant for this review</i>	and March 2013. Incident breast cancers diagnosed before May 2022 were included.	389 events (cases) in participants with a FDR with BC	53.90 (±9.87) years for the sub-cohort. <i>*This relates to the entire study population, mean age of family hx subgroup not reported separately</i>	>3 months to 10 years. Model performances were estimated at the end of each year of follow-up.
Eriksson 2023a	Women randomly selected from a Swedish screening cohort (KARMA study cohort) together with all incident breast cancers ('cases') that were diagnosed in the underlying cohort* <i>* Of these, only participants in the family hx subgroup relevant for this review</i>	Participants underwent mammographic screening between October 2010 and March 2013. Incident breast cancers identified in January 2022 register update	Sweden (with co-authors from USA)	988 participants with a first degree relative with breast cancer in sub-cohort/ 355 events (cases) 730 participants with a 2nd/3rd degree relative with BC/168 events (cases) Age 40–74 years. At baseline, the mean age was 56.52(±9.53) years for cases and 53.88 (±9.86) years for the sub-cohort* <i>*This relates to the entire study population, mean age of family hx subgroup not reported separately</i>	Case-cohort 1, 2, 3, 4, 5, 6, 8, 10-year risk The follow-up period was over 10 years and the time from the mammogram at enrolment to breast cancer diagnosis ranged from 3 months to over 10 years Outcome identified in the January 2022 register update The risk model performances were estimated at the end of each year of follow-up.
Lamb 2023	High risk women undergoing screening MRI Participants were considered high-risk based on one or more of the following: Genetic mutation carrier (high-penetrance genes [eg, <i>BRCA1/2</i>,	Participants underwent screening MRI from September 2017 to September 2020	USA	2250 MRI examinations (5-year & life-time) # participants unclear/ # events unclear Median age was 54 years (IQR, 48–60 years)	Retrospective study using data from women undergoing MRI screen- 5-year, lifetime risk Median/Mean FU not reported It was an inclusion criterion that participants needed to have at least 1 year of follow-up.

Table 3. Tyrer-Cuzick/International Breast Cancer Intervention Study (IBIS) model: overview of synthesis and included studies (Continued)

PTEN, TP53, CDH1] or moderate-risk genes [eg, *CHEK2, ATM, PALB2, BARD1*]],

- History of chest radiation,
- Personal history of breast cancer,
- history of high-risk lesion
- Family history of breast cancer.

Note: 76.9% [of examinations rather than individual participants] had a family hx of breast cancer

25.4% [of examinations rather than individual participants] had a personal hx of BC

ing examination

after screening MRI

BC: breast cancer; **FDR:** first-degree relative; **FU:** follow-up; **hx:** history; **IQR:** interquartile range; **KARMA:** Karolinska Mammography Project for Risk Prediction of Breast Cancer; **LR:** lifetime risk; **MRI:** magnetic resonance imaging; **NA:** not applicable; **NR:** not reported; **ProF-SC:** Breast Cancer Prospective Family Study Cohort; **RLR:** remaining lifetime risk; **SD:** standard deviation; **SDR:** second-degree relative; **VCU:** Virginia Commonwealth University

Table 4. Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm (BOADICEA) model: overview of synthesis and included studies

Study	Characteristics of included studies							
	Participants	Study dates	Country	Number of participants/ number of events	Age	Source	Prediction horizon	Follow-up
Model development study								
Antoniou 2002	Population-based series of breast cancer cases diagnosed at < 55 years identified through Anglican Cancer Registry (ABC study) AND UK families with multiple breast cancer cases, at least one of which was diagnosed at age < 50 years (B families)	ABC study: women diagnosed with BC between 1 January 1991 and 30 June 1996 ‘B families’: NR	UK	ABC study: 1484 participants/1484 events (case-only) B Families: 156 families (# individuals not reported)/618 events	ABC study: Mean age or age range NR All participants were diagnosed with BC age <55 years B families:	Combined dataset of population based series of breast cancer cases AND multiple case	NR	NR

Table 4. Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm (BOADICEA) model: overview of synthesis and included studies (Continued)

					NR	families in the UK		
BOADICEA model studies included in meta-analyses of both C statistic & O/E ratio								
Li 2021b	Australian women recruited to the Melbourne Collaborative Cohort Study (MCCS) Of these, only the sub-group of women with a family history of BC are applicable for this review.	Baseline (1990-1994) and up to 2 additional waves of active follow-up (in 1995-1998 and/or in 2003-2007).	Australia, UK, USA	697 participants (with family history) in full case cohort 598 in sub-cohort with PRS/ 116 events (cases)	50 to 75 years at start of follow-up	Cohort (case cohort design)	10-year risk	Mean/median FU NR FU ended at 1) diagnosis of invasive BC, 2) follow-up time reaching 10 years, 3) age 80 years (maximum age for estimating risk in BOADICEA), or 4) censor date of June 30, 2016, whichever came first.
Stahlbom 2012	Elevated risk (at least 17% lifetime risk from Claus tables) women attending a Swedish cancer genetics service who were breast cancer free before the start of observation (had a normal mammogram within 1 year before start of observation). Though of these, 27 had had a previous (but not current) breast cancer.	Recruited from January 2002 to June 2006	Sweden	640 participants/ 30 events (25 invasive breast cancers and five DCIS)	≤ 60 years	Retrospective cohort study	Risk predictions adjusted to observation period	Max. FU between Jan 2002 and September 2010 (<8 years)
Terry 2019	Women enrolled in the ProF-SC cohort This cohort is enriched for familial risk of breast cancer	Recruited between March 17, 1992, and June 29, 2011	USA, Canada, Australia	15732 participants/619 events	20-70 years	Cohort	10-year risk	Median follow-up of 11.1 years (IQR 6.0-14.4)
BOADICEA model studies included in meta-analyses of C statistic only								
Ming 2020	Women who attended a Swiss onco-genetics unit due to concerns about their family history	Recruitment from 1998 to 2017	Switzerland, USA	45,110 participants/ 4911 events (breast	49.82 (±11.02) years	Retrospective cohort/	Lifetime risk	Duration of FU unclear- not explicitly reported

Table 4. Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm (BOADICEA) model: overview of synthesis and included studies (Continued)

	Mixture of breast cancer affected and BC unaffected women.			cancer patients)		database study		
BOADICEA model studies included in meta-analyses of O/E ratio only								
Evans 2016 Chapter 2 (study 1a)	Women who had been referred to and assessed at the Family History Clinic (FHC) at the University Hospital of South Manchester	Assessed between 1987 and 2013	UK	6268 participants / 197 events	Age range 14.5–81.3 years; interquartile range (IQR) 33.9–46.0 years; median 39.5 years at entry	Cohort	10-year risk	Median/Mean FU NR 70,293 years of FU
Laitman 2013	Jewish Israeli high-risk women who tested negative for the predominant mutations in BRCA1 and BRCA2 in Jewish women 57.2% were of Ashkenazi origin.	Women counselled from 1 June 1996 to 31 May 2000.	Israel	358 participants/ 15 events	20–75 years (mean age 46.76±9.8 years)	Cohort	10-year risk	Average FU 13.57±1.45 years (range 11–16 years). total of 4861 person-years of follow-up.
BOADICEA model studies included in narrative synthesis								
Kenan 2018	Women counselled and genotyped at an Israeli oncogenetics unit	2000 and 2005 for oncogenetic testing. Follow-up last updated for the calendar year 2016 in 2017.	Israel	374 participants/ 25 events	Age range at counselling was 19–85 years (mean age 50.9±11.4 years)	Cohort	5-year, 10-year & LR	Follow-up between 12 and 17 years (12 years for those counselled in 2005, 17 for those counselled in 2000)
Li 2021a	Australian women recruited to the Melbourne Collaborative Cohort Study (MCCS) aged. Of these, only	Baseline (recruited) 1990–1994	Australia, UK, USA	1720 participants/ 110 events	50 to 65 years at start of follow-up	Cohort	15-year risk	Median/Mean FU NR Start of follow-up 2003–2007 FU

Table 4. Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm (BOADICEA) model: overview of synthesis and included studies (Continued)

	subgroup with family history of BC applicable for this review							ended when (i) diagnosis of invasive breast cancer, (ii) follow-up time reaching 15 years, (iii) age 80 years (maximum age for estimating risk in BOADICEA), or (iv) censor date of 1 March 2019, whichever came first.
Li 2017	Women enrolled in the Breast Cancer Family Registry (BCFR) cohort (from 6 sites: 4 in US, Australia & Canada) and the Kathleen Cuninghame Consortium Foundation for Research in Familial Breast Cancer (kConFab)	Women enrolled in BCFR since 1995, Women enrolled in kConFab since 1997	USA, Australia	2599 participants/205 events		Cohort (BCFR and kConFab familial BC cohorts)	10-year risk	Median FU = 6.2 years Mean FU = 7.4 years
MacInnis 2013	Whole cohort: Women of European ancestry unaffected at baseline from case and control families recruited to the Australian Breast Cancer Family Registry, a cohort of Australian women over-sampled for family history Subgroup analysis: subset of women from above cohort with at least 1 first degree breast cancer-affected relative.	Unclear Index case and controls ascertained between 1992 & 1999- unclear when family members recruited	Australia, UK, USA	Whole cohort: 4176 participants/ 115 events Subgroup of women with at least 1 first degree affected relative 2456 participants/ 81 events		Cohort (Australian Breast Cancer Family Registry (ABCFR))	10-year risk	FU of family of the index case was average 9.2 years for case families and 9.6 years for control families
MacInnis 2020	Women from Prof-SC cohort Cohort enriched for women with family hx of breast cancer (96.1% had an affected relative)	NR	Australia, USA, Canada	14657 participants/482 events	20 to 70 years at baseline	Cohort	10-year risk	Median FU 10 years (interquartile range = 6.1–10.0 years)

Table 4. Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm (BOADICEA) model: overview of synthesis and included studies (Continued)

MacInnis 2021	Women from Prospective Family Study Cohort (ProF-SC). Cohort is enriched for women with a family history of BC.	Enrolment between 1992 & 2011	Australia, USA, Canada	14,657 participants/482 events	20 to 70 years at baseline	Cohort	5-year risk, LR, RLR	Median FU 10 years
Ming 2019 (Swiss)	Women who received genetic evaluation or testing at a Swiss cancer genetics clinic or first degree relatives of males who received evaluation at the clinic	Unclear clinic has been running since 1998	Switzerland	2481 participants/ 886 events	50.78 ± 12.77 (13–89)	Retrospective dataset	Lifetime risk	NR
Quante 2015	Women recruited to the New York site of the Breast Cancer Family Registry (BCFR). Participants enriched for a family history of breast cancer.	Participants recruited to BCFR since 1995. Study dates for this particular cohort unclear	USA, Germany, Australia	1764 participants/95 events (though of these only 79 developed BC within the 10-year FU timespan of interest for prediction of 10-year risk)	20 to 70 years	Cohort	10-year risk	Unclear

BC: breast cancer; **DCIS:** ductal carcinoma in situ; **FDR:** first-degree relative; **FU:** follow-up; **hx:** history; **IQR:** interquartile range; **KARMA:** Karolinska Mammography Project for Risk Prediction of Breast Cancer; **LR:** lifetime risk; **MRI:** magnetic resonance imaging; **NA:** not applicable; **NR:** not reported; **ProF-SC:** Breast Cancer Prospective Family Study Cohort; **PRS:** polygenic risk score; **RLR:** remaining lifetime risk; **SD:** standard deviation; **SDR:** second-degree relative; **VCU:** Virginia Commonwealth University

Table 5. BRCAPRO model: overview of synthesis and included studies

Study	Characteristics of included studies							
	Participants	Recruitment	Country	Number of participants/ number of events	Age	Source	Prediction horizon	Follow-up
Model development study								

Table 5. BRCAPRO model: overview of synthesis and included studies (Continued)

Parmigiani 1998	Utilised a range of different published data sets that included different types of participants to build the model	Unclear - different study dates for each dataset used in model development	USA	NR	NR	Previously published data from several sources including: Penetrance estimates from Easton et al. 1995 & Easton et al. 1997 Cancer Incidence functions from Moolgavkar et al. 1979 & Yancik 1993 Frequency of BRCA1 mutations from Ford and Easton 1995. Prevalence of BRCA2 mutations from Andersen 1996	NR	NR
BRCAPRO model studies included in meta-analyses of both C statistic and O/E ratio								
Amir 2003	Women who attended a UK Family History clinic for evaluation	Data collected from 1987 to January 2001	UK	3150 participants (Total Study Population)/ 64 events	21-73 years median 44 years	Cohort	Predicted 10-year risk adjusted to no. of years of FU	5.27 years (range 0.10–15.00 years)
Li 2021a	Australian women recruited to the Melbourne Collaborative Cohort Study (MCCS) aged. Of these, only	Baseline (recruited) 1990-1994	Australia, UK, USA	1720 participants/ 110 events	50 to 65 years at start of follow-up	Cohort	15-year risk	Median/Mean FU NR

Table 5. BRCAPro model: overview of synthesis and included studies (Continued)

	subgroup with family history of BC applicable for this review							Start of follow-up 2003-2007 FU ended when (i) diagnosis of invasive breast cancer, (ii) follow-up time reaching 15 years, (iii) age 80 years or (iv) censor date of 1 March 2019, whichever came first.
McCarthy 2021	Women who underwent screening mammography. Of these, subgroup of women with a family history of breast cancer relevant for this review	Women presented for screening mammography between 2006-2015	USA	# participants in family history sub group unclear approx.. 24,511 #Events = 394	Aged 40-84 at baseline	Cohort	5-year risk	Median/Mean follow-up NR All participants had at least 5 years of FU.
Terry 2019	Women enrolled in the ProF-SC cohort This cohort is enriched for familial risk of breast cancer	Recruited between March 17, 1992, and June 29, 2011	USA, Canada, Australia	15732 participants/619 events	20-70 years	Cohort	10-year risk	Median follow-up of 11.1 years (IQR 6.0-14.4)
Guan 2023	Women enrolled in the Cancer Genetics Network (CGN). This cohort represents a higher-risk population enriched for family history of cancer	Enrolment began in 1999 and ended in 2010	USA	7,314 participants/ 112 events	Median age 47 (IQR 38-57) years	Cohort	5-year risk	Median follow-up 7.3 years (IQR 6.0-8.3)
BRCAPro model studies included in narrative synthesis only								
Kenan 2018	Women counselled and genotyped at an Israeli oncogenetics unit	2000 and 2005 for oncogenetic testing. Follow-up last updated for the calendar	Israel	374 participants/ 25 events	Age range at counselling was 19-85 years	Cohort	5-year, 10-year & LR	Follow-up between 12 and 17 years (12 years for those counselled in 2005, 17 for those counselled in 2000)

Table 5. BRCAPro model: overview of synthesis and included studies (Continued)

					year 2016 in 2017.	(mean age 50.9±11.4 years)		
MacInnis 2020	Women from Prof-SC cohort Cohort enriched for women with family hx of breast cancer (96.1% had an affected relative)	NR	Australia, USA, Cana- da	14657 partici- pants/482 events	20 to 70 years at baseline	Cohort	10-year risk	Median FU 10 years (interquartile range = 6.1–10.0 years)
McCarthy 2020	Women presenting for mammogra- phy screening at Newton-Wellesley Hospital Of these, only sub-group of women with family hx of breast can- cer are applicable for this review.	Recruited be- tween Feb- ruary 1, 2007 and Decem- ber 31, 2009	USA	12,994 partici- pants (with at least one FDR or SDR with breast or ovarian cancer)/ 304 events	Aged 40–84 years	Cohort	6-year risk	Mean FU 6.7 years (interquartile range = 6.3– 7.1 years) * This relates to the entire cohort. Not reported separate- ly for family history sub-group
Guan 2022	Women recruited to the Cancer Ge- netics Network (CGN) cohort. GN GCN counselees were ascertained based on having a family history of cancer 42.9% of participants had at least one female first-degree relative with breast cancer	Participants recruited be- tween 1999 and 2010	USA	7489 partici- pants/ 114 events	Median age 47 years IQR (38 – 57)	Cohort Data from the Can- cer Genet- ics Network (CGN) co- hort	5-year risk	~5-year follow-up

BC: breast cancer; **DCIS:** ductal carcinoma in situ; **FDR:** first-degree relative; **FU:** follow-up; **hx:** history; **IQR:** interquartile range; **KARMA:** Karolinska Mammography Project for Risk Prediction of Breast Cancer; **LR:** lifetime risk; **MRI:** magnetic resonance imaging; **NA:** not applicable; **NR:** not reported; **ProF-SC:** Breast Cancer Prospective Family Study Cohort; **PRS:** polygenic risk score; **RLR:** remaining lifetime risk; **SD:** standard deviation; **SDR:** second-degree relative; **VCU:** Virginia Commonwealth University

Table 6. Risk of bias (PROBAST) assessment of the Gail model

Model	Study	Prediction model Risk Of Bias ASsessment Tool (PROBAST)				
		Domain 1: participants	Domain 2: predictors	Domain 3: outcome	Domain 4: analysis	Overall judge- ment
Gail	Meta- analysed c- statistic	Amir 2003	Low	Low	Low	High

Table 6. Risk of bias (PROBAST) assessment of the Gail model *(Continued)*

	Clendenen 2019	High	High	Low	High	High
	Glynn 2019	Low	Low	Low	High	High
	Li 2021a	Low	Low	Low	High	High
	Mayo 2001	High	Unclear	High	High	High
	McCarthy 2021	Low	Low	Low	Unclear	Unclear
	Ming 2019 (USA)	High	High	Unclear	High	High
	Sevdalis 2022	High	Low	Unclear	High	High
	Terry 2019	Low	Low	Low	High	High
	Guan 2023	Low	Low	Low (overall CGN cohort) Unclear (strong family hx subgroup)	High	High
Meta-analysed O/E	Bondy 1994	Low	Low	Unclear	High	High
	Evans 2016 Chapter 2 (study 1a)	Low	Low	Low	High	High
	Glynn 2019	Low	Low	Low	High	High
	Li 2021a	Low	Low	Low	High	High
	McCarthy 2021	Low	Low	Low	Unclear	Unclear
	Terry 2019	Low	Low	Low	High	High
	Guan 2023	Low	Low	Low (overall CGN cohort) Unclear (strong family hx subgroup)	High	High
Narrative	Gong 2014	Low	Low	Unclear	High	High
	McCarthy 2020	Low	Low	Low	Low	Low
	Quante 2012	Low	Unclear	Unclear	High	High

Table 6. Risk of bias (PROBAST) assessment of the Gail model *(Continued)*

Rockhill 2001	Low	Low	Low	High	High
Ho 2023	Low	Low	Low	High	High
Lamb 2023	Low	Low	Unclear	High	High

CGN: Cancer Genetics Network; **hx:** history

Table 7. Applicability assessment for all models with external validations

Model	Study	PROBAST-Applicability			
		Domain 1: participants	Domain 2: predictors	Domain 3: outcome	Overall judgement
Gail	Meta-analysed C statistic	Amir 2003	Low	Low	Low
		Clendenen 2019	Low*	Low*	Low*
		Glynn 2019	Low*	Low*	Low*
		Li 2021a	Low*	Low*	Low*
		Mayo 2001	Unclear	Low	Unclear
		McCarthy 2021	Low*	Low*	Low*
		Ming 2019 (USA)	Unclear	Low	Unclear
		Sevdalis 2022	Low	Low	Unclear
		Terry 2019	Low	Low	Low
		Guan 2023	Low	Low	Low
	Meta-analysed O/E	Amir 2003	Low	Low	Low
		Bondy 1994	Low	Low	Low
		Evans 2016 Chapter 2 (study 1a)	Low	Low	Low
		Glynn 2019	Low*	Low*	Low*
		Li 2021a	Low*	Low*	Low*
		McCarthy 2021	Low*	Low*	Low*
		Terry 2019	Low	Low	Low
		Guan 2023	Low	Low	Low
	Narrative	Gong 2014	Low	Low	Low
		McCarthy 2020	Low*	Low*	Low*
		Quante 2012	Low	Low	Low
		Rockhill 2001	Low*	Low*	Low*
		Ho 2023	Low*	Low*	Low*
		Lamb 2023	High	Low	Unclear
Tyrer-Cuzick	Development	Tyrer 2004	Low	Low	Low

Table 7. Applicability assessment for all models with external validations (Continued)

Meta-analysed C statistic	Evans 2016 Chapter 3 (study 4)	Low	Low	Low	Low
	Glynn 2019	Low*	Low*	Low*	Low*
	Li 2021b	Low*	Low*	Low*	Low*
	McCarthy 2020	Low*	Low*	Low*	Low*
	Sevdalis 2022	Low	Low	Unclear	Unclear
	Terry 2019	Low	Low	Low	Low
	Warwick 2014	Low	Low	Low	Low
	Yala 2019	Unclear*	Low*	Low*	Unclear*
	Brooks 2023	Low	Low	Low	Low
	Guan 2023	Low	Low	Low	Low
Meta-analysed O/E	Evans 2016 Chapter 2 (study 1a)	Low	Low	Low	Low
	Glynn 2019	Low*	Low*	Low*	Low*
	Laitman 2013	Low	Low	Low	Low
	Li 2021b	Low*	Low*	Low*	Low*
	McCarthy 2020	Low*	Low*	Low*	Low*
	Terry 2019	Low	Low	Low	Low
	Brooks 2023	Low	Low	Low	Low
	Guan 2023	Low	Low	Low	Low
	Mabey 2024	Low	Low	Low	Low
Narrative	Amir 2003	Low	Low	Low	Low
	Gong 2014	Low	Low	Low	Low
	Hughes 2021	Unclear	Low	Low	Unclear
	Kenan 2018	Low	Low	Low	Low
	Li 2021a	Low*	Low*	Low*	Low*
	MacInnis 2020	Low	Low	Low	Low
	MacInnis 2021	Low	Low	Low	Low
	Quante 2012	Low	Low	Low	Low
	Quante 2015	Low	Low	Low	Low

Table 7. Applicability assessment for all models with external validations (Continued)

		Eriksson 2023b	Low*	Low*	Low*	Low*
		Eriksson 2023a	Low*	Low*	Low*	Low*
		Lamb 2023	High	Low	Unclear	High
BOADICEA	Develop- ment	Antoniou 2002	Unclear	Low	Low	Unclear
	Meta- analysed C statistic	Li 2021b	Low*	Low*	Low*	Low*
		Ming 2020	Unclear	Low	Low	Unclear
		Stahlbom 2012	Unclear	Low	Low	Unclear
		Terry 2019	Low	Low	Low	Low
		Meta- analysed O/ E	Evans 2016 Chapter 2 (study 1a)	Low	Low	Low
	Laitman 2013		Low	Low	Low	Low
	Li 2021b		Low*	Low*	Low*	Low*
	Stahlbom 2012		Unclear	Low	Low	Unclear
	Terry 2019		Low	Low	Low	Low
	Narrative	Kenan 2018	Low	Low	Low	Low
		Li 2021a	Low*	Low*	Low*	Low*
		Li 2017	Unclear	Low	Low	Unclear
		MacInnis 2013	Low	Low	Low	Low
		MacInnis 2020	Low	Low	Low	Low
		MacInnis 2021	Low	Low	Low	Low
		Ming 2019 (Swiss)	Unclear	Low	Low	Unclear
		Quante 2015	Low	Low	Low	Low
BRCAPRO	Develop- ment	Parmigiani 1998	Unclear	Low	Low	Unclear
	Meta- analysed C statistic	Amir 2003	Low	Low	Low	Low
		Li 2021a	Low*	Low*	Low*	Low*
		McCarthy 2021	Low*	Low*	Low*	Low*
		Terry 2019	Low	Low	Low	Low
		Guan 2023	Low	Low	Low	Low

Table 7. Applicability assessment for all models with external validations (Continued)

	Meta-analysed O/E	Amir 2003	Low	Low	Low	Low
		Li 2021a	Low*	Low*	Low*	Low*
		McCarthy 2021	Low*	Low*	Low*	Low*
		Terry 2019	Low	Low	Low	Low
		Guan 2023	Low	Low	Low	Low
Narrative		Kenan 2018	Low	Low	Low	Low
		MacInnis 2020	Low	Low	Low	Low
		McCarthy 2020	Low*	Low*	Low*	Low*
		Guan 2022	Low	Low	Low	Low
Claus	Narrative	Amir 2003	Low	Low	Low	Low
		McCarthy 2020	Low*	Low*	Low*	Low*
BCSC	Narrative	McCarthy 2021	Low*	Low*	Low*	Low*
		Gard 2024	Low*	Low*	Low*	Low*
Manual (adjusted Claus)	Narrative	Amir 2003	Low	Low	Low	Low
		Evans 2014	Low	Low	Low	Low
		Evans 2016 Chapter 2 (study 1a)	Low	Low	Low	Low
CARE	Narrative	Boggs 2013	Low*	Low*	Low*	Low*
iCARE-Lit	Narrative	Li 2021a	Low*	Low*	Low*	Low*
Palmer et al Model A	Narrative	Palmer 2021	Low*	Low	Low	Low*
Palmer et al Model B	Narrative	Palmer 2021	Low*	Low	Low	Low*
BR-CAPRO-BCRAT	Narrative	Li 2021a	Low*	Low*	Low*	Low*
		McCarthy 2021	Low*	Low*	Low*	Low*
		Guan 2023	Low	Low	Low	Low

*for subgroup with family history of breast cancer

Table 8. Risk of bias (PROBAST) assessment of the Tyrer-Cuzick model

Model		Study	Prediction model Risk Of Bias ASsessment Tool (PROBAST)				
			Domain 1: participants	Domain 2: predictors	Domain 3: outcome	Domain 4: analysis	Overall judge- ment
Tyrer-Cuzick	Development	Tyrer 2004	Unclear	Unclear	Unclear	Unclear	Unclear
	Meta-analysed c-statistic	Evans 2016 Chapter 3 (study 4)	Low	Low	Low	High	High
		Glynn 2019	Low	Low	Low	High	High
		Li 2021b	Low	Low	Low	Unclear	Unclear
		McCarthy 2020	Low	Low	Low	High	High
		Sevdalis 2022	High	Low	Unclear	High	High
		Terry 2019	Low	Low	Low	High	High
		Warwick 2014	Low	Low	Low	High	High
		Yala 2019	Unclear	Low	Low	High	High
		Brooks 2023	Low	Low	Low	Unclear	Unclear
		Guan 2023	Low	Low	Low (overall CGN cohort) Unclear (strong family hx subgroup)	High	High
	Meta-analysed O/E	Evans 2016 Chapter 2 (study 1a)	Low	Low	Low	High	High
		Glynn 2019	Low	Low	Low	High	High
		Laitman 2013	Low	Unclear	Low	High	High
		Li 2021b	Low	Low	Low	Unclear	Unclear
		McCarthy 2020	Low	Low	Low	High	High
		Terry 2019	Low	Low	Low	High	High

Table 8. Risk of bias (PROBAST) assessment of the Tyrer-Cuzick model *(Continued)*

	Brooks 2023	Low	Low	Low	Unclear	Unclear
	Guan 2023	Low	Low	Low (overall CGN cohort) Unclear (strong family hx subgroup)	High	High
	Mabey 2024	Low	Low	High	High	High
Narrative	Amir 2003	Low	Low	Low	High	High
	Gong 2014	Low	Low	Unclear	High	High
	Hughes 2021	High	High	High	High	High
	Kenan 2018	Low	Unclear	Low (5- and 10-year) High (lifetime)	High	High
	Li 2021a	Low	Low	Low	High	High
	MacInnis 2020	Low	Low	Low	Unclear	Unclear
	MacInnis 2021	Low	Low	Low (5-year) High (lifetime)	High	High
	Quante 2012	Low	Unclear	Unclear	High	High
	Quante 2015	Low	Unclear	Unclear	High	High
	Eriksson 2023b	Low	Low	Low	High	High
	Eriksson 2023a	Low	Low	Low	High	High
	Lamb 2023	Low	Low	Unclear	High	High

CGN: Cancer Genetics Network; **hx:** history

Table 9. Risk of bias (PROBAST) assessment of the BOADICEA model

Model	Study	Prediction model Risk Of Bias ASsessment Tool (PROBAST)
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Table 9. Risk of bias (PROBAST) assessment of the BOADICEA model *(Continued)*

			Domain 1: participants	Domain 2: predictors	Domain 3: outcome	Domain 4: analy- sis	Overall judge- ment
BOADICEA	Development	Antoniou 2002	High	Low	High	High	High
	Meta-analysed c-statistic	Li 2021b	Low	Low	Low	Unclear	Unclear
		Ming 2020	High	Unclear	Unclear	High	High
		Stahlbom 2012	High	Unclear	Unclear	High	High
		Terry 2019	Low	Low	Low	Low	Low
		Evans 2016 Chapter 2 (study 1a)	Low	Low	Unclear	High	High
	Meta-analysed O/E	Laitman 2013	Low	Unclear	Low	High	High
		Li 2021b	Low	Low	Low	Unclear	Unclear
		Stahlbom 2012	High	Unclear	Unclear	High	High
		Terry 2019	Low	Low	Low	Low	Low
	Narrative	Kenan 2018	Low	Unclear	Low (5- and 10-year risk) High (lifetime)	High	High
		Li 2021a	Low	Low	Low	Unclear	Unclear
		Li 2017	Low	Low	High	High	High
		MacInnis 2013	Low	Low	Low	High (for subgroup with ≥ 1 FDR) Unclear for whole cohort	High (for subgroup with ≥ 1 FDR) Unclear for whole cohort
		MacInnis 2020	Low	Low	Low	Unclear	Unclear
		MacInnis 2021	Low	Low	Low (5-year) High (lifetime)	High	High

Table 9. Risk of bias (PROBAST) assessment of the BOADICEA model *(Continued)*

	Ming 2019 (Swiss)	High	High	Unclear	High	High
	Quante 2015	Low	Unclear	Unclear	High	High

FDR: first-degree relative

Table 10. Risk of bias (PROBAST) assessment of the BRCAPRO model

Model	Study	Prediction model Risk Of Bias ASsessment Tool (PROBAST)				
		Domain 1: participants	Domain 2: predictors	Domain 3: outcome	Domain 4: analysis	Overall judgement
BRCAPRO	Development	Parmigiani 1998	High	Unclear	Unclear	High
	Meta-analysed c-statistic	Amir 2003	Low	Low	Low	High
		Li 2021a	Low	Low	Low	Unclear
		McCarthy 2021	Low	Low	Low	Unclear
		Terry 2019	Low	Low	Low	Low
		Guan 2023	Low	Low	Low (overall GCN cohort) Unclear (strong family hx subgroup)	Low (overall GCN cohort) High (strong family hx subgroup)
	Meta-analysed O/E	Amir 2003	Low	Low	Low	High
		Li 2021a	Low	Low	Low	Unclear
		McCarthy 2021	Low	Low	Low	Unclear
		Terry 2019	Low	Low	Low	Low
		Guan 2023	Low	Low	Low (overall GCN cohort) Unclear (strong family hx subgroup)	Low (overall GCN cohort) High (strong family hx subgroup)

Table 10. Risk of bias (PROBAST) assessment of the BRCAPro model *(Continued)*

Narrative	Kenan 2018	Low	Unclear	Low (5- and 10-year risk)	High	High
	High (Lifetime)					
	MacInnis 2020	Low	Low	Low	Unclear	Unclear
	McCarthy 2020	Low	Low	Low	High	High
	Guan 2022	Low	Low	Low	Unclear	Unclear

CGN: Cancer Genetics Network; **hx:** history